Joint Pell certificates and coefficient sharing for ZFC consistency

Round 23 research report and reproducible artifacts

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Abstract

This round improves several explicit polynomial presentations of syntactic ZFC consistency, while keeping their domains and size objectives separate. The principal unrestricted-integer quartic has 210 variables, 150 quadratic residuals, 5276 collected monomials and coefficient height 2^{24}. A 209-variable quartic trades a larger height for one fewer coordinate.

A new unweighted joint Pell certificate authenticates signed indices and power requests with two successive integrality arguments. Its selected joint equation reaches 113 variables at degree 16. A separate unfused monic-elimination control reaches 138 variables at degree eight; the smaller count has a substantially larger expanded support. Constructor fusion and inlining give a height-one quartic in 484 unrestricted integers, and reduce support at 493 variables from 24828 to 17058. Two complete ten-positive-integer circuits have exact degree/gate pairs 202/99 and 190/120, using the unchanged coherent Start-only coefficient tuple. Their lower degrees cost additional gates.

The report gives the unbounded witness arguments, precise inherited calibration boundary, complete native equations or finite substitution specifications, actual coefficient artifacts, and reproducible finite checks. It compares all nine incoming packages and selected current repository sources at a pinned snapshot. Ordinary ZFC proves that each selected equation's unsolvability on its stated domain is equivalent to Con(ZFC) through the identified source calibration. No truth of consistency, globally simplest equation, external refereeing or new end-to-end proof-assistant build is claimed.

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1 Selected results and what changed

The useful results are four distinct improvements, not a single ordering of "simplicity." Every coefficient, coordinate and arithmetic gate is fixed in the supplied artifacts. All statements of equivalence use ordinary unbounded proofs composed with the identified inherited logical calibration.

1.1 Unrestricted integer quartics

All four selected direct equations are sums of squares of quadratic polynomials on $\mathbb{Z}$ and have exact degree four.

Endpoint	Variables	Rows	Terms	Height
Principal	210	150	5276	2^{24}
Dimension	209	149	5278	2^{46}
Balanced	211	151	5401	65537
Lower height	212	152	18381	5200

The principal has the same dimension and row count as incoming Pell Growth 210, but six fewer collected terms and a height reduction from 2^{40} to 2^{24}. The 209-variable point is a dimension tradeoff, not a universal improvement: its height is larger. The balanced 211-variable point improves the incoming balanced 214-variable control's dimension, rows, support and height. The 212-variable point does not beat the retained Round22 heights 4096 and 2304; those remain genuine alternatives.

The new arithmetic work replaces a positive quadratic horizon by a signed affine horizon whose sign is recovered from an existing power request. Threshold four needs new local-budget and odd-selector completeness arguments. A separate, explicitly certified affine-atom frame and difference orientation lower coefficients. Monic growth splitting then makes a finite variable/height tradeoff. Appendix A prints every residual of the principal equation.

1.2 Joint pooling and exact elimination

The selected reciprocal equations also have unrestricted integer domains. Their degree is not four.

Endpoint	Variables	Rows	Degree	Terms
Joint dimension	113	53	16	1030202
Sparse companion	114	54	16	127818
Lower degree	138	78	8	12099

The two degree-16 heights are 16250018885420533678080; the degree-eight height is 2^{50}. The million-term endpoint is not advertised as sparse. The complete coefficients are supplied despite the compact proof/circuit presentation. The new theorem avoids a premature sign assumption by using one small integer sum, then a second integer sum whose nonzero summands have a common sign. A retained-frame-row numerator change allows one further monic elimination without increasing the selected degree.

The 138-variable point is an unfused monic-elimination control, not a saving credited to the new joint theorem. These endpoints use the conservative 209-variable square-frame source, not the weak-frame quartic source above. Appendix B prints that exact source, and the following finite ledger specifies the pools and all ordered substitutions. The incoming Mixed Base 112-variable point has degree 24; its smaller count does not make it a degree-16 comparison. Its 143-variable degree-eight point has a lower reported height bound than the new 138-variable equation, so the latter is not a uniform size improvement.

1.3 Coefficient height one

The new minimum has 484 variables and 24131 terms, versus the retained 493-variable, 24828-term point. At exactly 493 variables, the new support is 17058, a reduction of 7770 terms, about 31.3%. The new catalog also includes 533/11985, 600/8241 and 696/7113 variable/support tradeoffs. Every endpoint has exact degree four, unrestricted integer coordinates, constant one and all nonzero collected coefficients equal to ± 1.

The new operations fuse available constant and scale products into source rows, prune unused monic constructors, inline low-use constants and expand rare scales. Their exact source maps precede the unchanged nonnegative network energy. All nine selected Round22 height-one endpoints are dominated by members of the new finite catalog; no exhaustive frontier is claimed.

1.4 Ten strictly positive inputs

The complete new scalar equations have these exact live-circuit metrics:

Center and transport	Variables	Degree	Gates
Scalar quotient center, $L = bV$	10	202	99
Whole-product center, $L = UEb^2$	10	190	120

Both keep all four authentic Start-only 4127 numerical ports, totaling 97046 raw bytes. The new factor UE makes the squared transport valid even at the pre-rank auxiliary index-three case. Exact cancellation in the full norm defect, followed by actual leading-form substitution, establishes the degrees. The retained 214/87 and incoming 206/90 and 210/89 points remain lower-cost alternatives. None of these is a ten-unrestricted-integer result.

The source snapshot, complete comparison inventory, proof boundaries and rebuild commands appear later. In particular, a finite verification receipt is evidence about the emitted algebra and bytes, not a proof that ZFC is consistent or a replacement for an unbounded source theorem.

2 Logical meaning and exact polynomial definitions

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$ and $\mathbb{Z}$ is unrestricted. The domain belongs to an equation's definition. An equation over positive integers is not an unrestricted-integer equation; a natural-domain chart needs a separate sign-restoration proof.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected polynomial P on its stated domain D, the ordinary claim is

$$\exists x \in D^n \; P(x) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{1}$$

The unbounded arithmetic transformations below compose with the explicitly identified inherited calibration. Formalizing this proof chain in ZFC yields

$$\mathrm{ZFC} \vdash \left[\forall x \in D^n \; P(x) \neq 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

This is an equivalence, not a proof of either side. It does not assert consistency, decide solvability, give a globally smallest polynomial, or provide a new end-to-end proof-assistant kernel check.

2.1 The retained calibration boundary

For the direct arithmetic families, the named logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The first relates the fixed numerical calculus to a finite ZF refutation; the second transports between ZF and ZFC refutations. Their exact sources, immutable historical pins and ordinary arithmetic bridges are retained in the combined source closure. The coefficient-height-one families have separately identified original quadratic sources and transfer their inherited zero-set calibrations through the explicit graph maps below. They are not silently rebased to the newest direct system.

The positive scalar branch uses its own machine, native-encoding and positive-P66 calibration at one complete tuple of fixed coefficients. Its source theorem is not inferred merely from the sizes of those numerals. The whole tuple, exact finite language, coefficient recipes and unbounded native proof are supplied; replacing a single port in isolation is not licensed. The source package's old pin and this round's current comparison pin are different records and are never substituted for each other.

There are four different kinds of evidence here. The logical calibration relates finite syntax to the original arithmetic source. Unbounded proofs then establish Pell classification, cofinal witness construction, CRT compatibility and sign recovery. Exact algebra transfers the source zero set to finite new arrays or circuits. Finally independent finite checkers reconstruct those actual arrays, not merely an abstract theorem instance. A hash proves byte identity, not a theorem. Coefficient reconstruction proves an identity, not an unbounded Pell assertion. Successful finite replay is not an end-to-end formal derivation of (1).

2.2 Literal arrays and graphs

For each residual endpoint, the ordered `variables` array in `equation.json` fixes coordinates $x_0, \dots, x_{n-1}$. A term $[c, [i_1, \dots, i_d]]$ denotes $c \prod_j x_{i_j}$; an empty index list denotes the constant c. Repeated indices mean ordinary powers. If the listed residuals are $R_0, \dots, R_{m-1}$, the literal polynomial is

$$P(x) = \sum_{j=0}^{m-1} R_j(x)^2. \tag{3}$$

The sum of squares is zero exactly when every row is zero, on each of the stated integer domains. The degree of a merged reciprocal row need not be two; the advertised total degree is separately checked after collection. The quartic endpoints have degree-at-most-two rows and a nonzero collected quartic coefficient. Adjacent collected arrays give every actual integer coefficient, including a nonzero constant. There is no variable exponent, factorial, division, radical or sequence primitive in those arrays.

The height-one endpoints are defined by their complete collected coefficient arrays. Their graph certificates establish a nonnegative energy identity with an explicit integral lift and inverse projection. An independent checker reconstructs every energy coefficient and compares the complete output, including equality and unit terms.

A scalar circuit is a finite directed acyclic graph whose leaves are the stated independent variables or supplied fixed integer constants. Every internal gate is binary addition, subtraction or multiplication. It therefore denotes a unique ordinary polynomial over $\mathbb{Z}$. Registers are abbreviations, not new quantified unknowns. Fixed powers in the printed formula stand for the specified multiplication gates; no quotient or square-root gate is used. All numerical ports are supplied in full, not replaced by a hash or a promise that a suitable large integer exists.

2.3 Metrics and finite comparisons

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$,

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes the constant if nonzero. Residual count is not support. Every independent coordinate must be live after collection. Scalar cost counts live binary arithmetic gates, including

squaring and multiplication by fixed constants; reading a leaf and copying a register are free. Dead gates do not count. Circuit gate count is not bit complexity. When no full scalar expansion is supplied, its support and full coefficient height are unmeasured, not zero and not the maximum literal leaf.

Pareto comparisons hold only among the listed, certified presentations on compatible domains, with degree fixed where stated. A point dominates another if it is no worse in every compared measure and strictly better in at least one. Smaller dimension, lower degree, sparser support, coefficient height one and fewer circuit gates are distinct objectives. No worldwide priority or global minimality conclusion follows from this finite catalog.

3 The retained proof-calculus architecture

The numerical calculus has truth code 0, falsity code 1, modus ponens, generalization, eight logical axiom schemata, and a guarded seed schema. Its literal square-shell constructors are

$$\begin{aligned} I(a,b) &= 2((a+b)^2+a), & M(a,b) &= I(a,b)+1,\\ A(a,b) &= 2((a+b)^2+3a+4b+3), & E(a,b) &= A(a,b)+1. \end{aligned}$$

They encode implication, membership, universal quantification, and equality. Put $N(a) = I(a,1)$ and $X(a,b) = N(A(a,N(b)))$. The pairing maps $Q_0 = (a+b)^2+a$ and $Q_1 = (a+b)^2+3a+4b+3$ enumerate complementary square shells; parity supplies the four constructors. The inherited decoding theorem transports finite refutations between these codes and the original Cantor-coded C_{222} calculus, fixing 0 and 1 [7].

For clarity the retained logical codes are

$$\begin{aligned} r_1 &= X(a,E(a,b)),\\ r_2 &= I(A(a,I(b,c)),I(A(a,b),A(a,c))),\\ r_3 &= I(E(a,b),I(M(c,a),M(c,b))),\\ r_4 &= I(A(a,A(b,c)),A(b,A(a,c))),\\ r_5 &= I(E(c,a),I(M(c,b),M(a,b))),\\ r_6 &= I(X(a,A(a,b)),b),\\ r_G &= I(E(c,a),I(E(c,b),A(M(a,b),E(a,b)))),\\ r_8 &= I(I(I(c,d),a),I(I(a,c),I(b,c))). \end{aligned}$$

Define

$$\begin{aligned} \mathrm{Req}(b,k,m) &= A(b,I(m,N(A(b,I(m,N(M(b,k))))))),\\ S(a,b,k) &= A(k,I(M(k,a),M(k,b))),\\ T(a,b,k,m) &= N(A(a,I(S(a,b,k),\mathrm{Req}(b,k,m)))),\\ \mathrm{Clos}(a,b,k,m) &= A(b,I(M(b,k),N(A(a,I(m,T(a,b,k,m)))))),\\ \mathrm{nks}(a,b,c) &= A(c,I(\mathrm{Clos}(a,b,c,c),N(\mathrm{Req}(b,c,c)))),\\ \mathrm{RV}(a,b,c) &= I(M(a,b),A(c,M(a,b))),\\ r_9 &= N(I(\mathrm{RV}(a,b,c),\mathrm{nks}(a,b,c))). \end{aligned}$$

The last schema is allowed when $c \neq a,b$. Its arithmetic guard is

$$G = (c-a)^2(c-b)^2-(d+1)^2 = 0, \qquad a,b,c,d \in \mathbb{N}.$$

Such a d exists exactly when the two disequalities hold. The parameter d can separately be an ordinary formula parameter in the r_8 branch. The alternatives do not require both interpretations simultaneously.

The proof transcript is encoded by positive weights $w(j) = 1 + Fj$. A backward axiom step has $S'w(r) = S$, modus ponens has $S'w(b) = Sw(a)w(I(a,b))$, and generalization has $S'w(A(a,b)) = Sw(b)$. Truth uses $S' = S$. Let $D_r = S'w(r) - S$ and define the analogous MP and generalization differences. The retained nonnegative local polynomial is

$$\mathcal{L} = (S' - S)^2 \prod_{r \in \mathcal{R}} D_r^2 (D_{r_9}^2 + G^2) D_{\mathrm{MP}}^2 D_{\mathrm{gen}}^2 + B_0^2 + B_1^2.$$

Here $\mathcal{R} = \{r_1, r_2, r_3, r_4, r_5, r_6, r_G, r_8\}$, and B_0, B_1 are the two beta-remainder equations. Thus $\mathcal{L} = 0$ means that one valid proof-rule branch and both beta equations hold. The retained Round22 reference construction uses this local polynomial unchanged. The new source deltas below identify precisely which arithmetic rows change.

4 Formula bounds without a range test

The old arithmetic source separately required a derived sum of formula labels to lie below the proof horizon in every local row. That required its own numerator power, binomial divisibility test, and Pell index. The replacement needs only a fixed, much more generous syntactic bound.

Lemma 4.1 (Constructor-depth bound)**.** *For $t \geq 16$, every formula label needed by a local rule with $0 \leq a, b, c, d < t$ is smaller than $t^{2^{20}-1}$. In particular it is at most $L = t^{2^{20}}$.*

Proof. For a positive integer s, if integer inputs $0 \leq x, y < s$, then $x, y \leq s - 1$ and each of $I(x,y), M(x,y), A(x,y), E(x,y)$ is less than $8s^2$. For the largest constructor, $A(s-1, s-1) = 8s^2 - 2s$, and E adds one. If the inputs are less than t^e, $e \geq 1$, each output is consequently less than t^{2e+1}. Starting with exponent one, a term of constructor depth h is bounded by $t^{2^{h+1}-1}$.

The exact syntax trees of the nine axiom codes have depths

$$4, 3, 3, 3, 3, 5, 4, 3, 19.$$

The MP and generalization labels have depth one; parameter labels have depth zero. The depth certificate checks the trees against the pinned literal constructor source, including all occurrences of N and X. The maximum depth nineteen proves the result. No asymptotic estimate or coefficient rounding enters this argument. □

Lemma 4.2 (Sharp product separation)**.** *Let L, t, F be positive integers with $L \geq t \geq 1$ and $F \geq L^t$. Consider two finite lists of positive integer labels at most L. Suppose the first has at most t entries and the second at most $2t + 1$ entries. If their products of weights $1 + Fj$ agree, their label multisets agree.*

Proof. First the lists have equal length. If their lengths differ, let the smaller length be $r \leq t$. The product for the smaller list is at most $L^r(F+1)^r$, while the larger is at least $(F+1)^{r+1}$. But $F + 1 > L^t \geq L^r$, a contradiction.

Write their common length as $m \leq t$. For $1 \leq k \leq m$, each elementary symmetric coefficient e_k of the labels satisfies

$$1 \leq e_k \leq \binom{m}{k} L^k \leq m^{m-k} L^k \leq L^m \leq F.$$

The middle estimate uses $\binom{m}{k} = \binom{m}{m-k} \leq m^{m-k}$. Thus the corresponding coefficients of the difference of $\prod(1 + jX)$ have absolute value strictly less than F; the constant coefficient is zero. If the polynomial difference were nonzero, divide its value at F by the lowest power with nonzero coefficient and reduce modulo F. A nonzero integer of absolute value less than F would be divisible by F, impossible. The two polynomials coincide. Their linear factors over $\mathbb{Q}$ recover every label with multiplicity. The empty-list case is immediate. Labels zero, if present, contribute unit factors and are discarded before this argument. □

The boundary $F = L^t$ is permitted. Bounding coefficient differences by twice the largest coefficient would miss this useful fact: both coefficients are positive, so their difference has absolute value at most $F - 1$.

The Round22 weighted family uses fixed positive integers c, h with $h \geq 16$, $4 \mid h$ and $ch^3 \geq 2^{20}$, and the semantic choices

$$t = h + 4z^2, \qquad T_2 = t^2, \qquad K = cT_2^2, \qquad F = t^K. \tag{4}$$

Then $K \geq 2^{20}t$, so $F \geq L^t$ exactly as required. The selected weighted source has $(c, h) = (32, 32)$. The dimension companion instead uses

$$t = 1024 + 4z^2, \qquad K = 1024t^2, \qquad F = t^K.$$

Again $K \geq 2^{20}t$, including equality at the smallest horizon. The separate T_2 coordinate is not needed in this companion. All horizons are cofinal multiples of four; all other arguments need only $t \geq 16$ and $F \geq t^t$. The actual K, not an obsolete exponent expression, is included in the square frame and orientation ledger. Only the polynomial horizon and exponent relations are direct rows; the power relation is certified by the retained Pell mechanism.

5 The factorial-free arithmetic front

Use the following semantic quantities:

$$B = \binom{F+t}{t}, \qquad N = B^x, \tag{5}$$

$$y = F(2v+1), \qquad C = 2c_0, \tag{6}$$

$$H = 2 + 4(C + a + b + c + d + S + S' + u + v_q), \qquad q = (BH)^x. \tag{7}$$

The nine large CRT representatives are $a, b, c, d, S, S', u, v_q, C$; their local residues encode one proof row. The independent names c_0 and v in (6) are unrestricted integer witnesses, not extra local schema parameters.

The endpoints and row-index congruence are

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + [1 + (t+1)y]u_t, \tag{8}$$

$$B\eta = C + F + 1. \tag{9}$$

For a local row with index $i = C \bmod p^{ex}$, the beta equations are

$$x = S + [1 + (i+1)y]u, \qquad x = S' + [1 + (i+2)y]v_q.$$

The retained ranges are $a, b, c, d, C < t$, $S, S' \leq y$, and $u, v_q < x$ on local decoded residues. There is no longer a range test on a derived formula-label sum.

5.1 Why one binomial modulus covers every row

Here is the inherited selector theorem in the form used by this construction [2]. Let $\Lambda_t = \operatorname{lcm}(1, \ldots, t)$. If $R \geq \Lambda_t^2$ and $B = \binom{R+t}{t}$, then:

1. for every $1 \leq i \leq t$, some full prime power $p^e \| B$ divides $R + i$ and satisfies $p^e > t$;
2. every full prime power $p^e \| B$ divides $R + i$ for at least one such row i.

Both directions matter. The first authenticates each row; the second makes the reverse CRT construction compatible at every prime of B.

For completeness, the elementary proof is short. The valuation formula shows $t\binom{t-1}{i-1} \mid \Lambda_t$. Since $R+i > \Lambda_t^2$, some prime has $w = v_p(R+i) > 2a$, where $a = \lfloor \log_p t \rfloor$. Other factors $R+j$ have the same p-valuation as $j-i$. Consequently

$$v_p(B) = w - v_p\left(t\binom{t-1}{i-1}\right) > a, \qquad v_p(B) \le w,$$

which proves the first statement. For the second, let $w = \max_i v_p(R+i)$. In Legendre's formula for $v_p(B)$ every summand is zero or one, and all levels above w vanish. Hence $v_p(B) \le w$; a row attaining the maximum is compatible.

The standard elementary estimate $\Lambda_t \le 4^{t-1}$ gives $F \ge t^t \ge 16^t > \Lambda_t^2$. Therefore $R = F$ is valid. Unlike the parent, this construction has no equation requiring a variable to equal a factorial.

5.2 The range lemma and precision

For a prime p, integers $A \ge 0$ and $k, e \ge 1$,

$$p^e \mid \binom{A}{k} \quad \Longrightarrow \quad A \bmod p^e < k. \tag{10}$$

If none of the k consecutive factors $A-k+1, \dots, A$ is divisible by p^e, the valuation formula has at most $e-1$ nonzero levels, a contradiction. A factor divisible by p^e gives the claimed residue. The convention $\binom{A}{k} = 0$ for $A < k$ covers the boundary.

At a selector $p^e \| B$, the range for C and (9) give the exact local index $i-1$ in $0, \dots, t-1$. The other ranges recover the bounded local tuple modulo p^{ex}. The retained local bound is

$$0 \le \mathcal{L} < (2+t+x+y+F)^{2^{23}} < 2^x \le p^{ex}. \tag{11}$$

Its first inequality is the existing literal C_{222} circuit majorant [6]; changing the size of F does not change that polynomial bound. Once signs are recovered, $x \ge 2+(t+1)y$ and $y \ge F$ imply $2+t+x+y+F < 2x$ and $x \ge F+1 > 2^{32}$. If $2^n \le x < 2^{n+1}$, $n \ge 32$, then $2^{23}(n+2) < 2^n \le x$, proving the second strict inequality in (11). Thus $N \mid \mathcal{L}$ forces each selected local value to be zero.

6 A freely chosen CRT index as the Pell anchor

The incoming anchor change is independent of the detailed syntax of C_{222}. An integer used as a CRT representative can be enlarged by whole multiples of its modulus without changing the finite transcript it represents. We use that freedom to make C the largest requested Pell index.

Write $X_j(A), Y_j(A)$ for the Pell polynomials, defined by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2-1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

Signed indices have $X_{-j} = X_j$, $Y_{-j} = -Y_j$. The square frame is $W = 2 + \sum \alpha^2$, where every actual power base, index, and proposed output occurs among the α. The retained odd Pell frame proves $|A| > W^W$ and $|A-1| > 4W^3$.

The fourteen nominal indices are

$$t,\ F+t,\ a,b,c,d,\ S,\ y+1,\ S',\ u,\ x,\ v_q,\ C,\ K. \tag{12}$$

The former derived-label pair is reassigned to K; the existing C pair becomes the anchor. With anchor coordinates (U, Y), retain auxiliary Pell pairs (J_X, J_Y) and (V_X, V_Y) and comparison center D satisfying

$$U^2 - (A^2-1)Y^2 = 1, \qquad\qquad J_X^2 - (A^2-1)J_Y^2 = 1,$$

$$V_X^2 - (D^2-1)V_Y^2 = 1, \qquad J_Y = Y^2(2e+1),$$
$$D = 1 + 4J_Y(2d_0+1), \qquad D = A + J_X f_0,$$
$$V_X = U + J_X g_0, \qquad V_Y = C + 4Y h_0.$$

The squared bound is

$$Y^2 = C^2 + 1 + \sum_{i \neq \text{anchor}} Y_i^2 + r^2 + s^2 + w(w+1). \tag{13}$$

All these variables are unrestricted integers. Multiplication wires turn the displays into quadratic residuals; the final emitted data contain the actual wires, not a higher-degree shorthand.

Proposition 6.1 (Anchor substitution)**.** *The retained signed anchor and pooled orientation proof applies with nominal index C in place of the derived formula-label index. No monotonicity or syntactic interpretation of C is required in soundness. For completeness, C can be chosen even and larger than all other indices in* (12) *while preserving every required CRT residue.*

Proof. Equation (13) gives $|Y| > |C|$, so $Y \neq 0$. Pell classification and the retained squared-divisibility argument give an absolute index $i > 0$ with $|Y| = Y_i(|A|)$ and $i = \pm C \pmod{2|Y|}$. Since $1 \leq i \leq |Y|$ and $|C| < |Y|$, one obtains $C \neq 0$ and $i = |C|$. Every other classified index is strictly smaller than $|C|$ by the same squared bound.

The pooled orientation row is, modulo its retained norm identities,

$$\sum_i \big((X_i - 1)^2 + (Y_i - k_i)^2\big) = (A-1)Q_0.$$

Modulo $A-1$, write $s_i \in \{-1, 1\}$ and r_i for the small abscissa and ordinate residues of each classified pair. The absolute value of r_i is its classified index, hence at most $|C| < W$. Thus $|s_i - 1| \leq 2$ and $|r_i - k_i| < 2W$. For the fourteen pairs their squared-error sum satisfies

$$0 \leq \sum_i \big((s_i - 1)^2 + (r_i - k_i)^2\big) < 4 \cdot 14(W^2+1) < 4W^3 < |A-1|.$$

It must vanish, identifying every pair with the canonical signed pair at its nominal index. This includes the new K pair and the C anchor. These are exactly the hypotheses and steps of the inherited signed-anchor proof [5]; the name or origin of the nominal index never enters.

For the reverse construction, first fix t, K, F, y, x, B, N and the prescribed residue classes of the nine representatives. None of these choices depends on the eventual representative C. Since t is divisible by four and $F = t^K$, $v_2(F) = Kv_2(t) > \lfloor \log_2 t \rfloor$. Therefore $v_2(F+i) = v_2(i)$ for $1 \leq i \leq t$, and B and N^2 are odd. Each prescribed class modulo N^2 has cofinally many positive even representatives, with step $2N^2$. Choose the other eight representatives first, then choose C in its class above all the other fixed indices. This also preserves $B \mid C + F + 1$, since $B \mid N^2$. Only afterward are H, q, W, A chosen. There is consequently no cyclic size requirement.

For canonical pairs at a positive sufficiently large A, the strict index inequality makes the slack in (13) nonnegative. Since C is even, $Y_C(A)$ is even and $X_C(A)$ odd. The inherited explicit auxiliary choice at index $CY_C(A)$ makes $J_Y/Y_C(A)^2$ odd; a two-modulus CRT then constructs D and the remaining comparison coordinates. Every nonnegative slack has the ternary representation $r^2 + s^2 + w(w+1)$. Thus all anchor witnesses exist. □

7 Power certificates and the noncircular sign argument

The polynomial retains sixteen power requests: three individual requests for $F = t^K$, $N = B^x$, $q = (BH)^x$; ten at base $q+1$ for the binomial numerators; and three at base q for q^t, q^{y+1}, q^x. The incoming square frame has thirty-five actual atoms, including duplicates where the same

integer has two roles. Its coefficient identity is independently checked against every requested base, index, and output.

For a block at nonzero base n, let $Z_i = AY_i$ and let m_i be the proposed output. The inherited pooling mechanism uses just two rows:

$$K_n - 2\sum_i Y_i(m_i - X_i) - (2A - n)L_n = 0,$$
$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK_n + L_n = 0.$$

They imply divisibility of the sum of squared power errors by $T_n = 2An - n^2 - 1$. To handle signed indices, put $b = \lfloor \sqrt{W - 1} \rfloor$ and multiply by n^{2b}. Canonical Pell congruences then give

$$T_n \mid E = \sum_i (m_i n^b - n^{b+k_i})^2, \qquad 0 \le E \le 4rb^{4b} < W^W < |T_n|.$$

Here $r \le 15$, every exponent $b + k_i$ is nonnegative, and $\gcd(n, T_n) = 1$. Consequently every output is exact; at $|n| \ge 2$ the index is nonnegative, since a negative index would force an integer times a positive power of $|n|$ to equal one. The individual requests use the same canonical power argument. All displayed exponentials occur in this proof, not as operations in the final polynomial.

It remains essential to justify the bases before interpreting these requests. The order is as follows.

1. The known $t \ge 16$ and positive K give $F = t^K > 0$, even. Since $C = 2c_0$, equation (9) has odd right side, so $B \ne 0$. Independently, $H \equiv 2 \pmod 4$ for every signed representative tuple. Thus $|BH| \ge 2$.

2. The individual BH request gives $x \ge 0$ and $q = (BH)^x$. Equation $y = F(2v + 1)$ gives $|y| \ge F$. If $x = 0$, the last endpoint would make the integer $1 + (t + 1)y$, whose absolute value exceeds one, divide -1. Therefore $x \ge 1$, $|q| \ge 2$.

3. The q block is now legitimate. Its $y + 1$ index is nonnegative. A negative y would be at most $-F < -1$, so $y \ge F$. The first endpoint excludes $u_0 < 0$, and the last excludes $u_t \le 0$. Thus $x \ge F + 1$ and $x \ge 2 + (t + 1)y$.

4. Hence $|q + 1| \ge 2$. Its block recovers the nonnegative signs of all nine representatives and all their exact numerator powers. Now $H \ge 2$. No earlier step needed these signs.

5. The first binomial digit authenticates $B > 0$, as proved next. It follows that $q > 0$ and $N = B^x > 0$, completing the sign recovery.

7.1 Exact recovery of the digits

Let $M = q^x$. Ten normalized remainder rows share

$$M^2 = 1 + 4\sum_{i=1}^{10} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma + 1). \tag{14}$$

For a denominator $D_i = q^{k_i}$, the ratio equation $D_i a_i = M$ and normalized digit equation imply $r_i = \rho_i / a_i \in \mathbb{Z}$ and the usual unscaled division equation. The norm gives $|r_i| < |D_i|/2$, even before signs of the radix are known. The retained Round22 reference source uses this coefficient-four sphere exactly. Later source variants must authenticate their own literal norm. A true low binomial remainder with absolute value below $|D_i|/32$ must equal r_i, by congruence and the strict interval width.

For the first digit, every coefficient of $(q + 1)^{F+t}$ through degree t is at most $(F + t)^t \le 2^{t(F-1)}$. On the other hand, $|q| \ge 2^x \ge 2^{(t+1)F+2}$. The true low remainder and the t-th coefficient are

therefore smaller than their needed denominators divided by 32. The normalized first row gives $B \equiv \binom{F+t}{t} \pmod{q}$. Also $|B| \le |BH| < |q|/2$, since $x \ge 3$; the positive binomial coefficient is below $|q|/32$. This proves the exact identity $B = \binom{F+t}{t} > 0$.

For every other numerator with upper argument $a \ge 0$ and requested digit level $k \le x$, $\binom{a}{j} \le \max(1, a)^x \le H^x$. Since $q = B^x H^x$ and $B \ge 2$, the same strict remainder bound follows. There is no derived-label numerator needing a separate degree estimate here. The ratios prove the necessary denominator divisibilities, and the decoded digit equations yield the nine retained range predicates.

8 Soundness and completeness of the retained reference front

8.1 From an integer zero to a finite refutation

The previous sections recover every sign, power, binomial digit, and local range from an arbitrary integer zero. The selector theorem and (11) supply exactly t valid local proof steps. Their beta states join because the same x, y determine the unique remainders modulo $1 + (i + 1)y$.

The possible boundary $y = F$ must not be overlooked. Then the first beta remainder is zero, whereas the last is one. Every valid local weight equation propagates a zero state to zero, since all weights are positive. This contradiction proves $y > F$. Thus the initial and terminal states are respectively $F + 1 = w(1)$ and 1.

Multiplying the local equations and cancelling the nonzero intermediate states gives equality between the multiset of conclusion weights and the initial falsity weight together with all premise weights. There are at most t nonunit conclusion factors and at most $2t+1$ nonunit premise factors. All local labels are at most L by Lemma 4.1. Lemma 4.2 gives equality of the corresponding label multisets.

Finally, a balanced single-conclusion rule flow implies a finite derivation of its distinguished target. To see this without assuming the flow is acyclic, close the finitely many displayed rules under derivability, starting with the valid axiom and truth conclusions. If falsity were outside that closure, every rule with an outside conclusion would have at least one outside premise. Summing "outside conclusions minus outside premises" over all occurrences would be at most zero. The recovered balance makes the sum exactly one, contributed by the target falsity. Contradiction. The finite closure computation itself extracts a derivation.

8.2 From a finite refutation to an integer zero

Start with a finite backward proof-tree expansion. Choose the cofinal horizon $t = h + 4z^2$ (or $t = 1024 + 4z^2$) larger than its length and every local schema parameter, including all freshness witnesses. There is no need to make t exceed the full expanded formula codes. Set $K = ct^4$ (or $K = 1024t^2$) and $F = t^K$; form all finite pending-multiset products with weights $1 + Fj$, and pad the transcript with truth steps to length t.

Let O be the odd part of $t!$. Choose an arbitrarily large positive odd multiple w of O and put $y = Fw$. Since $v_2(F) = Kv_2(t) \ge 2t > v_2(t!)$, one has $t! \mid y$. Choose w large enough that $y > F$ and every pending state is at most y. The moduli $1 + (i + 1)y$, $0 \le i \le t$, are pairwise coprime: a common prime would divide both y and one of these moduli after taking an index difference at most t. CRT gives x with the prescribed state remainders; its residue class is cofinal, so all needed endpoint and quotient inequalities can be met.

Now set $B = \binom{F+t}{t}$ and $N = B^x$. At every full prime power $p^e \| B$, select one compatible row and choose all nine representatives to match that row modulo p^{2ex}. The generalized CRT theorem justifies this construction at every prime, including primes not used as the distinguished selector of any row. Polynomial evaluation preserves the local congruences. For a range with local value $w < k$, the falling factorial contains the factor $A - w$, divisible by p^{2ex}; the denominator loses at most $v_p(k!) \le k - 1 < x \le ex$ powers. Thus the required binomial coefficient is divisible by p^{ex}.

Choose the eight ordinary representatives first. Then choose C even, positive, and larger than every other nominal index, in its unchanged class modulo the odd integer N^2. Define H and q from their actual values, so enlargement of C strengthens the radix bounds. The ten true normalized remainders are each less than $M/32$. Their common slack in (14) is nonnegative, because $4 \cdot 10/32^2 < 1$. Every integer $n \geq 0$ has the form $r^2+s^2+w(w+1)$: apply the three-square theorem to $4n+1$, whose representation has one odd and two even coordinates.

Choose the large positive Pell center from the square frame, all canonical pairs, and the even-C auxiliary witnesses of Proposition 6.1. The canonical congruences fill the individual and pooled power quotients. This constructs an integer zero of the complete retained quadratic source with the chosen growth rows. The exact projections below carry it to the selected final equation.

9 Exact power pooling and integral algebra

9.1 The inherited two row certificate

For $|A| > 1$ let $(\chi_A(k), \psi_A(k))$, $k \in \mathbb{Z}$, be the canonical signed Pell pair. Its recurrence matrix is $\left(\begin{smallmatrix} A & A^2-1 \\ 1 & A \end{smallmatrix}\right)$, with initial pair $(1, 0)$. Its determinant is one, so all signed indices are defined. Given a nonzero base n, put

$$T_n = 2An - n^2 - 1.$$

Then $\gcd(n, T_n) = 1$, and

$$\chi_A(k) + (n - A)\psi_A(k) \equiv n^k \pmod{T_n} \qquad (k \in \mathbb{Z}). \tag{15}$$

Indeed $(n - A)^2 \equiv A^2 - 1 \pmod{T_n}$; evaluating the fundamental unit at $n - A$ gives n. Positive powers follow by multiplication, and negative powers by invertibility. The Pell functions are proof notation, not primitives in the final polynomial.

For r requests (k_i, m_i) at this base, retain the existing coordinates $X_i = \chi_A(k_i)$, $Y_i = \psi_A(k_i)$ and $Z_i = AY_i$. Introduce only two new integer coordinates K, L, and impose

$$K - 2\sum_i Y_i(m_i - X_i) - (2A - n)L = 0, \tag{16}$$

$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK + L = 0. \tag{17}$$

Every term has degree at most two. Neither T_n nor any displayed summation abbreviation introduces another independent coordinate.

Lemma 9.1. *The two rows imply*

$$\sum_i (m_i - X_i + (A - n)Y_i)^2 = T_n \left(L - \sum_i Y_i^2 \right).$$

Conversely, when each error on the left is divisible by nonzero T_n, there are integers K, L satisfying both rows.

Proof. Set $D_i = m_i - X_i + AY_i$ and $H = \sum_i Y_i(m_i - X_i)$. Eliminating K gives $\sum_i (D_i^2 - Y_i^2) - 2nH = T_n L$. Expanding $\sum_i (D_i - nY_i)^2$ subtracts exactly $T_n \sum_i Y_i^2$, proving the identity. For the converse choose

$$L = \sum_i Y_i^2 + T_n^{-1} \sum_i (D_i - nY_i)^2, \qquad K = 2H + (2A - n)L.$$

The divisibility assumption makes L integral and direct substitution proves both rows. No sign of K or L is required. □

Theorem 9.2 (Simultaneous exactness from a square frame)**.** *Suppose $W \geq 258$, $1 \leq r \leq 15$, $n \neq 0$, $|A| > W^W$, and $n^2, k_i^2, m_i^2 < W$. If the pairs are the specified canonical signed Pell pairs, the two quadratic rows force $m_i = n^{k_i}$ for every i. For $|n| \geq 2$ they also force $k_i \geq 0$; for $n = \pm 1$ all signed indices are allowed. Conversely all these exact powers admit K, L.*

Proof. Put $b = \lfloor \sqrt{W-1} \rfloor \geq 16$. Every absolute base, index and proposed output is at most b. Multiply the aggregate congruence by n^{2b}, using (15), to get

$$T_n \mid E, \qquad E = \sum_i (m_i n^b - n^{b+k_i})^2 \geq 0.$$

All exponents are nonnegative. Since $b \geq 2$, $|m_i n^b - n^{b+k_i}| \leq b^{b+1} + b^{2b} \leq 2b^{2b}$. Consequently

$$E \leq 4r b^{4b} < W^{2b+1} \leq W^W,$$

because $4r \leq 60 < W$, $b^2 < W$ and $2b + 1 \leq b^2 + 1 \leq W$. On the other hand $|T_n| \geq 2|A| - W > |A| > W^W$. Divisibility forces $E = 0$, so every summand vanishes. A nonnegative k_i gives the asserted power by cancelling n^b. If $k_i = -j < 0$, cancellation gives $m_i n^j = 1$; this excludes nonunit bases and gives the stated power at the two units. For the converse each exact error is divisible by T_n, so apply Lemma 9.1. □

The absolute-value argument includes negative radices and the unit case; assuming a positive radix before decoding its sign would not suffice. The actual square-frame ledger, canonical-pair classification and request ports are bound to the literal source by the finite checker. A different affine frame that only gives $|a| < W$ cannot silently be substituted for this square-frame hypothesis.

9.2 Exact integral projections along quadratic radicals

Write each residual as $Q_i = q_i + \ell_i + c_i$, with homogeneous quadratic part q_i, linear part ℓ_i, and integer Hessian H_i. For an integer vector v with $H_i v = 0$ for every i, exact expansion gives

$$Q_i(y + sv) = Q_i(y) + s\ell_i(v). \tag{18}$$

The mixed term is $sy^T H_i v = 0$ and the pure term vanishes since $2q_i(v) = v^T H_i v = 0$.

Theorem 9.3. *Suppose $v_p = 1$ and one slope $a_r = \ell_r(v)$ is ± 1. With $y_p = 0$, let*

$$R_i(y) = Q_i(y) - a_i a_r Q_r(y), \quad i \neq r, \qquad a_i = \ell_i(v).$$

These integer quadratics have a common zero set in bijection with that of all Q_i, with one fewer variable and residual. The lift is $x = y - a_r Q_r(y) v$, and the inverse is $y = x - x_p v$.

Proof. Put $s = -a_r Q_r(y)$. Equation (18) gives $Q_r(y + sv) = 0$ and $Q_i(y + sv) = R_i(y)$ for $i \neq r$. Conversely an old zero gives $y_p = 0$ and $Q_r(y) = -x_p a_r$, so the displayed lift restores precisely x_p. All other $R_i(y)$ vanish by the same formula. The two maps are inverse and integral; the new rows are constant combinations of restricted quadratics. No division condition remains. □

A primitive direction alone would not justify division by a nonunit slope. The actual certificates require the two literal unit conditions. These unrestricted-integer coordinate changes need not preserve a positive domain.

Lemma 9.4 (Affine consequence elimination)**.** *Suppose a nonzero constant integer combination $L = \sum_i \lambda_i Q_i$ has degree at most one and its primitive affine normalization has coefficient ± 1 on x_p. Solve it as $x_p = S(\widehat{x_p})$. Substitute in every row and omit any row r with $\lambda_r \neq 0$. The resulting quadratic system has an integer zero set in bijection with the original system.*

Proof. Every original zero satisfies $L = 0$, hence $x_p = S$ and every retained substituted row. Conversely, the substitution makes L identically zero. At a zero of the retained rows, $\lambda_r Q_r = 0$; the integers have no zero divisors, so $Q_r = 0$. Insertion of S and deletion of x_p are inverse. A unit λ_r is not needed; a unit solved variable coefficient after primitive normalization is essential. □

An elementary integer row operation $Q_i \leftarrow Q_i + kQ_j$, $i \neq j$, has inverse with $-k$. A triangular integral affine coordinate change has its explicit integral inverse. A monic quadratic definition can be eliminated only when its complete substitution retains the claimed degree. Each released certificate records these conditions and the exact index compaction. A sum of squares of the final rows is generally a *different* polynomial, with the same zero-existence statement.

10 The incoming Pell growth arithmetic interface

This section retains the complete incoming Pell Growth 210 arithmetic argument, rather than claiming it as a new result of this round [9]. Its source and independent fresh replay are identified in Section 22. The later Round23 affine horizon and coefficient changes compose with this argument at explicitly checked rows.

We describe the semantic coordinates before the exact projections. They are not all separate coordinates after projection; the incoming finite residual array is supplied in its source closure.

The positive horizon has the cofinal form

$$t = 16 + 4z^2. \tag{19}$$

The coordinate F provides formula weights $1 + jF$. The quantity $B = \binom{F+t}{t}$ will be recovered from a radix digit, and $N = B^x$ is the generalized CRT modulus. The beta spacing is $y = F(2v + 1)$ and the beta code x satisfies

$$x = F + 1 + (y + 1)r_0, \qquad x = 1 + [1 + (t + 1)y]r_t. \tag{20}$$

Eight independent CRT representatives encode the four local parameters, two pending states, and two beta quotients. The ninth representative C encodes the row index. It is explicitly even and satisfies

$$B\eta = C + F + 1. \tag{21}$$

The shared Pell machinery has its own large center A_* and a square frame containing every actual requested base, exponent and output. Its inherited signed index theorem first authenticates the principal Pell pairs at their nominal, initially signed indices. Its distinguished anchor is the pair for C, and its cap proves

$$C \neq 0, \qquad |k| < |C| \quad \text{for every other retained nominal index } k. \tag{22}$$

After the new growth pair is removed there are 13 retained principal pairs and 15 exact-power requests. These include the requests at base $q + 1$ with indices $F + t$, all eight ordinary representatives, and C; the requests at base q with indices t, $y + 1$, and x; and the two singleton requests with exponent x and bases B and the radix base.

For any integer base b with $|b| \geq 2$, the inherited signed power theorem forces a nonnegative requested exponent k and the exact output b^k. This theorem is used only after the main pairs have been oriented. The frame and anchor theorem does not require the front coordinates to have recovered their arithmetic signs. The new independent Pell witnesses will not be members of this frame; their existing output coordinate F remains a frame atom.

The incoming source changes to this interface are checked by independently reconstructing all 34 retained square-frame atoms, all 13 principal norms and multiplication wires, the whole orientation sum and the anchor cap.

10.1 A sharper bound for the same local syntax

The source-bound constructor audit gives maximum depth 19 for a local formula label. The nine axiom terms have depths

$$4, 3, 3, 3, 3, 5, 4, 3, 19.$$

Every leaf parameter is strictly below t. The following sharper majorant uses the same expression trees; it does not change the calculus.

Lemma 10.1 (A capacity bound for constructor trees)**.** *Put $d = 2^{19}$ and $L = 8^{d-1}t^d$. Every local formula label is strictly less than L.*

Proof. For $0 \le u, v \le m$, the largest constructor is bounded by $E(m, m)$, and

$$E(m, m) + 1 = 8m^2 + 14m + 8 \le 8(m + 1)^2.$$

A depth-zero leaf has value plus one at most t. The recurrence $H_0 = t$, $H_{r+1} = 8H_r^2$ therefore bounds value plus one at depth r. Its explicit solution is $H_r = 8^{2^r-1}t^{2^r}$. At depth 19 it equals L. Unequal child depths are handled by monotonicity. Thus a label has value at most $L - 1$, as asserted. □

11 Growth from three quadratic equations

The old exact request $F = t^{2^{20}t}$ is stronger than the proof extraction needs. The required interface is a sufficiently large even nonzero F, eventually positive, with a cofinal completeness construction. Introduce three integer coordinates X, Z, u and impose

$$\boxed{Z = tF, \qquad X^2 - 16Z^2 - 8ZF = 1, \qquad F = 2^{19}t(2u + 1).} \tag{23}$$

These are quadratic equations. They replace five coordinates of the old growth request: its principal Pell triple, its modulus and its quotient. The obsolete growth index is removed from the frame. The old principal pair's norm, multiplication wire, orientation term and cap term are removed exactly as specified by the source certificate; F remains a frame atom.

11.1 Pell recurrence facts

For $a \ge 2$ define $X_n(a), Y_n(a)$ by

$$\begin{aligned} X_0 &= 1, & Y_0 &= 0, \\ X_{n+1} &= aX_n + (a^2 - 1)Y_n, & Y_{n+1} &= X_n + aY_n. \end{aligned}$$

The least positive Pell unit at discriminant $a^2 - 1$ is $a + \sqrt{a^2 - 1}$: the ordinate of a nontrivial positive integral solution is at least one, and $(a, 1)$ is a solution. The standard Pell classification therefore gives, for every integer solution with $Y \ne 0$, $|X| = X_n(a)$ and $|Y| = Y_n(a)$ at a unique $n \ge 1$.

We need three elementary consequences:

$$Y_n(a) \equiv n \pmod{a - 1}, \tag{24}$$

$$v_2(Y_n(a)) = v_2(n) \qquad (a \text{ odd}, \ n \ge 1), \tag{25}$$

$$Y_n(a) \ge (2a - 1)^{n-1} \qquad (n \ge 1). \tag{26}$$

The first follows by specializing the integer polynomial recurrence at $a = 1$. For the second, all $X_n(a)$ are odd and $Y_n(a)$ has the parity of n when a is odd; the duplication identity $Y_{2m} = 2X_mY_m$ adds exactly one factor of two at each doubling. For the third, Y_n is increasing and $Y_{n+1} = 2aY_n - Y_{n-1} \ge (2a - 1)Y_n$.

Theorem 11.1 (Sign-free growth)**.** *Every integer solution of* (19) *and* (23) *has even nonzero* F *and*

$$|F| \geq L^t, \qquad L = 8^{2^{19}-1} t^{2^{19}}.$$

In particular $|F| \geq t^t > t$.

Proof. The last equation makes F nonzero. Eliminating Z gives

$$X^2 - ((4t+1)^2 - 1)F^2 = 1.$$

Pell classification at the odd center $a = 4t+1$ gives $|F| = Y_n(a)$ for $n \geq 1$. Put $d = 2^{19}$. Since $4t \mid F$, (24) gives $4t \mid n$. Moreover,

$$v_2(n) = v_2(F) = 19 + v_2(t).$$

Consequently the positive integer $n/(4t)$ has exact valuation 17, and $n \geq dt$. Formula (26) gives

$$|F| \geq (8t+1)^{n-1} \geq (8t)^{dt-1} \geq \left(8^{d-1} t^d\right)^t.$$

The last ratio is exactly $8^{t-1}/t \geq 1$. This proves the result with no assumption about the sign of F. □

The flexibility of this certificate is useful. It does not insist that F be one specified power. Section 13 will choose among its Pell ordinates to meet an additional binary-residue condition. Without that choice, merely proving a growth lower bound would not establish completeness.

12 The dominant CRT index supplies the radix

Replace the old base $B[2 + 4(C + \sum A_i)]$ by

$$D = 2(C + F + 1), \qquad q = D^x. \tag{27}$$

The power in this display is still represented by the retained polynomial certificate. It is not an operation in the final polynomial. The definition of D is affine, so its old multiplication coordinate and defining residual can be eliminated without raising degree. Crucially, the actual affine expression is substituted into the main square frame as well.

Equation (21) gives $B \mid D$ directly. Once the signs have been established, the existing anchor supplies $D > 2C$ and bounds every needed binomial upper argument by C. These two roles replace the old explicit factor B and the old sum-majorant.

12.1 Recovering the signs in the correct order

The main shared-Pell frame and C-anchor theorem apply first, while the front coordinates are signed. Theorem 11.1 independently gives even nonzero F with $|F| > t$. Because C is even, $C + F + 1$ is odd and nonzero; thus $|D| \geq 2$ and the request $q = D^x$ forces $x \geq 0$.

Since $y = F(2v+1)$, $|y| \geq |F|$. If $x = 0$, the terminal endpoint in (20) would make $1 + (t+1)y$ divide -1. This is impossible:

$$|1 + (t+1)y| \geq (t+1)|y| - 1 > 1.$$

Hence $x \geq 1$ and $|q| \geq 2$. The retained power at base q and exponent $y+1$ now forces $y + 1 \geq 0$. Negative y would be at most $-|F| < -1$. Therefore $y \geq |F| > 0$.

The value $x = 1$ needs a separate exclusion. The first endpoint would give $-F = (y+1)r_0$, whereas $0 < |F| \leq y < y + 1$. This is impossible. Thus $x \geq 2$, and the terminal endpoint gives

$$x \geq 2 + (t+1)y. \tag{28}$$

It follows that $|q| \geq 2^x \geq 4$ and $|q+1| \geq 3$. The base-$(q+1)$ requests now force all actual CRT representatives and $F+t$ to be nonnegative. If F were negative, then $F+t \leq t-|F| < 0$; hence $F > 0$. The already nonzero C is positive as well. Thus $D > 0$ and $q > 0$.

Finally, if $r_0 \leq -1$, the first endpoint gives $x \leq F - y \leq 0$. Therefore $r_0 \geq 0$ and $x \geq F+1$. We have recovered every required sign without adding a sign witness. In particular, the old assumption of a positive growth output has been replaced by a proved consequence of retained equations.

12.2 One binomial bound for the whole front

By (22), the initial upper argument $F+t$ and all ordinary representatives are below C; the anchor argument itself is C. The lower cutoffs $t, y+1, x$ are at most x. Also

$$D = 2(C+F+1) > 2C, \qquad q = D^x > (2C)^x.$$

For every binomial coefficient in any required low digit, including B,

$$0 \leq \binom{a}{j} \leq C^x < q/2^x \quad (0 \leq a \leq C,\ 0 \leq j \leq k \leq x). \tag{29}$$

Coefficients with $j > a$ are zero; $j = 0$ is included since $C > t \geq 16$. As $x \geq F+1$, this implies $C^x/(q-1) < 1/32$.

For clarity, the retained remainder certificate uses $M = q^x$, denominators q^k, integer ratios $a_k = M/q^k$, and normalized remainders ρ_k. Its exact division equations imply $a_k \mid \rho_k$. Its ternary norm budget

$$M^2 = 1 + 4\sum_k \rho_k^2 + r^2 + s^2 + w(w+1)$$

gives $|\rho_k| < M/2$, hence the actual integer remainder $r_k = \rho_k/a_k$ satisfies $|r_k| < q^k/2$. The genuine binomial low tail is less than $q^k/32$ by (29). Congruence modulo q^k therefore forces the two remainders to agree exactly. This is an equality argument, not an approximation.

The first digit consequently gives $B \equiv \binom{F+t}{t} \pmod q$. Equation (21), with nonzero integer η, also gives

$$|B| \leq C + F + 1 = D/2 < q/2.$$

The true binomial coefficient is positive and below $q/32$, so

$$B = \binom{F+t}{t} > 0. \tag{30}$$

It is at least two. The retained request authenticates $N = B^x$, and

$$q = D^x = (2B\eta)^x = N(2\eta)^x.$$

Thus $N \mid q$, the exact modulus relation required by the generalized range compiler. The same tail argument authenticates all remaining range digits. The inherited local arithmetic bound below 2^x continues to hold: it needs $F \geq t^t$, $t \geq 16$ and (28), all now proved.

13 A cofinal choice with an odd binomial selector

Soundness alone is not enough. The old exact power supplied high powers of two automatically. The new ordinate has a prescribed finite valuation in our completeness construction, so we must choose its other binary digits.

Lemma 13.1 (Pell permutation modulo powers of two)**.** *If $a \equiv 1 \pmod 4$, the map $s \mapsto Y_s(a)$ induces a permutation of $\mathbb{Z}/2^r\mathbb{Z}$ for every $r \geq 1$. Each odd residue has arbitrarily large positive odd indices realizing it.*

Proof. All $X_s(a) \equiv 1 \pmod 4$ and $Y_s(a) \equiv s \pmod 2$. Put $h = 2^r$. Equation (25) gives $v_2(Y_h) = r$. The norm and $8 \mid a^2 - 1$ imply $v_2(X_h^2 - 1) \geq 2r + 3$. Since $X_h + 1$ has valuation one, $X_h \equiv 1 \pmod{2^{2r+2}}$. The addition identity then gives

$$Y_{s+h} - Y_s = (X_h - 1)Y_s + X_s Y_h \equiv 2^r \pmod{2^{r+1}}.$$

The two index lifts s and $s + 2^r$ produce the two distinct output lifts. Induction from modulo two proves bijectivity and periodicity. Positive lifts in the resulting residue class are cofinal and preserve odd parity. □

Theorem 13.2 (Compatible cofinal growth witnesses)**.** *There are cofinally many horizons of the form* (19) *for which* (23) *has arbitrarily large positive solutions F such that $\binom{F+t}{t}$ is odd.*

Proof. Choose the free horizon parameter z divisible by 2^{12}. These horizons remain cofinal and satisfy $t \equiv 16 \pmod{2^{26}}$. Thus $v_2(t) = 4$ and the binary digit of t at position 23 is zero.

Put $d = 2^{19}$, $n_0 = dt$, $y_0 = Y_{n_0}(4t + 1)$ and $a_0 = X_{n_0}(4t + 1)$. The congruence modulo $4t$ and (25) show that $b_0 = y_0/(dt)$ is an odd integer. Moreover $a_0 \equiv 1 \pmod 4$, since n_0 is even and $X_{2m} = 2X_m^2 - 1$ with X_m odd.

Choose $\ell > 26$ with $2^\ell > t$, and set $r = \ell - 23$. The odd factors $t/16$ and b_0 are units modulo 2^r. By Lemma 13.1, choose a positive odd s, as large as desired, such that

$$(t/16)b_0 Y_s(a_0) \equiv 1 \pmod{2^r}.$$

Set $n = n_0 s$ and

$$F = Y_n(4t + 1) = y_0 Y_s(a_0), \quad X = X_n(4t + 1), \quad Z = tF, \quad u = \frac{F/(dt) - 1}{2}.$$

Composition of Pell units proves the displayed identity for F, and the odd quotient makes u integral. These values satisfy all three equations and

$$F \equiv 2^{23} \pmod{2^\ell}. \tag{31}$$

Since $t < 2^\ell$ and its bit 23 is zero, the binary supports of F and t are disjoint. There are no carries in their addition. Writing $s_2(m)$ for the binary digit sum, the elementary factorial identity gives

$$v_2\binom{F+t}{t} = s_2(F) + s_2(t) - s_2(F + t) = 0.$$

Thus the binomial is odd. Cofinal positive lifts of s give arbitrarily large F at each chosen horizon. □

The congruence proof is finite arithmetic at an arbitrarily chosen modulus; no analytic inverse theorem or computation of an infinite 2-adic object is needed. The retained executable modular examples check this construction at the actual $d = 2^{19}$, but the theorem follows from the recurrence proof for every modulus.

13.1 The beta moduli need only prime support

The new construction has $v_2(F) = 23$ at the selected horizons. We must not reuse the old sufficient condition $t! \mid F(2v + 1)$ for arbitrary large t. Fortunately the beta CRT needs only every prime at most t to divide y.

Choose a sufficiently large positive odd w divisible by all odd primes at most t, and put $y = Fw$. The integer y is divisible by every prime at most t. If a prime p divided both $1 + iy$ and $1 + jy$ for $1 \leq i < j \leq t + 1$, it would be coprime to y and divide $j - i$. Then $p \leq t$, contradicting $p \mid y$. Thus the beta moduli are pairwise coprime. The odd multiplier can also make y exceed every finite pending state and F, and it has exactly the required form $2v + 1$.

This replacement of an unnecessarily strong factorial divisibility is part of the completeness proof. Keeping the old assertion would leave a real gap.

14 From a zero to a refutation, and back

14.1 Sharp separation of the weighted labels

Apply Lemma 4.2 to the sharper capacity bound. The equality case $F = L^t$ is permitted by its strict coefficient-difference argument.

The decoded local Horn rows have weighted updates

$$S_{i+1}(1 + Fv_i) = S_i(1 + Fp_i)(1 + Fq_i),$$

with missing premises filled by the truth label zero. The endpoint $y = F$ would give an initial beta residue zero and terminal residue one. Positive weights propagate a zero initial state to zero at every row, so this case is impossible. Hence $y > F$, the actual initial state is $F + 1$ and the final state is one. All states are positive. Multiplying the updates and cancelling states gives the equality between the conclusion multiset and the premise multiset plus the target label one. Theorem 11.1 and Lemma 10.1 give precisely the hypotheses of Lemma 4.2.

Cycles in a selected multiset of Horn rules do not cause a gap. Let U be the labels not derivable from the available axioms using those rules. Every selected rule whose conclusion belongs to U has at least one premise in U. If the target one belonged to U, multiplicity balance would say that the number of conclusions in U is one more than the number of premises in U, contradicting the preceding inequality. Thus there is a finite $\mathcal{C}_{222}$ refutation.

14.2 Soundness of the full arithmetic source

From an arbitrary integer zero, first classify the standalone Pell pair to obtain even, nonzero large F. Independently use the retained main frame, anchor and orientation theorem to recover the canonical signed indices. Section 12.1 then proves the front signs. The uniform digit argument recovers B, N, and all range binomials. The inherited generalized binomial-CRT theorem applies because $F \geq L^t \geq t^t$ and hence $F \geq \operatorname{lcm}(1, \ldots, t)^2$. Its two required conclusions are retained: each row has a full prime-power selector exceeding t, and every full prime power of B is assigned to some row. The local arithmetic bound below 2^x converts those congruences into actual local equations.

This decodes the beta states, valid local rules, and their strict parameter ranges. The preceding product and Horn-flow argument yields a finite $\mathcal{C}_{222}$ refutation. All semantic source changes have now been accounted for; the rest of the source decoding is the identified inherited theorem.

14.3 Constructive completeness and the order of choices

Start with a finite $\mathcal{C}_{222}$ refutation. The choices must be made in the following order; this prevents a circular size or congruence argument.

1. Choose a cofinal horizon as in Theorem 13.2, exceeding the refutation length and all finite local parameters; pad by truth steps. Choose positive F and its standalone Pell witnesses with the odd binomial selector property. This fixes $B = \binom{F+t}{t}$.

2. Form the finite weighted pending-state path. Choose the odd multiplier of F so y bounds this path and every prime at most t divides y. Finite CRT chooses an arbitrarily large positive beta code x satisfying the state residues and endpoints. Put $N = B^x$.

3. The generalized binomial CRT supplies the nine representative classes modulo N^2. Choose the eight ordinary representatives positive first. Because N is odd, the ninth class has cofinally many even representatives. Choose C even and larger than every other nominal index and representative. The prescribed index class gives $B \mid C + F + 1$.

4. Set $D = 2(C + F + 1)$ and $q = D^x$ only now. Then $N \mid q$ and $D > 2C$. Evaluate the true requested power outputs, then choose the main square frame and the positive canonical main

Pell pairs after all requested atoms have been fixed. The inherited cofinal anchor completion supplies the private comparison witnesses, orientation quotient and cap slack with one fewer principal pair.

5. Take the normalized low remainders. The estimates of Section 12 satisfy the retained common norm budget. Every nonnegative slack has the inherited representation $r^2 + s^2 + w(w+1)$, supplying the remaining integer witnesses.

This constructs a zero of the changed arithmetic source from every finite refutation. No guessed universal-machine parameter or unknown constant is required.

14.4 The precise logical dependency

Combining the two directions with the square-shell numerical-calculus calibration of Section 2 proves the same syntactic ZFC-refutation equivalence. The model-theoretic bridge is internal to the background theory; it does not assume an externally well-founded transitive model. Selected logical sources are retained, but no complete transitive Lean or Rocq build is claimed.

15 Signed affine horizons and smaller frame coefficients

The starting point is the fully authenticated incoming Pell Growth 210 system described above. This round changes its cofinal horizon, and for explicitly selected profiles also changes the frame and one orientation error. These are semantic source changes with proofs in both directions; they are not asserted to be pointwise bijections with the old quadratic horizon. Later elimination, factor splitting and chart operations are exact integer zero-set bijections.

15.1 Recover the sign of an affine horizon

Write $d_0 = 2^{19}$ and replace the horizon by

$$t = m(2z+1), \qquad m \in \{4, 16\}, \quad z \in \mathbb{Z}. \tag{32}$$

Before classifying the standalone Pell equation, its divisibility row gives $F = d_0 t(2u+1)$, hence

$$|F| \geq d_0|t| \geq d_0 m, \qquad F \neq 0, \quad 2 \mid F, \qquad |y| \geq |F|.$$

The conservative square frame contains the actual F, so $W \geq F^2 + 2 > 258$. The retained anchor theorem therefore supplies canonical signed main Pell pairs before recovering front signs. Since C, F are even, $D = 2(C + F + 1)$ is nonzero and even. Its retained power request gives $x \geq 0$ and $q = D^x$.

As $|t+1| \geq m - 1$ and $m \geq 4$,

$$|1 + (t+1)y| \geq (m-1)d_0 m - 1 > 1.$$

The terminal beta equation excludes $x = 0$. Thus $x \geq 1$, $|q| \geq 2$. The retained request at base q and nominal index t now forces $t \geq 0$, so (32) gives $t \geq m$. This is the literal `pow4.output` request at `index0`, not an imagined q^{t+1} request. Nothing used the sign of t before this step.

For $m = 16$ all remaining incoming arithmetic inequalities apply unchanged. Completeness uses the incoming cofinal horizons $t \equiv 16 \pmod{2^{26}}$, which belong to the new progression $t \equiv 16 \pmod{32}$. This gives the conservative 209-coordinate source used independently by the reciprocal family.

15.2 The smaller threshold has its own estimates

For $m = 4$, positivity gives $t \geq 4$ and $4 \mid t$. The constructor inequality $E(a, a) + 1 \leq 8(a + 1)^2$ is valid for every $a \geq 0$, and constants zero and one are below t. The same literal depth-19 trees are therefore bounded by

$$L = 8^{d_0-1} t^{d_0}.$$

Pell classification at $4t + 1$, the congruence $Y_n(4t + 1) \equiv n \pmod{4t}$, and the exact valuation $v_2(Y_n) = v_2(n)$ again give $n \geq d_0 t$. Hence

$$|F| \geq (8t + 1)^{d_0 t-1} \geq L^t.$$

The last comparison uses $(8t)^{d_0 t-1}/L^t = 8^{t-1}/t \geq 1$. These actual bounds give $|F| > 2^{64}$ and $|F| > 16^t$ already at $t = 4$. The remaining incoming bootstrap proves $F > 0$, $y \geq F$ and $x \geq 2 + (t + 1)y$. Therefore $x > 2^{32}$ and

$$2 + t + x + y + F < 2x.$$

The original literal local bound $\mathcal{L} < (2x)^{2^{23}} < 2^x$ is valid with these repaired hypotheses. Also $\operatorname{lcm}(1, \ldots, t) \leq 4^{t-1}$ and $F > 16^t$ supply the selector radius. No use of an unproved $t \geq 16$ remains in these steps. The radix parity and remainder norm are unchanged.

15.3 A cofinal binary residue construction at threshold four

Choose arbitrarily large $t \equiv 4 \pmod{2^{24}}$ above the finite proof's length and local parameters. Such a t belongs to (32), has valuation two, and has binary bit 21 equal to zero. Put

$$n_0 = d_0 t, \quad y_0 = Y_{n_0}(4t + 1), \quad a_0 = X_{n_0}(4t + 1), \quad b_0 = y_0/(d_0 t).$$

The incoming congruence and valuation proof gives odd integral b_0 and $a_0 \equiv 1 \pmod 4$. Choose $\ell > 24$ with $2^\ell > t$, and set $r = \ell - 21$. The Pell permutation lemma gives a positive odd s, as large as desired, for which

$$(t/4) b_0 Y_s(a_0) \equiv 1 \pmod{2^r}.$$

Set

$$F = y_0 Y_s(a_0), \quad X = X_{n_0 s}(4t + 1), \quad Z = tF, \quad u = (F/(d_0 t) - 1)/2.$$

All growth rows hold, and $F \equiv 2^{21} \pmod{2^\ell}$. Since $t < 2^\ell$ and bit 21 of t is zero, $F + t$ has no binary carries; $B = \binom{F+t}{t}$ is odd. The prime-support beta spacing, finite CRT, even cofinal C, collapsed radix and final main-Pell completion now follow in the same acyclic order as the incoming construction. Thus threshold four has full completeness, not just a sign or growth argument.

15.4 An optional compressed affine frame

This further source change is used only where the endpoint certificate says so. A literal 34-atom ledger first checks the exact incoming frame $W = 2 + \sum_j a_j^2$, containing every actual base, index and proposed output of the fifteen requests. After substituting (32), each a_j is an integer affine form in ordinary front coordinates, not private Pell coordinates.

For an affine form $L = c + \sum_i a_i u_i$, the integer inequality $|a_i u_i| \leq u_i^2 + \lfloor a_i^2/4 \rfloor$ yields

$$|L| \leq \sum_{a_i \neq 0} u_i^2 + |c| + \sum_i \lfloor a_i^2/4 \rfloor.$$

The largest constant in the complete ledger is $m^2 + m + 1$, from $t + 1$. Replace the frame by

$$W = m^2 + m + 2 + \sum_{\text{distinct underlying front coordinates } u_i} u_i^2. \tag{33}$$

All actual affine forms stay in their power and index requests. The ledger proves $|a_j| < W$ and the stronger bound $|a_j| \le (3m+1)\sqrt{W}$, since $3m+1$ is the maximum absolute coefficient sum including the constant, attained at $t+1$ or $F+t$. The underlying coordinate F is retained, so already $W > (d_0 m)^2 \ge 2^{42}$.

The old squared-atom power theorem is not reused silently. Set $s_0 = \lceil \sqrt{W} \rceil$, $b_0 = (3m+2)s_0$. Then every actual atom has absolute value at most b_0, and

$$b_0 < W, \qquad 4b_0 + 1 \le W, \qquad b_0^2 \le 10000W.$$

These follow from $s_0 \le 2\sqrt{W}$, $m \le 16$ and $W \ge 2^{42}$; for example $4b_0 + 1 \le 400\sqrt{W} + 1 < W$. The exact negative-index clearing used in Theorem 9.2 now gives, for at most fifteen requests,

$$0 \le E_* \le 60 b_0^{4b_0} < W^{4b_0+1} \le W^W.$$

The modulus still satisfies $|T_n| \ge 2|A| - b_0^2 - 1 > W^W$, because $|A| > W^W$ and $W^W > 10000W + 1$. Thus $E_* = 0$, giving exact powers and excluding negative indices at bases of absolute value at least two. The same argument with one summand proves each singleton request. No fourfold enlargement of the center is required.

15.5 Use a difference error in orientation

The anchor comparison first identifies the classified index $|C|$, using $|C| < W$; the cap bounds every other classified index by $|C|$. Replace only the orientation square

$$(Y_{F+t} - (F+t))^2 \quad \text{by} \quad (Y_{F+t} - Y_t - F)^2,$$

retaining $(Y_t - t)^2$ and every abscissa error. The common zero set of the nonnegative error forms is unchanged: the retained t error vanishes first, then the difference error gives the original $F + t$ equality.

Modulo $A - 1$, the changed error has absolute value below $3W$, each other ordinate error below $2W$, and each abscissa error at most two. For thirteen principal pairs the nonnegative error sum is therefore

$$< 57W^2 + 52 < 4W^3 < |A - 1|.$$

Its divisibility forces every error to vanish, recovering the canonical orientation. The exceptional anchor branch is handled by the retained cap and nonzero anchor error before invoking a comparison at an exceptional modulus. The exact quadratic delta is propagated through the literal emitted basis and independently checked; it is not merely an informal replacement of a proof expression.

Completeness fixes all arithmetic-front values first, then the new W and a sufficiently large positive shared center with its private anchor witnesses. Canonical errors are divisible by $A - 1$, and the exact power identities supply their quotients. No earlier CRT modulus depends on the eventual W. Thus both the frame and orientation changes preserve the ordinary source calibration.

15.6 Exact elimination and coefficient splitting

In the modified source, substitute $t = m + 2mz$ everywhere and delete its monic horizon row and coordinate. The inverse inserts that same affine expression. This exact integral bijection includes the frame, cap, orientation, power and growth rows, not just the visible arithmetic front.

For nonnegative integers $a_1, \ldots, a_r$ summing to 19, replace the growth divisibility row by successive monic definitions

$$w_1 = 2^{a_1} t, \quad w_2 = 2^{a_2} w_1, \quad \ldots, \quad F = 2^{a_r} w_{r-1}(2u+1).$$

The wires have unique integer values. Forgetting and restoring them in order are inverse zero-set maps, and every residual stays quadratic. The chosen splits are finite certificate data, not new equation parameters. Subsequent integer shears, translations and row additions have their recorded integral inverses. These operations can lower coefficient height at a variable cost; there is no general monotonicity claim.

15.7 The frozen charts and their finite maps

The selected endpoints use threshold four and the independently checked compressed frame. The factor exponents and final row operations are:

Endpoint	Split of 19	Residual rows	Final row additions
dimension209	no extra wire	149	0
principal210	$(8, 11)$	150	0
balanced211	$(4, 8, 7)$	151	4
low212	$(3, 5, 6, 5)$	152	426

Every metric in Section 1 is obtained from full independent coefficient collection of these actual arrays. The final balanced height is 65537; earlier exploratory values and splits are not retained in this table. The exact conservative $m = 16$ source remains a separate 209-variable, 149-row object for reciprocal compilation, with its own complete printed array in Appendix B.

The unbounded source proofs and exact finite maps compose with the named C222/ZF/ZFC calibration to prove (1) for each final quartic. This does not assume consistency, restrict codes to externally standard ones, or claim a fresh end-to-end formal-kernel derivation.

16 Unweighted joint index authentication and power pooling

This theorem combines the incoming source 176 raw-index orientation argument with the mixed-base two-integrality argument [11, 10]. Neither ingredient is credited as new here. The new synthesis removes the base weights from a joint orientation and power certificate. It needs no common sign or pairwise coprimality of the bases, and remains valid at zero and unit bases. The emitted source instances separately authenticate their actual ports.

16.1 The complete joint residual

Let $W \geq 258$, $|A| > W^W$, and let $b \geq 16$ be an integer with $12b + 10 \leq W$. Suppose $p \leq 15$ raw principal Pell pairs have classified representations

$$(X_i, Y_i) = \varepsilon_i(X_{\nu_i}(A), Y_{\nu_i}(A)), \qquad \varepsilon_i \in \{-1, 1\}, \quad |\nu_i| \leq b.$$

The requested signed indices obey $|k_i| \leq b$. There are at most five base blocks and at most 17 total requests. Every actual base n_j and proposed output m_i has absolute value at most b. Requests may reuse a principal pair. The retained gates give $Z_i = AY_i$. Define

$$\begin{aligned} d &= A - 1, \qquad T_j = 2An_j - n_j^2 - 1, \\ \Omega &= \sum_{i=1}^{p} [(X_i - 1)^2 + (Y_i - k_i)^2], \\ H_j &= \sum_{i \text{ in block } j} Y_i(m_i - X_i), \\ S_j &= \sum_{i \text{ in block } j} [(m_i - X_i + Z_i)^2 - Y_i^2], \qquad E_j = S_j - 2n_jH_j. \end{aligned}$$

The unweighted joint residual, with one new unrestricted integer z, is

$$\Omega \prod_j T_j + d \sum_j E_j \prod_{h \neq j} T_h - zd \prod_j T_j = 0. \tag{34}$$

Products and sums here are fixed finite polynomial operations. The display introduces no reciprocal gate or unspecified integer parameter.

Theorem 16.1. *Under these hypotheses, an integer z satisfying* (34) *exists if and only if every principal pair is canonical at its requested signed index and every request at a nonzero base satisfies $m_i = n_j^{k_i}$ in $\mathbb{Q}$. A base with $|n_j| \geq 2$ forces $k_i \geq 0$. Unit bases allow signed indices. Zero-base requests impose no condition.*

Proof. Put $M_0 = W^W$. As A is integral and $|A| > M_0$, $|d| \geq M_0$ and d has the sign of A. At $n_j = 0$, $T_j = -1$. Otherwise

$$\frac{|T_j|}{|n_j|} \geq 2|A| - \frac{n_j^2 + 1}{|n_j|} \geq 2|A| - b - 1 > M_0,$$

and $n_j T_j$ has the sign of A. In particular all denominators used in this proof are nonzero.

Modulo d, the signed Pell recurrence gives $X_\nu \equiv 1$ and $Y_\nu \equiv \nu$. Therefore $\Omega \equiv o \pmod{d}$, where

$$o = \sum_i [(\varepsilon_i - 1)^2 + (\varepsilon_i \nu_i - k_i)^2], \qquad 0 \leq o \leq 60(1 + b^2) \leq 120b^2.$$

The indices here are bounded *classified* indices, not the requested ones whose authenticity remains to be proved.

For each request put $\delta_i = m_i - X_i + (A - n_j)Y_i$. Exact expansion gives

$$\sum_{i \text{ in } j} \delta_i^2 = E_j - T_j \sum_{i \text{ in } j} Y_i^2. \tag{35}$$

The Pell recurrence, evaluated at the square root $n_j - A$ modulo T_j, gives for nonzero n_j

$$X_i + (n_j - A)Y_i \equiv \varepsilon_i n_j^{\nu_i} \pmod{T_j}.$$

Negative powers exist in this residue ring because $\gcd(n_j, T_j) = 1$.

First integrality step. Divide (34) by $d \prod_j T_j$ and use (35). It follows that $\Omega/d + \sum_j \sum_i \delta_i^2 / T_j$ is integral. Remove the zero-base blocks; those summands are already integers. Let J be the remaining blocks and define

$$C_* = (\prod_{j \in J} n_j)^b, \qquad L_* = \prod_{j \in J} n_j,$$

with empty products one. For a request in J, define the integer

$$e_i = C_* m_i - \varepsilon_i (C_*/n_j^b) n_j^{b+\nu_i}, \qquad D_j = \sum_{i \text{ in } j} e_i^2, \quad D = \sum_{j \in J} D_j.$$

All exponents are nonnegative. The recurrence congruence makes $C_* \delta_i \equiv e_i \pmod{T_j}$. Hence

$$R_* = C_*^2 o/d + \sum_{j \in J} D_j / T_j$$

is an integer. Since $|C_*| \leq b^{5b}$ and $|e_i| \leq 2b^{6b}$,

$$\begin{aligned} C_*^2 o + D &\leq 120 b^{10b+2} + 68 b^{12b} \leq 188 b^{12b+2} \\ &< b^{12b+5} < W^{12b+5} \leq W^W = M_0. \end{aligned}$$

Here $188 < b^3$, $b < W$, and $12b + 5 \leq W$ by hypothesis. Each active denominator has magnitude at least M_0, so $|R_*| < 1$. Thus $R_* = 0$, although its summands may have different signs.

Second integrality step. The two exact rational identities

$$\frac{2A}{d} = 2 + \frac{2}{d}, \qquad \frac{2A}{T_j} = \frac{1}{n_j} + \frac{n_j^2 + 1}{n_j T_j}$$

split $2AR_* = 0$ as $U_* + V_* = 0$, where

$$U_* = 2C_*^2 o + \sum_{j\in J} D_j/n_j,$$

$$V_* = 2C_*^2 o/d + \sum_{j\in J} \frac{D_j(n_j^2+1)}{n_j T_j}.$$

The integer L_*U_* makes $L_*V_* = -L_*U_*$ integral. The denominator estimate also gives

$$\frac{n_j^2+1}{|n_jT_j|} \le \frac{1+n_j^{-2}}{2|A|-b-1} < \frac{2}{M_0}.$$

Using $|L_*| \le b^5$ yields

$$\begin{aligned}|L_*V_*| &\le \frac{240b^{10b+7}+136b^{12b+5}}{M_0} \le \frac{376b^{12b+7}}{M_0}\\ &< \frac{b^{12b+10}}{M_0} < \frac{W^{12b+10}}{M_0} \le 1.\end{aligned}$$

We used $376 < b^3$, $b < W$, and $12b+10 \le W$. Thus $L_*V_* = 0$; since $L_* \ne 0$, $V_* = 0$. Every nonzero summand of V_* has the sign of A: $o, D_j \ge 0$, d has that sign, and so does n_jT_j. Consequently $o = 0$ and every $D_j = 0$. The first forces $\varepsilon_i = 1$ and $\nu_i = k_i$ for every principal pair. The second gives $e_i = 0$, hence $m_i = n_j^{k_i}$. For $|n_j| \ge 2$, negative k_i would force the impossible integer equation $m_i n_j^{-k_i} = 1$. If J is empty, the same proof has $C_* = L_* = 1$ and $V_* = 2o/d$, so it still authenticates every pair.

Conversely, canonical pairs give $d \mid \Omega$. True nonzero-base powers make every δ_i divisible by T_j, so (35) gives $T_j \mid E_j$. At zero bases this is automatic because $T_j = -1$. Thus $z = \Omega/d + \sum_j E_j/T_j$ is integral and satisfies the residual. □

16.2 Square and affine frame corollaries

For the exact square frame, $b = \lfloor\sqrt{W-1}\rfloor$ has $b \ge 16$ and $12b + 10 \le b^2 + 1 \le W$: the first difference is $b^2 - 12b - 9$, positive at 16 and increasing thereafter. The original square-frame interface is therefore a special case.

A centered frame could instead supply $b \le 14\sqrt{W}$ and $W \ge 2^{42}$. Then $12b+10 \le 168\sqrt{W}+10 \le W$. This is a conditional extension: every actual base, output, nominal index and raw classified index must be independently bounded at that same b. It is not an automatic port transfer to an unaudited weak frame. The selected reciprocal endpoints retain the conservative exact-square source.

16.3 Restore the source before recovering its signs

The source's orientation quotient is Ω/d, with any explicitly recorded retained-row correction restored. For a singleton whose rows are $T = T_j$ and $\delta_i + TQ = 0$, restore $T = T_j$, $Q = -\delta_i/T_j$. For an aggregate restore

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

These are integers also at zero or unit bases. Substitution in the exact identities restores every original row. Only then apply the original semantic bootstrap, which proves the actual arithmetic bases nonzero and positive in its own established order. No circular assumption that q is already positive is needed for the joint theorem.

One may partition the base blocks and fuse orientation into one part using (34). That part authenticates *all* principal pairs, even if it contains only some requests. The other parts then use the incoming unweighted mixed-base theorem at already canonical pairs. Its proof is the two integrality steps above with $o = 0$, so no new sign or coprimality premise is introduced. All quotient restorations are performed before source decoding.

16.4 Authenticating actual ports and the retained frame

The weighted reference has 14 principal pairs, three singleton requests and two aggregate blocks. Its certificate checks every norm and multiplication gate, all private coordinates, and the exact orientation identity

$$\Omega - (R_{\mathrm{orient}} + (A-1)q_{\mathrm{orient}}) = \sum_{i=1}^{14} R_{\mathrm{norm},i}.$$

The original squared C cap bounds every raw classified index by $|C| \le b$ *before* orientation. Its frame contains the actual growth index and all requested bases, indices and outputs. Its inherited $q \equiv 1 \pmod 3$ restriction belongs to that frozen source's completeness construction, not to Theorem 16.1 itself.

The new affine-source instance instead binds the conservative $m = 16$ 209-coordinate source of Section 15, with 13 principal pairs and four remaining power blocks. It checks the actual affine ports and all retained-row corrections from the centered norm/orientation chart. This is the same theorem with fewer active pairs, not an assumption that similar coordinate names denote equal polynomials. In particular, the weak affine-atom frame discussed there is *not* substituted into the square-frame hypothesis of the joint theorem.

16.5 A retained row changes the numerator chart

The smallest selected endpoint uses a further exact algebraic adjustment, not a weaker-frame theorem. Let $R_W = 0$ be the retained frame residual of the conservative source and replace the orientation numerator by

$$\Omega' = \Omega + R_W.$$

All retained source zeros have $R_W = 0$, so the joint zero-set argument is unchanged. In the actual centered source coordinates this addition cancels the quadratic x^2 term. A later monic substitution for x can then have degree four without making the selected pool row exceed degree eight. The certificate checks the exact coefficient cancellation and restores the retained row before invoking Theorem 16.1. It is not a claim that Ω' is itself nonnegative away from those retained zeros.

For the 113-variable endpoint the four base blocks are partitioned as (B) and $(D, q, q+1)$, with orientation fused into the first part. The source has thirteen principal pairs and fifteen power requests, not the fourteen-pair weighted reference. The 202-coordinate, 142-row joint seed then undergoes 89 exact unit-pivot eliminations, leaving 113 coordinates and 53 residuals of degree at most eight. Its sum of squares has exact degree 16, support 1030202 and height 16250018885420533678080.

The sparse 114-variable companion keeps orientation separate and partitions the power blocks as (B, D) and $(q, q+1)$. It has 54 residuals, exact degree 16, 127818 terms and the same height. This is a useful sparseness tradeoff, not another use of the fused orientation theorem. The 138-variable companion has 78 residuals, exact degree eight, 12099 terms and height 2^{50}. The 138-variable point uses direct monic elimination from the conservative source, without a new pool, and is not credited to the joint theorem. All three full collected polynomials are supplied; their supports are not estimates inferred from circuit sizes.

16.6 The finite substitution specification

For clarity, a unit-pivot step at the current residual row has the form

$$Q_r = \epsilon x_v + f(\widehat{x_v}), \qquad \epsilon \in \{-1, 1\},$$

with no other occurrence of x_v in Q_r. Substitute $x_v = -\epsilon f$ in every other residual, then delete row r and coordinate v. This is an integral zero-set bijection: insertion of that polynomial is its

exact inverse, and the deleted equation becomes an identity. Later restorations compose in reverse order. Every step's resulting degree is checked after full exact coefficient collection; a generic degree-propagation bound is not used as an attained degree certificate.

The appendix gives the complete conservative source, literal principal-pair and request ports, exact removed rows, fixed pool partitions, and each ordered triple (v, r, ϵ). Starting with the stated seed, set σ_0 to the identity and successively apply

$$\sigma_{j+1} = \sigma_j\big[x_v \mapsto -\epsilon\,\sigma_j(Q_r)|_{x_v=0}\big].$$

Here the substitutions mean applying the current pivot replacement to every current row; the recorded order removes any ambiguity about composition. The final polynomial is the sum of squares of all retained substituted rows, in the explicitly listed retained coordinate order. This is a finite ordinary polynomial definition with no existential register or division. The supplied full residual and coefficient arrays independently agree with this specification.

The matched unfused control uses the same source and admissible monic pivot strategy. It isolates the extra elimination enabled by the joint numerator chart. The sparse 114-variable endpoint and the low-degree 138-variable endpoint are separate controls with their own partitions and metrics; they are not falsely reported as a simultaneous improvement in all objectives. The unbounded joint theorem, exact substitutions and inherited source calibration together establish (1) for every selected endpoint on $\mathbb{Z}$.

17 Unit homogenized equality at coefficient height one

The source compiler acts on a finite quadratic system over unrestricted integers. The inherited Round 20 construction already supplies certified normalization, opposite-occurrence fusion, polarization, anchored equality and internal-flow contraction. We give its complete energy and witness argument to establish the interface, then retain the monic product and chained-scale constructors before proving this round's additional substitutions. Unit-homogenized cycles are retained from Round21. Every height claim concerns the entire collected polynomial, including all equality penalties.

17.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_cX_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_is_{c,j}$ otherwise. The cover condition proves that one option always exists. The retained deterministic greedy cover starts with the loop vertices and repeatedly selects a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and t_cX_i, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus normalization preserves both witness maps. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j:m_{rj}=m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

17.2 Share a coefficient across both monomial factors

The retained split-factor method supplies a further exact normalization step. For a quadratic occurrence cxy, choose positive integers a, b with $|c| = ab$. Let $S(1, x) = x$; for $d > 1$ introduce a shared coordinate $S(d, x)$ with monic definition $S(d, x) - K_d x = 0$, where $K_d = d$ is fixed by the retained triangular constant chain. Replace the occurrence by

$$\operatorname{sgn}(c) S(a, x) S(b, y).$$

It remains a degree-at-most-two monomial of coefficient ± 1. Its unique constructor substitution is exactly cxy. If $x = y$ the same scale coordinate may occur twice, giving its ordinary square. Repeated definitions are shared; no unused scale coordinate is introduced.

This is an unconditional integral source equivalence. Given a source tuple, its monic constants and scales have unique values and every normalized row pulls back to its original row. Conversely those monic rows force the values and recover every original equation. There is no variable division or hidden divisibility assumption. The positive integer factor choices are frozen certificate data, not equation parameters.

The finite selection enumerates divisor pairs in both orientations and accepts a reduction of the number of used scales plus directly needed constants, then builds the actual constant chain. That cost is only a search heuristic. Exact final source pullbacks and coefficient expansion, not a presumed optimum of the heuristic, certify the delivered counts. The changed normalization alters the eventual graph namespace; its bijection with the old graph goes through the common arithmetic source, not an identity on arbitrary final tuples. The graph-energy and universal lift proof below applies unchanged to either normalized source.

17.3 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical

factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v\in B_r} b_v = T_r(y), \qquad \sum_{v\in B_0} b_v = -\sum_r T_r(y). \tag{36}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (36) is unchanged by fusion. This is the inherited occurrence-boundary identity.

17.4 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2}\sum_v d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - ab^3 = (a - b)^2(a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell}(u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{37}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The retained square-alias transformation below preserves this energy identity and revises a selected equality penalty with its exact zero set intact.

17.5 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (36) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2}\left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x)\right)^2\right). \tag{38}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (38) composes unchanged through affine refactorization. One must not replace (38) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (37) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

17.6 Square polarization removes equality dummies

Suppose a logical value currently has exactly two factor aliases a, b, and its equality penalty uses a fresh dummy d:

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

The reason for retaining this dummy can be a prescribed square a^2 inside the incidence graph: directly substituting the two-alias penalty $Q(a, b)$ may collide with its degree-four diagonal contribution. The retained Round 19 operation first changes selected prescribed squares into products of same-logical aliases, $a^2 \mapsto ab$ (or $b^2 \mapsto ab$). It then replaces $C(a, b, d)$ by $Q(a, b)$ and deletes the dummy coordinate d.

Theorem 17.1. *Make any finite selection of such same-logical prescribed-square changes, and replace each affected two-alias equality cycle by Q. If the full collected final polynomial has coefficient height one, it is a height-one quartic with an integer zero set bijective to the original arithmetic source. Each deleted equality dummy reduces its dimension by one.*

Proof. Changing a prescribed edge's label from a^2 to ab leaves its two incidence endpoints and orientation unchanged. The graph energy therefore remains exactly $\frac{1}{2}\sum_v d_v^2$, with its divergences

evaluated using the modified labels. It is still nonnegative, and all labels still have degree at most two. Also

$$Q(a,b) = (a-b)^2(a^2+ab+b^2) \geq 0,$$

and $Q(a,b) = 0$ if and only if $a = b$: the second factor equals $(a+b/2)^2 + 3b^2/4$ and vanishes only at $a = b = 0$.

At any zero of the final energy, every nonnegative equality term is zero, so same-logical aliases agree and units equal one. Consequently every polarized label ab has exactly the value previously specified by a^2. The occurrence-boundary identity is unchanged after logical substitution. Vanishing vertex divergences therefore give all normalized source rows, and monic reconstruction gives an actual arithmetic source zero.

Conversely, for an original source zero assign both a and b their common logical value. All polarized edge labels then have their old values, all new Q terms vanish, and the same unique subtree flow assignment makes every vertex divergence zero. Deleted dummies have unique old value $d = a = b$, so extension and deletion are inverse on the zero sets. This also proves that the source-polynomial lift identity (38) survives universally, not only at source zeros.

The degree is at most four by construction. Exact degree four and coefficient height one are finite properties of the entire collected polynomial and are explicitly checked. They do not follow merely from the simplicity of the incidence graph after monomial substitution. The operation is accepted only after this full collection check, including all equality and unit terms. Thus possible collisions do not invalidate the claim. □

The theorem makes no assertion that an arbitrary polarization is safe for height one, nor that a chosen finite sequence is optimal. It separates the unconditional witness proof from the coefficient test selecting a particular literal endpoint.

17.7 Absorbing equality dummies into anchored products

The retained anchored operation removes an equality dummy without changing any prescribed square label. Suppose two aliases a, b of one non-unit logical value are tied by a fresh dummy d through

$$C(a,b,d) = a^2 + b^2 + d^2 - ab - bd - da.$$

Assume d occurs nowhere else. Let u be an existing unit alias already anchored to one by U. Substitute $d = au$ and delete the coordinate d. The new equality penalty is

$$\widetilde{C}(a,b,u) = a^2 + b^2 + a^2u^2 - ab - abu - a^2u. \tag{39}$$

It has degree four and is the literal polynomial $C(a,b,au)$.

Theorem 17.2. *Under the stated incidence restrictions, replacing $C(a,b,d)$ by* (39) *preserves the integer zero set by a polynomial bijection after deleting one coordinate. It also preserves the universal source lift identity* (38)*. If the entire resulting collected polynomial has height one, exact degree four and all declared coordinates present, it is a certified height-one quartic with one fewer independent integer variable.*

Proof. For arbitrary real a, b, u,

$$2\widetilde{C} = (a-b)^2 + (b-au)^2 + (au-a)^2 \geq 0.$$

At a zero of the complete energy, its unchanged unit penalty forces $u = 1$. The new equality penalty then vanishes exactly when $a = b$. Thus the old dummy has the unique value $d = a = b$, while every other summand is unchanged. Extending with $d = a$ maps a new zero to an old zero. Conversely, at an old zero $d = a = b$ and $u = 1$, so deleting d gives a new zero. These maps are inverse polynomial maps on the zero sets.

For a canonical source lift, all unit aliases equal one and all ordinary aliases equal their logical values. Consequently $au = a = d$ identically on that lift, including away from source zeros. No prescribed incidence label or divergence has changed, so the universal lift identity is retained.

Polynomial substitution preserves nonnegativity but not automatically coefficient height. The term a^2u^2 may collide with graph-energy terms, and several absorbed dummies may share a unit alias. The acceptance test therefore collects the whole energy after every proposed substitution, including all other penalties. It explicitly checks the finite claims about coefficients, degree and the live coordinate set. No extra unit coordinate is introduced: u was already present and anchored. □

Several such absorptions can be composed, using the same or different existing unit aliases, provided the full polynomial test accepts the frozen final choices. Unlike Round 19 polarization, this operation leaves the prescribed square monomials in the incidence graph untouched. Unlike a formal substitution $d = a$ alone, it does not insert a coefficient -2 into the equality row before the complete energy is considered. Its quartic product with an already anchored unit provides additional distinct monomials that can retain height one.

17.8 Contracting an internal flow edge

The retained operation changes the divergence energy rather than only the alias penalties. It removes a flow coordinate on an internal tree edge and identifies its endpoints. It is therefore not an alias identification or an instance of square polarization. Its soundness is unconditional; coefficient height one is a separate exact acceptance test.

Theorem 17.3. *Let an incidence presentation have the energy form* (37)*, with every internal edge belonging to a tree inside one row block or the hub block. Choose an internal edge from u to v, with its own flow coordinate w. Delete that edge, identify its vertices, and replace their two divergences by their sum. Retain all prescribed occurrence labels and equality and unit penalties. Then the old and new integer zero sets are in polynomial bijection, preserving the original arithmetic coordinates. The new presentation has one fewer flow coordinate and degree at most four. If full collection of its polynomial has height one, exact degree four, and every listed variable occurs, it is a certified height-one quartic on that smaller coordinate set.*

Proof. Write the old two divergences as

$$d_u = a_u + \epsilon w, \qquad d_v = a_v - \epsilon w, \qquad \epsilon \in \{-1, 1\},$$

where a_u, a_v are integer polynomials independent of w. No other divergence or penalty depends on w, because it is the independent flow coordinate of this one edge. The merged divergence is $d_{uv} = a_u + a_v$. Thus the new polynomial is exactly

$$P' = \frac{1}{2}\left((a_u + a_v)^2 + \sum_{t \neq u,v} d_t^2\right) + \sum C + \sum Q + U.$$

This expression proves nonnegativity and the degree bound without any claim that the old and new polynomials themselves agree.

At an old zero, nonnegativity forces $d_u = d_v = 0$ and every other summand to vanish. Deleting w therefore gives $a_u + a_v = 0$ and a new zero. Conversely, at a new zero all its displayed summands vanish. Set

$$w = \epsilon a_v.$$

Then $d_v = 0$, while $d_u = a_u + a_v = 0$; all other old summands are unchanged. Indeed the stronger identity $P|_{w=\epsilon a_v} = P'$ holds for every remaining tuple: the two old divergences become $a_u + a_v$ and zero. This is an integral polynomial restoration, and it is unique on zeros because ϵ is a unit.

Restoration and deletion are mutually inverse on the zero sets. All equality and unit arguments, and hence all recovered logical coordinates, are identical on the two sides.

The edge's endpoints belong to one block. Summing the contracted vertex divergences in that block therefore gives the same boundary sum as before contraction. In particular (36) continues to recover exactly the same normalized source row; no source occurrence is deleted. Contracting an edge of a tree leaves a tree. The remaining flows are still uniquely recovered by leaf elimination, or equivalently by composing the one-edge restoration above.

Finally, degrees and coefficient heights concern the polynomial after all monomial collisions, including equality and unit terms. Contracting vertices can create parallel prescribed incidences, so the initial simple graph argument alone is no longer a sufficient height certificate. The selection test reconstructs and collects the entire new energy and checks its actual integer coefficients. This establishes the stated height, exact degree and coordinate census only for the accepted finite instance. □

Corollary 17.4. *Any finite certified sequence of accepted internal-edge contractions preserves the calibrated integer zero set. Its inverse is obtained by restoring the deleted flows in reverse order. If all k deleted flows were distinct live coordinates and no other coordinate is removed, its independent-variable count decreases by exactly k.*

Proof. Compose the polynomial deletion and restoration maps of Theorem 17.3. Every step preserves the block boundary identity and leaves internal blocks as trees. The final independently collected array fixes the actual coordinate count and all measured polynomial invariants. □

This argument does not predict which edges pass the height test or minimize the support. A contraction can save a variable while creating more monomials; it can also fail the height test entirely. Adaptive selection is a construction method, not a proof of optimality. The released certificate freezes each accepted edge, so reproduction requires no heuristic search.

17.9 Unit homogenized equality cycles

The retained Round21 support reduction generalizes the anchored three-cycle without adding a coordinate. Let a, b be the two aliases of one non-unit logical class, and let $u_1, \dots, u_\ell$ be existing unit aliases, already anchored by the unchanged unit energy U. Choose three expressions f_1, f_2, f_3, each equal to a, b, au_j or bu_j, with both underlying aliases a and b represented. Replace that class's equality penalty by

$$C(f_1, f_2, f_3) = f_1^2 + f_2^2 + f_3^2 - f_1 f_2 - f_2 f_3 - f_3 f_1. \tag{40}$$

No other incidence label or unit relation is changed.

Theorem 17.5. *The replacement preserves the calibrated integer zero set and the universal source lift. It introduces no variable and has degree at most four. Any frozen instance whose complete coefficient collection has height one and exact degree four is a height-one quartic on the same independent coordinate set.*

Proof. The polynomial is nonnegative for every real tuple because

$$2C(f_1, f_2, f_3) = (f_1 - f_2)^2 + (f_2 - f_3)^2 + (f_3 - f_1)^2.$$

At a zero of the complete energy, the separately nonnegative unit summand gives $u_j = 1$ for every unit alias. The three expressions then reduce to copies of a and b, with both present. Their equality is therefore exactly $a = b$. Conversely $a = b$ and all $u_j = 1$ make the new penalty vanish. All other summands and the source-block divergence identity are unchanged. The old and new energies have literally the same zero set in the same coordinate space, so this step has the identity

map as both witness maps. The same source projection and unique tree-flow lift also remain inverse on zeros.

On the canonical lift of an arbitrary source tuple, all units are one and all aliases equal their logical values. Thus the new penalty vanishes identically there, not merely over source zeros. This proves that (38) is unchanged. Each f_i has degree at most two, so the new penalty has degree at most four.

These facts do not by themselves prove height one or decreased support. A product au_j can create monomials already present elsewhere. The complete energy is therefore collected after each proposed change, including incidence, other equality penalties and the unit path. The accepted frozen certificate must pass exact checks of every coefficient, degree, constant and live coordinate. No choice of f_i is assumed universally safe for height. □

This changes the polynomial away from its zeros, but retains the same source witnesses. The tested cycles save support through actual signed monomial collisions in the full energy, not through an uncollected count of terms in one penalty. The finite selection procedure is not an exhaustive search over height-one equations.

17.10 Product constructors for constants

The new normalization retains the old logical coordinate list, constant values, scale values and source rows. If a needed constant K_c has available earlier constants K_a, K_b with $1 < a \leq b < c$ and $ab = c$, replace its additive definition by

$$K_c - K_a K_b = 0.$$

The chosen pair is the lexicographically least (b, a) among the eligible pairs. This is fixed certificate data, not an unknown of the equation.

Proposition 17.6. *These replacements preserve the normalized integer zero set literally, and preserve the final network zero set by the canonical source lift and projection. They introduce no new normalization coordinate or sign branch.*

Proof. Constants are ordered increasingly. Both factors precede K_c, even when $a = b$. Induction through the unchanged additive and new multiplicative monic definitions forces $K_c = c$ uniquely. Conversely the intended values satisfy every definition. Every source row and scale has exactly the same pullback. Thus the normalized zero sets agree in the same coordinates. Each final network separately has the already proved inverse projection and unique lift through the common source, which supply the desired bijection even though the final aliases and flows may differ. The universal identity (38), including the hub term, remains valid. No division is used. Positivity of a forced constant is not a restriction on a quantified final coordinate. □

The monic row now has two signed monomial occurrences rather than three. This changes the graph search and can remove more than one eventual alias or flow; final savings are measured after complete coefficient collection.

17.11 Chained scale constructors

The smallest selected endpoint also replaces a scale definition $S(c, x) - K_c x = 0$ when an earlier scale $S(b, x)$ and a constant K_d exist with $d > 1$, $b < c$ and $db = c$. The new definition is

$$S(c, x) - K_d S(b, x) = 0.$$

Constants precede all scales and scales are ordered by (coefficient, source coordinate). Induction therefore forces $S(c, x) = d(bx) = cx$ for every integer x, including negative and zero values. Conversely those values satisfy the new rows. The monic pivot and predecessor on the *same*

original coordinate are essential; merely factoring the numerical coefficient without checking the base coordinate would not establish the identity. This again preserves both normalized zero sets and the final source-lift bijection without division. The deterministic choice is the lexicographically least available (d, b).

The independent checker removes row orientations and verifies the identical coordinate list and metadata, every changed and unchanged row, all earlier-factor conditions, every exact source pullback, and the complete collected final polynomial. It tests source/constant corruption classes and a separate chained-scale corruption. No heuristic claim of optimality is used to prove correctness or numerical metrics.

18 Pruning and inlining the arithmetic network

The new height-one improvements change the normalized arithmetic DAG before network compilation. They do not require a new incidence-energy identity. The graph construction, equality penalties, unit anchoring and universal lift (38) continue to apply to the changed normalization. The original four arithmetic source families remain those identified by `height_one/SOURCE_BINDINGS.json`; a lower direct-quartic dimension elsewhere is not silently substituted for one of these source theorems.

18.1 Fuse source products and prune unused constructors

Start with the retained triangular monic constant and scale definitions. For a constant source occurrence of value c, replace K_c by $K_a K_b$ when $ab = c$ and both factors are available. For a linear occurrence cx, use an available $S(c, x)$ or $K_a S(b, x)$ with $ab = c$. Signs are retained. The inherited factorization of quadratic source monomials is unchanged. Every replacement is one signed monomial of degree at most two and has the same exact pullback to the original source coordinates.

These are identities after constructor substitution, not arbitrary-tuple identities: K_6 and $K_2 K_3$ can differ away from constructor zeros. Retain only constructors reachable from the newly selected source terms, using the actual definition dependencies, and compact in inherited order. Every retained dependency is then retained, so monic triangular induction still uniquely evaluates all new constructors over every original tuple. Every deleted old constructor has a unique intended value as well.

Proposition 18.1. *The replacement and pruning preserve the normalized integer zero set by inverse projection and canonical restoration maps. The corresponding final network zero sets are bijective through their common original source.*

Proof. A zero of the old normalization projects to its original tuple, on which every replacement has the same pullback. The retained constructors have their intended values, so its projection is a new zero. Conversely, at a new zero monic induction restores the original source equations and all retained constructor values. Evaluate every deleted constructor at its unique intended value; then every old row vanishes. These maps are inverse on zero sets. Compose them with the already proved unique network lifts and projections. No sign restriction, division or congruence condition is added. □

The finite selector uses product-first ordering, then at most five ordered strict improvements of the number of reachable constructors and their total occurrence count. Those costs are heuristics, not the final support. The actual final collection determines the measured improvement.

18.2 Inline low use constants by addition

Let K_c be retained, and choose $a + b = c$ with $1 \le a \le b < c$. Use $K_1 = 1$ as notation, not a new coordinate. Other factors are earlier retained constants. Delete the defining row of K_c, substitute

$K_a + K_b$ in every other row, and expand distributively.

Every replacement remains degree at most two. If a quadratic monomial contains K_c twice, retain all four signed occurrences. In particular $a = b$ creates repeated occurrences, not a collected coefficient-one claim. Every remaining constructor keeps its own coefficient-one pivot, and its new dependencies are earlier, so triangular monic induction still gives its original intended value. Each source pullback is unchanged since $a + b = c$. A new zero uniquely restores $K_c = c$; all old rows and its deleted definition then vanish. Deletion and restoration are inverse over both $\mathbb{R}$ and $\mathbb{Z}$. Repeated inlining composes the restorations in reverse deletion order. The usage budget and pair order choose valid substitutions but do not prove monotone final support or optimality.

18.3 Expand rare scale coordinates

For a retained scale $S(c, x)$, $c > 1$, delete its monic defining row and replace every occurrence by the linear polynomial cx, represented as c signed copies of x. A quadratic occurrence containing the scale twice gives c^2 equal signed degree-two occurrences. All source coordinates precede constructors, so the other pivots stay monic and triangular. Their exact pullbacks are unchanged. Restore the deleted scale uniquely as cx, for negative, zero and positive x alike. The same inverse zero-set maps therefore apply. Prune newly unused constructors by Proposition 18.1.

The finite rule tests the precise increase of normalized occurrences against a budget. The subsequent graph may have more or fewer monomials; neither direction is a theorem. Repeated occurrences are represented as distinct bound source occurrences, with factor aliases chosen as needed. They are not confused with final collected coefficients. Height one is required and checked only for the entire collected polynomial.

18.4 The new finite catalog

Every entry below is a polynomial on unrestricted integers, has exact degree four, constant one, coefficient set $\{-1, +1\}$, and is nonnegative on all real tuples. There are no equality dummies and every stated coordinate is live. Support includes the constant. The source numbers are inherited family identifiers, not a claim that the normalized source has that dimension.

Endpoint	Variables	Terms	Aliases	Flows	Source
minimum484	484	24131	461	23	241
compact485	485	23698	462	23	241
compact486	486	20007	462	24	242
compact487	487	19748	463	24	242
compact488	488	19649	464	24	242
compact489	489	17729	462	27	242
compact493	493	17058	465	28	242
compact494	494	16557	464	30	242
compact495	495	16400	465	30	242
balanced531	531	12020	484	47	263
balanced533	533	11985	486	47	263
sparse599	599	8279	504	95	285
sparse600	600	8241	505	95	285
sparser695	695	7135	504	191	285
sparser696	696	7113	505	191	285

The new 484/24131 variable/support point improves both dimensions against the Round22 minimum 493/24828. The new 533/11985 point improves the retained 557/11991 tradeoff in both measures. Likewise 600/8241 improves 605/8305, and 696/7113 improves 701/7141. Every one of the nine selected Round22 endpoints is dominated by at least one of the selected Round23 endpoints in variable count and support, with degree and height fixed. At exactly 493 variables, support falls from the old 24828 to 17058, a reduction of 7770 monomials (about 31.3%). This is separate from the new minimum of 484 variables. The new 495/16400 point also improves the old

496/20548 point. This is a checked finite comparison, not an exhaustive global frontier. Within the new catalog the alternatives trade dimension for support.

18.5 Independent source semantics and final collection

The independent normalizer audit computes each coordinate's intended polynomial from metadata without importing the producer. It verifies every source row, with its certified sign, has exactly its pinned pullback; every constructor has identically zero pullback and exactly one monic triangular pivot; and every source row occurs once. A separate factor checker verifies all affine pullbacks to the byte-pinned original sources. Thus fusion, inlining and pruning are validated without trusting their selector algorithms or cost calculations.

The final graph audit independently reconstructs square differences, every occurrence and factor binding, every block boundary, each internal tree, all aliases and units, and the symbolic lift divergence. Exact expansion checks the entire height-one array, degree four, constant one and coordinate participation. The literal text form is independently parsed as well. Normal and optimized-Python replays, deliberate corruption tests and exact same-source/hub/seed controls bind these properties to the delivered bytes. A newly generated parameter-matched control is explicitly labelled new; it is not called a historical endpoint. Full normalization comparisons are not misrepresented as isolated one-operation experiments.

Combining the inverse maps with (38) and the named original source calibration proves (1) for each selected polynomial. Successful coefficient checks are not substituted for that unbounded argument.

19 The retained ten positive integer arithmetic interface

The three reference equations retained from Round22 have exactly ten independent inputs

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Their complete four-port tuple is the Start-only4127 specialization of Section 20. The arithmetic proof below is uniform under its authenticated scalar bounds; the whole-native bridge in that section licenses all four ports together. Strict Horizons supplies the free quotient and projection-first strategy; Trace Radical supplies the linear trace; Unsquared Factors and Linear Selectors supplies the inverse correction at k. The retained Round21 uniform signed bound licenses their composition without assuming $q \mid Y$ early and without the extra positive shift $T + \alpha$.

19.1 The three complete polynomial definitions

Put $h = \beta J$, $q = h + 1$, $C = q - F - \alpha$ and $k = \eta + \zeta$. Choose one of the following fixed charts. The symbols W_b, λ and D below are polynomial abbreviations, not independent inputs.

Chart	W_b	X	Y	λ	D
Free plain	qw	q^2w	s	q	q^7
Scaled inverse	w	qw	qs	1	$6q^6$
Free inverse	w	qw	s	q	$6q^7$

In every chart define

$$c = kY + \eta, \qquad A = \lambda Y(X+1) + 1, \qquad \Delta = A^2 - 1,$$
$$H = 4A - 5, \qquad v = X\lambda^2 Y^2, \qquad E = 1 + v(v+1)k^2,$$
$$R = [h(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J,$$
$$B = (K_0 + W_b)C - F, \qquad U = ic^2, \qquad a = \Delta U^2 E,$$

$$F_0 = 1 + a, \qquad G = EF_0, \qquad f = 1 + 2a,$$
$$Q = 4\Delta a F_0, \qquad b = cT - R, \qquad V = fb - c,$$
$$K = X(3c - 2X) + 2, \qquad L = bV, \qquad M = hDH.$$

In the plain chart put $e_c = k - R - 1 - q^4 s$; in either inverse chart put $e_c = 6(k - R - 1 - q^4 s) - vk(k^2 - 4)$. Finally set

$$Z = hD[H(n - LB) + K] - H[DB + he_c],$$
$$P = Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2). \tag{41}$$

The equation is $P = 0$. Fixed powers are implemented by binary multiplications. There is no radical or quotient gate. The graph's register sharing, not the typography of these abbreviations, defines the measured operation count.

For the proof only, let $t_* = B/h$, $\kappa = K/H$, $z = Z/M$ and $W_* = 1 + hL$. In the plain chart let $e = k - R - 1$, $\rho = e/q^7$. In the inverse charts let

$$e_* = k - R - 1 - \frac{vk(k^2 - 4)}{6}, \qquad \rho = e_*/q^d, \quad d = 6 \text{ or } 7.$$

In all charts put $\sigma = Y/q^3$. Direct expansion gives

$$P = (Q - 1)Z^2 - M^2 G(QV^2 - 1), \qquad z = n + \kappa + \sigma - W_* t_* - \rho. \tag{42}$$

These proof quotients are rational until integrality is established. The carrier denominator is nonzero for every positive input.

19.2 Root extraction before any transport sign

Positive inputs give $q \geq 32$, $X \geq q$, $Y \geq 1$, $k \geq 2$, $c \geq 3$, $A > q^2$, $\Delta > 1$, $U \geq c^2$ and $E > 1$. The exact identities are

$$f^2 = 1 + 4aF_0, \quad Q = \Delta(f^2 - 1) = 4\Delta^2 U^2 G, \quad \gcd(E, F_0) = 1. \tag{43}$$

Also $f > 2c$ and $Q > c^2$, hence $V \neq 0$ for every integral b. At a zero, valuations in (42) imply that $G = g^2$ for a positive integer g: every prime of G divides neither $Q - 1$ nor $QV^2 - 1$, so its valuation in the rational square z^2 is even. Coprimality then gives $E = \tau^2$, $F_0 = f_0^2$ and $g = \tau f_0$ with positive integer roots.

The inherited adjacent-coefficient descent makes $y = |z|/g$ integral. For completeness, a reduced denominator m satisfies $m^2 \mid Q - 1$. For $u = |V| > 1$, the norm-preserving transformation

$$u' = (2Q - 1)u - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qu$$

has a nonzero integral first coordinate with $|u'| < u$ and preserves the reduced denominator of the second. The first integrality follows from $(Q - 1)y \in \mathbb{Z}$; denominator preservation follows from $\gcd(2Q - 1, m) = 1$. The norm gives $u < y < (2Q - 1)u/[2(Q - 1)]$, proving $0 < u' < u$ after reflection. Descent reaches $u = y = 1$, forcing $m = 1$. Therefore

$$QV^2 - (Q - 1)y^2 = 1, \qquad z = \varepsilon g y, \quad \varepsilon \in \{-1, 1\}.$$

With $I = 2Ug$, equations (43) give $f^2 - \Delta I^2 = 1$, $Q = (\Delta I)^2$, $c^2 \mid I$ and $c \leq I$. The retained signed all-index folding theorem applies to $V \equiv -c \pmod f$ and gives $c = \psi_A(p)$ for a positive odd index p. Since $c \geq 3$, $p \geq 3$, and hence

$$c \geq c_3 := 4A^2 - 1 = 4\Delta + 3 > 4q^4, \qquad c > X, \qquad K > 0. \tag{44}$$

No sign of R, B, C, b, V or divisibility $q \mid Y$ was used.

19.3 A uniform bound removes the positive shift

If $b = 0$, the auxiliary norm would give $0 < y^2 - c^2 = (c^2 - 1)/(Q - 1) < 1$, impossible for integers. Thus $b \neq 0$, $\operatorname{sgn} V = \operatorname{sgn} b$ and $|V| > 1$. Put $S = \Delta I$, so $Q = S^2$. Pell classification of (SV, y) gives an odd auxiliary index: even traces are units modulo S, whereas SV is divisible by S. Index one would give $|V| = 1$. The index is therefore at least three, and $|V| \geq 4Q - 3$. Moreover

$$\frac{a^2}{g^2} = \frac{\Delta U^2 a}{1 + a} > \frac{\Delta U^2}{2} > 9,$$

so $f > 6g$. Using $c_3 \leq c$ and $f > 2c$ yields the essential bound, for either sign of V,

$$|b| \geq \frac{|V| - c}{f} \geq 4\Delta f - \frac{c + c_3}{f} > 4\Delta f - 1 > 4g. \tag{45}$$

The possible auxiliary index three is harmless; it is not excluded. The norm also gives $y < 3|V|/2$. Consequently

$$L = bV = |b||V| > 0, \qquad gy < 3L/8 < L/2, \qquad |b|/q^d < L/2 \quad (d = 6, 7).$$

This remains valid at $Y = 1$, before its cubic q-scale is recovered.

In the plain chart $e = k - R - 1 < b$ since $cT - k + 1 > 0$. In the inverse charts $k \geq 2$ gives $e_* \leq e < b$. Thus $\rho < |b|/q^d$ in all charts. If $B \leq -1$, then $W_* t_* < -L$, and (42) implies

$$n = \varepsilon g y - \kappa - \sigma + W_* t_* + \rho < L/2 - L + |b|/q^d < 0,$$

contradicting the positive domain. Hence $B \geq 0$ and $C > 0$. This handles all negative values of R and zero uniformly. Only now use the authentic whole-tuple mask bound $3q < R < q^4 - 2$. Equation (44) gives $c > 2R$, hence $b, V > 0$. The retained normalized strong-rank and signed auxiliary-index theorems apply to $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$; the latter uses $f - 1 = 2\Delta i^2 c^4 E$. With $c > 2p, 2R$ they yield $p = R$. Thus R is positive odd, and mask parity gives even q and odd J. Neither ρ nor σ has yet been assumed integral.

19.4 Authenticate the first index at every prime

The square $E = \tau^2$ gives $k = 2\psi_{1+2v}(m)$ with $m \geq 1$. The case $m = 1$ would give $k = 2$, $\eta = \zeta = 1$ and $c = 2Y + 1$ strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. Thus $m \geq 2$. The first base $1 + 2v$ exceeds A, since $\lambda Y(2X\lambda Y - X - 1) > 0$. Monotonicity and $c > k$ then give $m < R$.

In the free plain chart $v = q^4 w s^2$. Integral z and $\gcd(q, hH) = 1$ give $q^7 \mid e - q^4 s$, hence $q^4 \mid e$. The recurrence $k \equiv 2m \pmod{4v}$ shows $q^4 \mid 2m - R - 1$. Its absolute value is less than $R < q^4$; therefore $2m = R + 1$.

In either inverse chart $v = q^3 w s^2$. The inherited finite Lucas identity, valid at every index, is

$$k = 2m + 8vC_m + 32v^2 Z_m, \quad C_m = \binom{m+1}{3}, \quad Z_m \in \mathbb{N}.$$

Since k is even, $k(k^2 - 4)/6$ is integral: write $k = 2u$ and use $3 \mid u(u - 1)(u + 1)$. Thus e_* is integral. Clearing the carrier and cancelling its literal factor six, not inverting six modulo q, gives $q^d \mid hH(e_* - q^4 s)$, hence $q^4 \mid e_*$. Put $t_0 = 2m$, $d_0 = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + kt_0 + t_0^2 - 4$. Cubic subtraction gives exactly

$$3(e_* - d_0) = 4v^2(24Z_m - A_m S_m). \tag{46}$$

Indeed $t_0(t_0^2 - 4) = 48C_m$ and $k - t_0 = 8vA_m$. For any prime $p \mid q$, writing $a_p = v_p(q) \geq 1$, the identity gives

$$v_p(e_* - d_0) \geq 6a_p - v_p(3) \geq 4a_p.$$

This includes the prime three. Hence $q^4 \mid d_0$, and the strict size bound gives $d_0 = 0$. In all three charts,

$$m = (R+1)/2. \tag{47}$$

Full q^6- or q^7-divisibility of e_* has not been inferred here; the third-primary obstruction is repaired only after projection.

19.5 Projection precedes complete denominator recovery

The literal numerator and integral z give $H \mid hDK$. Cancel only the q-part: $H \mid d_1 hK$, where $d_1 = 1$ in the plain chart and $d_1 = 6$ in the inverse charts. Write $R = 2r+1$, so $r \geq 48$. The retained λ-uniform ratio theorem applies to the authenticated indices and strip $Y < c/k < Y+1$:

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

Modulo odd H, the universal scaled recurrence gives $3 \cdot 2^{R-1}\psi_A(R) \equiv 4^R - 1$. The retained projector yields

$$H \mid d_1 h(X - 2^R)(2^R X + 1).$$

The complete product, including d_1, has magnitude below H:

$$d_1 h|(X - 2^R)(2^R X + 1)| < 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H.$$

After division by X^{r+1} the two terms decrease for $X \geq 32$, $r \geq 48$; at $(32, 48)$ their sum is $2^{-117} + 2^{-25} < 1$. Thus $X = 2^R$. The universal projector itself then gives positive integral $\kappa = K/H$; no unproved cancellation of $d_1 h$ is used.

Now $q \mid X$ implies $q = 2^a$ with $a \geq 5$. Since $R > 3q \geq 6a$, one has $q^6 \mid X$. In the free charts $v = Xq^2Y^2$ is divisible by q^8. In the plain chart $4v \mid e$ gives $q^7 \mid e$; in the free inverse chart (46) with $d_0 = 0$ gives $q^7 \mid e_*$. In the scaled chart $Y = qs$, so $q^8 \mid v$ as well and the same identity gives $q^6 \mid e_*$. Thus ρ is integral everywhere.

The carrier now makes $\sigma - W_* B/h$ integral. Multiply by $q^3 h$ and reduce modulo q^3 to obtain $q^3 \mid Y$; reduce modulo h to obtain $h \mid B$, using $\gcd(q, h) = 1$ and $W_* \equiv 1 \pmod{h}$. Thus $t_* = B/h \in \mathbb{N}$. These are the exact retained native outer conditions. The fixed-minus parity and half-binomial theorems, uniform in λ, reconstruct fresh source witnesses at $A_{\text{source}} = Y(X+1)+2$, not at the present base. The retained whole-tuple native/refutation calibration then supplies a finite syntactic ZFC refutation.

19.6 A constructive positive lift with possibly negative correction

Conversely a finite refutation supplies canonical outer data $q = 2^a$, $X = 2^R$, $R > 3q$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, $C > 0$ and integral $t_* = B/h \geq 0$. Choose $w = X/q^2$ in the free plain chart and $w = X/q$ otherwise; take $s = Y$ in free charts and $s = Y/q$ in the scaled chart. All are positive integers. Construct fresh A, v and set

$$c = \psi_A(R), \quad k = 2\psi_{1+2v}((R+1)/2), \quad \tau = \chi_{1+2v}((R+1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The retained exact uniform ratio theorem gives $0 < \eta < k$. The same recurrence and power-of-two divisibility prove integral ρ; projection gives $0 < \kappa < Xc$, and σ is positive integral.

Choose a positive Pell matrix period at A modulo $c^2\tau$, obtaining a seed (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$. Then $a = f_0^2 - 1$, $G = (\tau f_0)^2$ and the doubled strong pair is $f = 1 + 2a$, $I = 2f_0 I_0$. Keep this seed fixed. The retained positive auxiliary reconstruction gives (V_*, y) with $V_* = f^2(cT_{\text{old}} - R) - c$. Define

$$T_{\text{new}} = \frac{V_* + c + Rf}{cf}.$$

This is integral: $f^2 \mid V_* + c$, $V_* \equiv -R \pmod{c}$, $f \equiv 1 \pmod{c}$ and $\gcd(c, f) = 1$. It is strictly positive, and $V_* = f(cT_{\text{new}} - R) - c$ exactly. This is a fresh witness construction, not an inverse shear on arbitrary positive tuples.

In inverse charts the correction may be negative. Since $k \geq R + 1$ and $vk^2 > 6$, one has $|\rho| < vk^3/3$. For every chosen $i \geq 1$,

$$g > \sqrt{\Delta}c^2E > c^2v^2k^2 > vk^3, \qquad g > 4Xc.$$

For the latter inequality use $\sqrt{\Delta} > A - 1 > X$ and $cE > 4$. Thus $\kappa + \sigma < g/2$ and $|\rho| < g/3$. In the plain chart $\rho \geq 0$, and the same bound on g suffices. In all cases

$$n = gy + (1 + hL)t_* + \rho - \kappa - \sigma$$

is a strictly positive integer. Its carrier has $z = gy$, so the literal polynomial vanishes. Every one of the ten independent inputs has been constructed in the actual positive domain at one coherent tuple.

19.7 Attained degrees and exact circuit counts

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$. Complete homogeneous collection gives

$$\begin{aligned}
[P_{\text{plain}}]_{240} &= -2^8 q_0^{112} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2, \\
[P_{\text{scaled}}]_{232} &= -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2, \\
[P_{\text{free}}]_{214} &= -9 \cdot 2^{10} q_0^{86} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2.
\end{aligned}$$

They have respectively $99, 202, 99$ collected monomials. In the free charts the leading part of b is $-q_0^3(q_0 - F)$; in the scaled chart the full degree tie $b_0 = q_0 B_0$ must be retained. Selecting $\eta^{32} F^2$ in the free cases gives nonzero coefficients $-2^8\beta^{112}$ and $-9 \cdot 2^{10}\beta^{86}$; selecting $\eta^{34} T^2$ in the scaled case gives $-9 \cdot 2^{10}\beta^{100}$. These prove attained degree at the authentic positive β.

In chart order, the degrees of (Q, M, G, V, Z) are $(82, 13, 52, 40, 60)$, $(80, 11, 50, 40, 57)$ and $(72, 12, 46, 36, 54)$. The unique dominant output term is $-QM^2GV^2$. The live binary-operation counts are respectively $82, 85, 87$. The independent checker verifies complete top forms of every register, not sampled lines. Historically, the scaled 232/85 point saved two operations against incoming Trace Radical's 232/87; the free 214/87 point saved eighteen degrees against that same incoming point. Neither comparison is a global bound or a claim about unrestricted integer variables.

20 The retained coherent coefficient tuple

The three ten-positive-variable equations retain their free-port arithmetic and their exact degree/gate pairs $(240, 82)$, $(232, 85)$, and $(214, 87)$. The Round22 result, retained here, is a coherent specialization of *all four* fixed ports to the Start-only complemented native encoding with cell length 4127. The complete unsigned coefficient payloads shrink from 295 388 to 97 046 bytes, a saving of 198 342 bytes (67.146%). The incoming Start-marker package had specialized its fifteen-positive P66 front to this tuple, but retained the old radix-67 tuple for its ten-positive 232/85 equation. That result proves the whole-native bridge for all three retained ten-positive charts. It does not lower their degrees or operation counts, or change the separate six-port integer11 family.

20.1 The complete fixed tuple and its semantic requirements

Put $\rho = 2^{47}$, $M = 2^{44}$, $L = 4127$, $B_0 = \rho^L$ and $\beta = B_0 - 1 = 2^{193969} - 1$. Let $t_j = (w_j, n_j, e_j, s_j)$, $0 \leq j \leq 942$, be the complete fixed tile list reconstructed from the retained 436-character machine

numeral. Its Start is $(2, 435, 3, 1)$ and tile942 is $(4, 437, 2, 437)$. Define

$$c_j = 1 + 2^{10}(1 + w_j + 2^8 n_j + 2^{17} e_j + 2^{25} s_j), \quad 0 \le j \le 942,$$

and, in addition,

$$\begin{aligned} c_{1002+j} &= 2^{10+j}, \quad c_{992+j} = 2^{27+j} \quad (0 \le j < 8), \\ c_{973+j} &= 2^{18+j}, \quad c_{943+j} = 2^{35+j} \quad (0 \le j < 9). \end{aligned}$$

All other c_e, $0 \le e \le 3296$, are zero. Write $G_m(x) = \sum_{j=0}^{m-1} x^j$ and

$$\begin{aligned} \Psi(x,t) &= x^{992}(x^{10} + t^{17})G_8(tx) + t^8 x^{943}(x^{30} + t^{17})G_9(tx) + t^{34}(2-t)x^{1031}, \\ \Phi(x) &= \sum_{j=0}^{942} c_j x^j + 2^{10}\Psi(x,2), \qquad \Lambda = \rho^{2042} + \rho^{3074}, \\ D_C &= \rho^{20} + \rho^{30} + \rho^{1031} + \Lambda\Phi(1/\rho), \\ K_0 &= D_C + B_0\rho^{10}, \\ M_C &= \beta + \rho - (\rho+3)\rho^{990} - G_{943}(\rho) - \Psi(\rho,1), \\ M_F &= B_0 + 3 + (M-2)\Lambda + \rho^{973}G_9(\rho) + \rho^{1012}G_8(\rho) + \rho^{1031} + 2^{29}\rho^{4124} - \rho^{4106}. \end{aligned}$$

These finite sums define integers: the reflected exponents are nonnegative. M_F is already the once-shifted source mask; the proof-only native mask is $M_F - \beta$, not a second independently selected coefficient. The complete numerical payloads, decoder, all 943 tiles and 3297 weights are included, rather than replacing coefficients by hashes.

The retained Start-only proof establishes more than scalar inequalities. Its physical support has 980 positions, with a separate zero-weight flag at 1031. The carry box is $0, \dots, 29$; clean clauses precede alignment. Eight Start label bits and the flag test leave offsets 0 and 32, and the actual complemented south color $511 - 437 = 74$ of tile942 excludes 32. The flag at a temporal predecessor equals the current Start selector, *not* occupancy. Positive complemented colors propagate occupation after alignment. Complete clause resets, origin masks and 875 padding bits then reconstruct the genuine marked blank computation. Every one of these semantic conditions is retained; matching port names or the inequalities $0 < M_C < \beta < M_F < 2\beta$ would not alone license this specialization.

For the converse the actual new controller has

$$d = 193969, \quad \lambda = 474491, \quad \mathrm{ord}_{47}(B_0) = 23, \quad \mathrm{ord}_{4127}(B_0) = 2063, \quad \mathrm{lcm}(23, 2063) = 47449.$$

The values of $1 + B_0K_0 + B_0^2$ modulo $47, 4127, 474491$ are respectively $9, 3487, 153928$, and $B_0^{\lambda-1} = 1 + 135285\lambda \pmod{\lambda^2}$. Full trial division and all prime-factor order tests discharge these finite premises. The whole cofinal rectangle/subset-sum proof uses this new K_0, not old controller residues or an old packed word.

20.2 Why all three arithmetic charts transfer

Write $h = q - 1$, $B_{\mathrm{tr}} = (K_0 + W)C - F$ and $t = B_{\mathrm{tr}}/h$. The retained ten-positive proof, with the new authenticated scalar boxes, recovers $X = 2^R$, $q^3 \mid Y$, $C > 0$ and integral $t \ge 0$. In the plain free chart $X = q^2 w$ and $W = qw$; in both inverse charts $X = qw$ and $W = w$. Thus in *every* chart $W = X/q$. Define fresh source coordinates

$$w_s = X/q, \qquad s_s = Y/q^3, \qquad z_s = t + 1.$$

They are positive integers and satisfy the exact transport, not merely a congruence:

$$(K_0 + w_s)C + (q - F) - z_s h = ht + q - (t+1)h = 1.$$

The rank $R = [h(q-F)+\alpha](q^2-1) + (M_C + qM_F)J$ and $C = q - F - \alpha$ agree literally with the new native interface. Projection still precedes full denominator recovery; no early inference from $q^3 \mid HY$ to $q^3 \mid Y$ is used.

For an explicit soundness bridge set $r = (R-1)/2$ and rebuild the unscaled source base, rather than identifying it with a free-chart base:

$$\begin{aligned}
a_s &= Y(X+1), & A_s &= a_s + 2, & \Delta_s &= A_s^2 - 1, & H_s &= 4a_s + 3,\\
c_s &= \psi_{A_s}(R), & D_s &= \chi_{A_s}(R), & P_0 &= 1 + 2XY^2,\\
k_s &= 2\psi_{P_0}(r+1), & \tau_s &= \chi_{P_0}(r+1),\\
\eta_s &= c_s - k_s Y, & \zeta_s &= k_s - \eta_s,\\
h_s &= (k_s - R - 1)/(XY), & \gamma_s &= (D_s - X - a_s c_s)/H_s.
\end{aligned}$$

The retained fixed-minus parity and exact half-binomial theorem prove $R = 3 \pmod 4$ and positivity and integrality of these witnesses. Take $m_s = c_s R$, $I_s = \psi_{A_s}(m_s)$, $f_s = \chi_{A_s}(m_s)$ and $i_s = I_s/c_s^2$. The Pell multiplication identity gives $c_s^2 \mid I_s$. The canonical auxiliary construction at $S_s = \Delta_s I_s$ supplies positive V_*, y_s, T_s with

$$V_* = f_s^2(c_s T_s - R) - c_s, \qquad S_s^2(V_*^2 - y_s^2) + y_s^2 = 1.$$

These fresh witnesses give every normalized P66 factor equal to1, at the *same complete new tuple*. The Start-only P66 soundness theorem therefore extracts a finite syntactic ZFC refutation. This establishes a semantic composition, not equality with the old specialized polynomial.

Conversely, the new native theorem constructs fresh canonical data from any finite refutation, with $q = 2^{dN}$, $X = 2^R$, $R > q(q^2-1)$, $R = 3 \pmod 4$, $\mathrm{pc}(R) = 3dN + 2$, $q^3 \mid Y$ and the actual integral transport. The rank margin permits $w = X/q^2$ in the plain free chart and $w = X/q$ in the inverse charts; take $s = Y$ for free charts and $s = Y/q$ for the scaled chart. The retained uniform ratio theorem, divisible strong period, trace-linear auxiliary reconstruction and positive-n bounds then produce all ten positive coordinates. The inverse correction may be negative; the original g bounds explicitly handle this. Thus each new polynomial has a positive zero exactly when a finite ZFC refutation exists. Negation is equivalent in ZFC to syntactic Con(ZFC), without deciding either side.

20.3 Exact degree and verification

With $q_0 = (2^{193969} - 1)J$, $k_0 = \eta + \zeta$ and $B_* = k_0 sT - q_0^2(q_0 - F)$, the full highest forms remain

$$\begin{aligned}
[P_{240}]_{240} &= -2^8 q_0^{112} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2,\\
[P_{232}]_{232} &= -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_*^2,\\
[P_{214}]_{214} &= -9 \cdot 2^{10} q_0^{86} w^{26} s^{58} i^{10} k_0^{32} (q_0 - F)^2.
\end{aligned}$$

Their selected nonzero coefficients establish actual degrees at these fixed numerals. Every original free-port gate is unchanged. Independent reconstruction decodes the complete machine numeral, all tiles and weights, and all coefficient bytes without importing the producer; it also checks full native geometry and controller premises. Normal and optimized-Python deep checks reject all 508 single-operator corruptions, and clean-copy replay reconstructs graphs and receipts byte-identically. These finite checks authenticate the artifacts; they are not an unbounded native/refutation proof or a new formal-kernel certificate.

The squared-transport variant is not silently transferred to the two free charts: their pre-rank auxiliary index-three case is allowed, and the proven bound relative to bV does not imply the required bound relative to b^2. Likewise no assertion of global minimality or a new signed11 degree floor is made.

21 Quotient centers with an index three safe transport

This round composes the retained free inverse chart with the quotient centers of source 178 [12]. The new ten-positive-variable degree/gate pairs are $(202, 99)$ and $(190, 120)$. The latter uses the transport weight $L = (UE)b^2$, rather than the unsupported bare b^2. The factor UE makes the transport safe before sign and rank recovery, including the possible main and auxiliary index-three cases. Both equations use exactly the whole Start-only 4127 tuple of Section 20; no numeral, mask shift or native controller is changed.

These are degree improvements with a gate cost, not simultaneous improvements in all metrics: the retained 214/87 and incoming 206/90 points have fewer gates. All ten inputs remain strictly positive. No statement about ten unrestricted integer variables or the separate signed eleven-variable family is inferred.

21.1 Both complete polynomial definitions

The independent coordinates are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Use the following polynomial abbreviations, in the displayed order:

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha, & X &= qw,\\
Y &= s, & N_0 &= qY, & k &= \eta + \zeta, & c &= kY + \eta,\\
A &= N_0(X + 1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5, & v &= XN_0^2,\\
\ell &= vk, & E &= 1 + \ell(\ell + k), & U &= ic^2, & a &= \Delta U^2 E,\\
F_0 &= 1 + a, & G &= EF_0, & f &= a + F_0, & Q &= 4\Delta a F_0,\\
b &= cT - R, & V &= fb - c, & K &= X(3c - 2X) + 2,\\
D &= 6q^7, & M &= hDH.
\end{aligned}$$

Here the earlier abbreviation R and the remaining abbreviations are

$$\begin{aligned}
R &= [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J,\\
B &= (K_0 + w)C - F, \qquad e_6 = 6(k - R - 1) - \ell(k^2 - 4),\\
S &= (kN_0)\big[\ell + k - (kN_0)(N_0 + 1)\big].
\end{aligned}$$

Equivalently, evaluate R before b; these displays specify one finite acyclic expression. The notation N_0 avoids confusion with the CRT modulus in the earlier direct family.

For the degree-202 equation choose

$$r = 2, \qquad L = bV, \qquad \Omega = (2AUE - US)V. \tag{48}$$

For the degree-190 equation choose

$$\begin{aligned}
&r = 4, && L = (UE)b^2, && e_2 = (kN_0)^2, && \theta = U^2 b,\\
&R_c = \theta(3E + e_2) - (b - c), && S_c = 4E^2 + 3S^2 + e_2^2,\\
&\Omega = U\big[2SR_c + A(4EV - \theta S_c)\big].
\end{aligned} \tag{49}$$

In either case the complete final equation is $P = 0$, with

$$\begin{aligned}
Z_0 &= hD[H(n - LB) + K] - H[DB + h(e_6 - 6q^4 s)],\\
Z &= Z_0 - M\Omega,\\
P &= Q[Z^2 - r^2 M^2 G V^2] + [r^2 M^2 G - Z^2].
\end{aligned} \tag{50}$$

Every displayed abbreviation is a polynomial, not an extra quantified unknown. The actual graph uses only binary $+, -, \times$ and integer leaves. In proof notation only, let

$$t_* = B/h, \quad \kappa = K/H, \quad \rho = (e_6/6)/q^7, \quad \sigma = Y/q^3, \quad z = Z/M, \quad W_* = 1 + hL.$$

Their exact rational identity is

$$z = n + \kappa + \sigma - W_* t_* - \rho - \Omega. \tag{51}$$

No quotient is assumed integral at this stage.

21.2 Root extraction before signs and rank

The valuation and rational adjacent-Pell descent of Section 19 apply to z/r. Multiplying the norm by the fixed square r^2 does not change any prime-valuation parity. At any zero there are positive integers τ, f_0, g, y and a sign ε with

$$E = \tau^2, \quad F_0 = f_0^2, \quad G = g^2, \quad z = \varepsilon r g y, \qquad QV^2 - (Q-1)y^2 = 1,$$
$$f^2 - \Delta I^2 = 1, \quad I = 2Ug, \quad c^2 \mid I, \qquad c = \psi_A(p), \quad p \text{ positive and odd.}$$

This uses rational descent, not an assumed integrality of z or y. As before, $q \geq 32$, $N_0 \geq q$, $A > q^2$, $c \geq 3$, and $p \geq 3$. Consequently

$$c \geq 4\Delta + 3 > 4q^4, \qquad c > X, \qquad K > 0.$$

The full signed auxiliary argument gives $b \neq 0$, equal signs of b, V, $f > 2c$, $f > 6g$, $y < 3|V|/2$, and

$$|b| > 4\Delta f - 1 > 3\Delta f, \qquad |V| \leq f|b| + c < \tfrac{3}{2} f|b|. \tag{52}$$

The auxiliary index-three branch is allowed. None of these facts uses $C > 0$, a sign for R, or any transport integrality. Also

$$(AUE)^2 - G = U^2E^2 - E > 0, \qquad \frac{(AUE)^2}{G} < \frac{A^2}{\Delta} < \frac{9}{4}, \qquad g < AUE < \tfrac{3}{2} g.$$

The factors U, E are strictly positive already from their definitions.

21.3 The scalar center and its pre rank bound

Regard A as a formal variable over $\mathbb{Z}[N_0, k, U, b, c]$. The identity $v = N_0(A - N_0 - 1)$ gives

$$E = e_2 A^2 + e_1 A + e_0,$$
$$e_2 = k^2 N_0^2, \qquad e_1 = k^2 N_0 (1 - 2N_0 - 2N_0^2),$$
$$e_0 = 1 + k^2 N_0 (N_0 + 1)[N_0(N_0 + 1) - 1].$$

Polynomial division gives $E = AS + e_0$ with exactly the displayed S. Since $X, N_0 \geq 32$,

$$S = k^2 N_0 [(X-1)N_0^2 - N_0 + 1] > 0, \qquad e_0 > 0, \qquad 0 < S < E/A.$$

The scalar center $C_2 = 2AUE - US$ therefore obeys $0 < C_2 < 2AUE < 3g$. For the degree-202 choice,

$$2gy + |\Omega| < 6g|V| < L/2.$$

Indeed (52) and $f > 6g$ give $|b| > 18\Delta g > 12g$, and $L = |b||V|$. For canonical positive V, Ω is positive as well.

21.4 The whole product center as an integer polynomial

Let $\mathcal{F}(A) = E(A)V(A)$, where $V = [1 + 2(A^2 - 1)U^2E]b - c$, and let Q_j be its formal polynomial quotient by A^j. The intended center is

$$\Omega = U[4AEV - 2Q_1 - Q_3/2]. \tag{53}$$

This is an integer polynomial: the degree-at-most-two term $(b - c)E$ does not contribute to Q_3, and the remainder $2U^2b(A^2 - 1)E^2$ is coefficientwise even.

Here is the full identity justifying the division-free evaluator. Writing quot for ordinary polynomial quotient in A, put

$$\begin{aligned} J_1 &= \mathrm{quot}(E^2, A) = S(2E - AS), \\ J_3 &= \mathrm{quot}(E^2, A^3) = e_2(2S - Ae_2), \\ H_1 &= \mathrm{quot}((A^2 - 1)E^2, A) = AE^2 - J_1, \\ H_3 &= \mathrm{quot}((A^2 - 1)E^2, A^3) = J_1 - J_3. \end{aligned}$$

Then $Q_1 = (b - c)S + 2\theta H_1$, $Q_3 = 2\theta H_3$, and

$$4H_1 + H_3 = 4AE^2 - 3J_1 - J_3 = A(4E^2 + 3S^2 + e_2^2) - 2S(3E + e_2).$$

Substitution proves (49) exactly as an integer-polynomial identity at every tuple. Division and halves occur only in the proof; there is no division gate in the emitted circuit.

21.5 A signed whole product bound and the new transport

For a polynomial in A, write a superscript $\#$ for absolute-coefficient evaluation at the positive A. Let $\delta = (N_0 + 1)/A < 1/16$. Then

$$\begin{gathered} |e_1| < 2e_2(N_0 + 1), \quad 0 < e_0 \le e_2(N_0 + 1)^2, \\ E^{\#} \le e_2(A + N_0 + 1)^2, \quad E \ge e_2A^2(1 - 2\delta), \quad E^{\#} < 2E. \end{gathered}$$

It follows that

$$\mathcal{F}^{\#} < 2E(|b| + c) + 8(A^2 + 1)U^2E^2|b|.$$

Use $|b| \ge 1$, $c + 1 < 2U^2E$, and $|V| > (f - c)|b| > \Delta U^2E|b|$. The last inequality follows from $a = \Delta U^2E > c$ and $f = 1 + 2a$. Hence

$$\frac{\mathcal{F}^{\#}}{|EV|} < \frac{4}{\Delta} + \frac{8(A^2 + 1)}{\Delta} = 8 + \frac{20}{\Delta} < 10.$$

Deleting low coefficients gives $|Q_j| \le \mathcal{F}^{\#}/A^j$. Equation (53) consequently gives

$$|\Omega - 4AUEV| < 10U|EV| \left(\frac{2}{A} + \frac{1}{2A^3}\right) < AU|EV|.$$

Thus Ω has the sign of V and $|\Omega| < 5AUE|V| < 7.5g|V|$. Therefore

$$4gy + |\Omega| < 13.5g|V| < 16g|V|.$$

The new transport is $L = UEb^2 > 0$. From (52),

$$\frac{g|V|}{L} < \frac{3Af}{2|b|} < \frac{A}{2\Delta} < \frac{1}{A}.$$

The final inequality uses $A^2 > 2$. Since $A > q^2 \ge 1024$,

$$4gy + |\Omega| < 16L/A < L/2. \tag{54}$$

This proof works for both signs of b, V and at the possible index-three case. It does not claim $gy < b^2/2$; the additional factor UE is essential to this pre-rank estimate.

21.6 Transport soundness and denominator recovery

The subtracted inverse-Lucas term is nonnegative since $k \geq 2$, so $e_6/6 \leq k - R - 1 < b$ for every integer R. Therefore $\rho < |b|/q^7$. Both weights have $L \geq |b|$, and hence $\rho < L/2$. If $B \leq -1$, then $W_* t_* = B/h + LB < -L$. Solving (51) for the positive input n, using $\kappa, \sigma > 0$ and the preceding center bounds, gives

$$n < L/2 - L + L/2 = 0,$$

a contradiction. Thus $B \geq 0$ and $C > 0$. Only now use the authenticated whole-tuple bounds $3q < R < q^4 - 2$. Since $c > 2R$, $b, V > 0$; the retained strong-rank and signed auxiliary-index arguments yield $p = R$, odd R, even q, and odd J.

The center change is an integer-polynomial multiple of M, and both L are integer polynomials. Every denominator step in the reference free inverse proof is consequently still valid, in the same order:

1. From $E = \tau^2$, obtain $k = 2\psi_{1+2v}(m)$ with $2 \leq m < R$. Cancel the literal factor six over the integers, not as a modular unit. Integral Z_0/M and $\gcd(q, hH) = 1$ give $q^4 \mid e_6/6$.

2. The exact cubic difference (46), with $v = q^3 w s^2$, proves $q^4 \mid [e_6/6 - (2m - R - 1)]$ at every prime, including three. The strict bound $|2m - R - 1| < R < q^4$ gives $m = (R + 1)/2$. Neither q^7 divisibility nor $q^3 \mid Y$ was assumed.

3. The literal projection gives $H \mid 6hK$. The full-product magnitude bound, without cancelling $6h$ through H, forces $X = 2^R$. Thus q is a power of two. Rank growth gives $q^6 \mid X$, then $q^8 \mid v$ and $q^7 \mid e_6/6$. Hence ρ and κ are integers.

4. Since $\Omega, z, n, \kappa, \rho$ are integral, (51) makes $\sigma - W_* B/h$ integral. Reduction modulo q^3 gives $q^3 \mid Y$; reduction modulo h gives $h \mid B$, because $W_* \equiv 1 \pmod{h}$ for either weight. Thus $t_* \in \mathbb{N}$.

These are exactly the same native outer conditions: $X = 2^R$, $q^3 \mid Y$, $C > 0$, and $(K_0 + w)C - F = ht_*$ with $t_* \geq 0$. The full reconstruction in Section 20 creates fresh unscaled Pell witnesses at $Y(X + 1) + 2$, and the native appendix decodes an actual marked blank computation and finite ZFC refutation. It does not infer native soundness from numerical boxes, mix separately calibrated ports, or shift M_F a second time.

21.7 Strictly positive completeness

Conversely the same whole Start-only native theorem supplies canonical outer data from a finite refutation. Use the reference construction to choose all positive coordinates except n: the strip coordinates, a sufficiently divisible strong Pell period, and the positive trace-linear auxiliary T. Then $b, V > 0$ and $t_*, \rho, \kappa, \sigma$ are integers with $t_* \geq 0$. The inherited bounds give $|\rho| < g/3$ and $\kappa + \sigma < g/2$, including possibly negative ρ. Both center proofs give $\Omega > 0$. Choose

$$n = rgy + \Omega + (1 + hL)t_* + \rho - \kappa - \sigma.$$

This is strictly positive since $r, y \geq 1$ and $g > \kappa + \sigma + |\rho|$. It makes $z = rgy$, so the literal polynomial vanishes. This is a constructive positive lift, not an assertion that an arbitrary inverse shear preserves the positive orthant. Both witness directions now prove (1) on the actual positive domain, without deciding consistency.

21.8 Exact cancellation degrees at the actual numerals

Use independent formal variables (A, N_0, k, U, b, c) with weights $(4, 2, 1, 5, 4, 2)$. The base degrees of (Q, M, G, V, UE) are $(72, 12, 46, 36, 19)$. For the degree-202 chart, $\deg Z_0 = 54$ and $\deg C_2 = 23$, while

$$C_2^2 - 4G = 4U^2 E e_0 + U^2 S^2 - 4E$$

has exact weighted degree 34 and leading form $4U^2A^2N_0^6k^4$. The output's dominant degree is $72 + 24 + 72 + 34 = 202$; its cross term has degree 197.

For the degree-190 chart, $\deg L = 27$ and $\deg Z_0 = 41$. Exact collection of all 594 monomials of $D_{\text{whole}} = \Omega^2 - 16GV^2$ gives weighted degree 94 and the entire highest component

$$[D_{\text{whole}}]_{94} = 8A^6N_0^8k^8U^6b^2(A^2 - 6N_0^4).$$

The center has 84 monomials and degree 59. The complete decomposition is

$$P = Q[Z_0^2 - 2MZ_0\Omega + M^2D_{\text{whole}}] + 16M^2G - (Z_0 - M\Omega)^2.$$

Its five summand upper degrees are $154, 184, 190, 70, 142$. Thus only QM^2D_{whole} can contribute degree 190. Naive gatewise degree propagation would miss the exact cancellations.

Set $q_0 = \beta J$, $k_0 = \eta + \zeta$. After substituting all actual leading forms in the ten original variables, the complete leaders are

$$[P_{202}]_{202} = 9 \cdot 2^{12}q_0^{80}w^{22}s^{56}i^{10}k_0^{32}(q_0 - F)^2,$$
$$[P_{190}]_{190} = 9 \cdot 2^{11}q_0^{74}w^{18}s^{52}i^{10}k_0^{32}(q_0 - F)^2(w^2 - 6s^2).$$

They contain 99 and 198 monomials. For example the P_{190} coefficient on $J^{74}F^2\eta^{32}w^{20}s^{52}i^{10}$ is $9 \cdot 2^{11}\beta^{74} \neq 0$ at the actual positive numeral β. Thus these are attained degrees, not only generic free-port bounds.

The graphs have 99 live operations $(60\times, 21+, 18-)$ and 120 live operations $(75\times, 24+, 21-)$. Every square, fixed-numeral multiplication and independent input is counted. Their entire 97046-byte tuple is unchanged. Independent checks authenticate every gate, exact quotient identity, all 594 defect coefficients, complete highest forms and coefficient bytes; normal and optimized runs reject all single-operator corruptions. These finite results supplement, rather than replace, the unbounded proof above.

22 Current inputs and fair comparisons

The current repository comparison is pinned to

`b42c796d1f5de9912a576bc2fd77b778ae7402bf`.

All nine incoming ZIPs were retrieved and byte-verified against that snapshot. The independently prepared `audit/INPUT_COMPARISON.json` identifies 13 sources and 45 selected profiles, including consolidated sources 175–178. This is not an exhaustive literature search or a full frontier. The older immutable pins inside each inherited proof remain unchanged.

Evidence codes in the tables are deliberately different. C denotes fresh independent, no-producer-import recollection of the complete expanded array. R denotes a freshly completed original root replay in both normal and optimized Python; it authenticates that source's finite verification chain, not a new unbounded proof. I denotes the previously sealed Round22 result, whose repacked incoming union matches its original 749-file inventory. The repack has separate part wrappers; their common-root union, not either wrapper in isolation, is the integrity target. This is not a missing-payload defect in the prior release. S denotes a source-declared comparison with no new complete replay asserted here. The three C controls are Pell Growth 210 principal and balanced, and Diagonal Growth 214. The Affine Horizons R replay completed all 18 stages in both modes.

22.1 Incoming explicit array controls

The table records the selected array-style controls from all nine incoming archives and sources 175–178. A dash means unreported here, not zero. All entries are inherited comparison inputs, not new Round23 endpoints.

Source and chart	Domain	n	Degree	Terms	Height	Evidence
Affine horizons / integer215 sparse	$\mathbb{Z}$	215	4	14578	16866	R
Affine horizons / integer215 lowheight	$\mathbb{Z}$	215	4	34700	4512	R
Affine horizons / natural202	$\mathbb{N}$	202	4	9353	1854167	R
Affine horizons / natural202 lowheight	$\mathbb{N}$	202	4	16927	416488	R
Affine horizons / natural200	$\mathbb{N}$	200	8	10392	13654244	R
Cofinal Pell / balanced	$\mathbb{Z}$	229	4	4477	7968	S
Cofinal Pell / compact	$\mathbb{Z}$	228	4	12001	7968	S
Cofinal Pell / split	$\mathbb{Z}$	207	16	94294	13607065661	S
Cofinal Pell / all5	$\mathbb{Z}$	206	24	–	–	S
Diagonal growth / Q214	$\mathbb{Z}$	214	4	5656	219813	C
Mixed base / recommended112	$\mathbb{Z}$	112	24	–	–	S
Mixed base / lowdegree143	$\mathbb{Z}$	143	8	–	–	S
Pell growth 210 / principal210	$\mathbb{Z}$	210	4	5282	2^{40}	C,R
Pell growth 210 / balanced214	$\mathbb{Z}$	214	4	5699	91679	C,R
Signed horizons / Q214	$\mathbb{Z}$	214	4	5875	655360	S
Signed horizons / Q212	$\mathbb{Z}$	212	4	–	–	S
Signed horizons / direct86	$\mathbb{Z}$	86	2097780	–	–	S
Round22 / principal215	$\mathbb{Z}$	215	4	7081	8880	I
Round22 / balanced215	$\mathbb{Z}$	215	4	7315	4096	I
Round22 / lowheight215	$\mathbb{Z}$	215	4	16077	2304	I
Round22 / dimension214	$\mathbb{Z}$	214	4	7078	1052028	I
Round22 / reciprocal212	$\mathbb{Z}$	212	10	55820	31104	I
Round22 / height one493	$\mathbb{Z}$	493	4	24828	1	I
175 contractions / integer191	$\mathbb{Z}$	191	6	13074	3268176223	S
176 oriented pool / joint seed	$\mathbb{Z}$	206	26	–	–	S
178 centers / natural203	$\mathbb{N}$	203	4	8237	799349232	S
178 centers / integer233	$\mathbb{Z}$	233	4	10379	12874	S

22.2 Incoming circuit controls

Source and chart	Domain	n	Degree	Gates	Evidence
Affine horizons / positive10 center	$\mathbb{Z}_{>0}$	10	206	90	R
Affine horizons / positive10 lower cost	$\mathbb{Z}_{>0}$	10	210	89	R
Affine horizons / integer14	$\mathbb{Z}$	14	85	110	R
Direct elimination / recommended85	$\mathbb{Z}$	85	16964	1788	S
Direct elimination / dimension84	$\mathbb{Z}$	84	2097732	1775	S
Pell carrier 65 / positive14	$\mathbb{Z}_{>0}$	14	150	65	S
Pell carrier 65 / natural14	$\mathbb{N}$	14	150	79	S
Pell carrier 65 / integer42	$\mathbb{Z}$	42	300	163	S
Pell growth 210 / compact11	$\mathbb{Z}_{>0}$	11	174	104	S
Signed horizons / P36	$\mathbb{Z}$	36	288	145	S
Signed horizons / positive14 lowdegree	$\mathbb{Z}_{>0}$	14	146	68	S
Round22 / positive10	$\mathbb{Z}_{>0}$	10	214	87	I
175 contractions / principal92	$\mathbb{Z}$	92	692	2428	S
176 oriented pool / integer90	$\mathbb{Z}$	90	32768	2306	S
176 oriented pool / integer121	$\mathbb{Z}$	121	16	2453	S
177 oriented powers / integer87	$\mathbb{Z}$	87	8388612	2447	S
177 oriented powers / positive10 center	$\mathbb{Z}_{>0}$	10	224	90	S
178 centers / positive10 wholeproduct	$\mathbb{Z}_{>0}$	10	208	116	S

22.3 What these comparisons do and do not imply

The incoming Direct Integer Elimination 85-variable equation has reported exact degree 16964 and 1788 live gates; its 84-variable alternative has degree 2097732. Those lower variable counts are not quartic improvements. Their maximum absolute literal is 1099511627780, while their expanded coefficient height bounds are respectively 2^{59805} and $2^{7342557}$. A literal maximum must not be substituted for an expanded height. Similarly, the Mixed Base 112- and 143-variable controls have

exact degrees 24 and 8, maximum literal 60, and reported expanded-height bounds strictly below 2^{79} and 2^{43}. Their supports are unreported here, not zero.

The Affine Horizons 202-variable quartics are over $\mathbb{N}$, not $\mathbb{Z}$. Their natural triangular-source restoration argument is a genuine domain proof, not an invertible affine map preserving every point of the orthant. Its 200-variable degree-eight point is neither another quartic nor an unrestricted-integer point. Positive Pell Carrier 65, unrestricted variants, and the positive scalar charts must likewise remain in separate domain rows.

The incoming Cofinal Pell construction's smaller frame uses an order-eight local square-class argument and a cofinal even-frame choice. Its completeness is not a pointwise inverse for arbitrary two-square witnesses. That source's 206-variable degree-24 and 207-variable degree-16 points are meaningful comparators to higher-degree direct systems, but not to fixed-degree quartics without reporting degree. The selected new reciprocal family uses its explicitly pinned conservative $m = 16$ square-frame 209-coordinate source. The weighted family remains a theorem instance and control, not the selected arithmetic source. A newer source is not transferred merely by renaming its requests.

The source 176 joint orientation theorem is a genuine mathematical input: it controls raw classified signed indices before canonical orientation. The unweighted joint theorem of Section 16 is not claimed as the invention of joint orientation itself. Source 178's whole-product centering is a distinct positive route with its own calibrated coefficient tuple. Its 208/116 (degree/gate) control is already dominated in those two measures by the incoming Affine Horizons 206/90 control at ten positive variables. Matching port names or coefficient-size boxes does not equate their calibrations.

The consolidated fetch is a selected, authenticated comparison subset, not an assertion that each extracted source subtree is independently standalone. The release source-closure maps distinguish retained prerequisites from comparison-only material. All ordinary mathematical reviews in this round are AI-assisted independent reviews; they are not external human refereeing or a new full Lean or Rocq dependency build.

23 Reproduction and the evidence boundary

The release is supplied as ordinary disjoint ZIP parts, each within the requested 4 MB limit and sharing one top-level directory. Extract all parts into the same parent directory. The combined directory is the standalone source closure; an individual part is not advertised as complete. No publication, repository commit or external formal-kernel execution is implied by the archive.

The normative definitions are complete finite arrays and arithmetic graphs. Each selected quartic includes its ordered variables, residuals and full collected polynomial. Each selected reciprocal endpoint includes its exact source/pool certificate, unit-pivot substitution ledger, complete residuals and any stated complete expansion. Height-one endpoints include collected JSON and literal-text polynomials with complete semantic graph certificates. Scalar endpoints include full binary-operation graphs and all four actual minimal unsigned coefficient payloads. A hash authenticates those bytes; it is never substituted for an absent coefficient.

23.1 Executable reconstruction

The component READMEs and the root release README give exact offline command lines and closure locations. The core finite workflows use Python's standard library and do not require the network or a proof assistant. Each checker is run normally and with assertions disabled by optimized Python where stated; its explicit failures remain active. The root receipt records terminal outcomes, not an optimistic command list. Integrity-only verification is labelled as such and is not confused with mathematical reconstruction.

For the direct quartics, the independent checks reconstruct the actual source changes, frame atoms, orientation deltas, growth splits and affine maps. They compare every changed and unchanged row, restore every deleted coordinate, collect the entire sum of squares, and verify

exact degree, height, support and live coordinate count. The ordinary Pell and cofinal arguments are separate proofs, not inferred from examples.

For the reciprocal family, independent checks authenticate every raw pair, request port, norm, wire, private quotient and source-row correction. They reconstruct both unweighted pool rows, every selected monic elimination and its integral inverse, then recollect the stated complete expansions. A matched control uses the same permitted pivot strategy with orientation unfused, isolating the new joint theorem rather than comparing unrelated search outputs. Larger exploratory residual systems with unmaterialized sum-of-squares expansions have no invented support or height.

The network checker does not import the producer. It checks each monic triangular constructor and original source pullback, all occurrence bindings, block trees, aliases, equality/unit squares, full coefficient arrays and the symbolic lift including the hub term. The final text parser is a separate representation check. Parameter-matched new controls are distinguished from unchanged historical controls.

The scalar checker independently reconstructs every gate cut and formal quotient identity, all 594 whole-product defect terms, both complete highest forms after actual leading substitution, and the complete coherent coefficient tuple. The 438 single-operator mutations are rejected; a separate algebra implementation confirms the 84-term center and 594-term defect. The inherited native proof and decoder are retained in their full authenticated source closure. These checks do not turn scalar numerical bounds into a new native soundness theorem.

23.2 Native report build and visible equations

From the combined root, rebuild this document with

```
sh report/build_report.sh
```

The script compiles native pdfLaTeX three times in a fresh temporary directory, with a local format fallback when necessary. It rejects overfull boxes, unresolved references and undefined citations. No screenshot or imported PDF is substituted for native mathematical source.

The principal residual appendix is generated from the literal residual array; its independent parser re-reads every displayed coefficient and exponent and compares all rows. The comparison and catalog tables are generated from frozen inventories and bound by their input and TeX digests. The selected scalar equations are printed in full with their acyclic polynomial abbreviations, while their operation-counted DAGs remain supplied verbatim. The reciprocal source and ordered pivot tables specify exact finite substitution rather than a generic promise to eliminate variables.

After the final three-pass compilation every PDF page is rendered and visually inspected. The report QA receipt binds the inspected PDF hash, page count, render inventory and successful compilation logs. Only the complete TeX/PDF, rebuild/check prerequisites and concise QA records appear in the report release whitelist; exploratory build products and stale reports do not.

23.3 What the release does not claim

The ordinary unbounded proofs use the explicitly named inherited arithmetic and logical interfaces. They are formalizable in ZFC, but no fresh complete transitive Lean or Rocq build, encoded object-language derivation, external human refereeing, consistency proof or global minimality theorem is asserted. All claimed new endpoints have full witness arguments and finite artifact bindings; the comparison-only inputs carry the narrower evidence labels in Section 22. The distinction survives successful replay.

24 Further questions with concrete proof obligations

1. Can the affine horizon use a smaller progression than the selected threshold-four family? Every proposed change must reprove the initial signed bootstrap, exclude exceptional zero or

unit bases, retain the actual q^t request, and construct an odd binomial selector cofinally. A formal coefficient saving without those conditions is not an equation for the same arithmetic relation.

2. Can a different integral basis for the affine frame lower coefficients without weakening the error budget too far? The relevant tests are the full actual-atom ledger, raw classified-index bounds, $|A| > W^W$, the orientation residue sum, and the appropriately strengthened power or joint theorem. Bounding only the nominal requests is not enough before orientation.

3. Can the unweighted joint certificate be composed with a lower-height source or another block partition before monic elimination? The matched unfused control isolates the current orientation saving; further apparent dimension improvements must report degree, collected support and height as well. At zero or mixed-sign bases, the second integrality step cannot be replaced by a direct nonnegative-sum argument.

4. Can exact elimination retain the joint theorem's small degree while avoiding its large expansion? A compact arithmetic DAG and a sparse expanded polynomial measure different resources. Any further deletion needs its unit pivot, acyclic restoration and attained final degree checked on the actual emitted source, not only on a symbolic model.

5. Can a joint search over constructor fusion, additive inlining, scale expansion and graph aliases improve the height-one catalog? The search cost is not final support, and individual changes can increase support. Every candidate still needs the complete triangular pullbacks, occurrence coverage, unit anchoring, integral energy and universal lift including the hub term.

6. Can the newer direct arithmetic fronts improve those height-one networks? Their smaller original dimensions do not imply fewer aliases or occurrences after normalization. Such a transfer needs a new measured graph, not a relabelled inherited source theorem.

7. Can higher quotient centers reduce the positive scalar degree at a reasonable gate cost? The key obligations are an exact division-free integer polynomial evaluator, an all-sign pre-rank center bound, a transport valid at the auxiliary index-three branch, and a strictly positive completeness lift. Bare b^2 is not licensed by the proved UEb^2 inequality.

8. Can these ordinary compositions be given a complete formal-kernel replay? This would require the actual transitive logical and arithmetic source closure and explicit port bindings. Finite coefficient reconstruction, mutation rejection and byte replay are useful prerequisites, not substitutes for such a formalization.

These are bounded research directions with named proof obligations. They are not claims of achievable savings, exhaustive negative results, global lower bounds or mathematical priority.

References

[1] ProveIt, *Pooling exact power certificates for Diophantine consistency*, source 158, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers`.

[2] ProveIt, *Removing factorial authenticity from Diophantine consistency*, source 151, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/151-factorial-free-binomial-crt`.

[3] ProveIt, *Linear CRT radii and semantic radix recovery*, source 157, 25 October 2026. Pinned alongside the preceding source; the copied `MAIN_PROOF.md` supplies the sign bootstrap and every-prime CRT completeness.

[4] ProveIt, *Quadratic projection*, source 159, 25 October 2026. Exact common-Hessian and affine projection algorithms, with an independent integer certificate checker.

[5] ProveIt, *Reusing a semantic Pell anchor over unrestricted integers* and *Recovering nonnegative arithmetic from signed Pell orientation*, source 135, retained in source 158's provenance. Copied as `PROOF_315.md` and `ORIENTATION_FIRST_PROOF.md`.

[6] ProveIt, *Aligning the amplification exponent with a strict beta quotient range*, source 135, and source 158's *Cofinal strict bounds and a linear beta base.*

[7] ProveIt, *Square-shell encoding proof* and the literal `shell_source.py` constructor source; identified by source 158's `SHELL_ENCODING_PROOF.md` supplement and exact source hashes.

[8] ProveIt, Lean sources `KCode222.lean`, `KSeedDirect222.lean`, `KEndpoint.lean`, and `ConstructibleChoice.lean`. Exact origins and inspected copies are listed in `PROVENANCE.json`; no new full Lean build is claimed.

[9] ProveIt, *Pell growth from divisibility*, incoming `ConZFC_Pell_Growth_210_2026-10-25.zip`, including the standalone growth proof, affine radix, ordinary reviews, exact equations and frozen ancestral closure. Retrieved at comparison commit `b42c796d1f5de9912a576bc2fd77b778ae7402bf`.

[10] ProveIt, *Mixed base pooling*, incoming `ConZFC_Mixed_Base_Pooling_2026-10-25.zip`, ordinary two-integrality theorem and exact source package at the same comparison commit.

[11] ProveIt, *Oriented pooling*, consolidated source 176, ordinary raw-classified-index joint theorem, at the same comparison commit. Its own retained source pins are separately preserved.

[12] ProveIt, *Polynomial centers*, consolidated source 178, ordinary signed whole-product bound and exact quotient algebra, at the same comparison commit. The selected scalar package retains the source texts under `scalar/source_centers/`.

[13] *CRT anchors and smaller coefficients for ZFC consistency*, Round22, 26 October 2026. Complete prior release and its incoming repack union; inherited coefficient tuple, arithmetic networks and exact reference charts.

A Every residual of the principal equation

The following is the complete finite definition of the stated polynomial. The source is `signed/principal210/equation.json`. The coordinate namespace is local to this appendix. Its independent coordinates are $x_0, \ldots, x_{209} \in \mathbb{Z}$, and

$$P(x_0, \ldots, x_{209}) = \sum_{i=0}^{149} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$\begin{aligned}
R_0 &= 4 - 5x_3 - 8x_3x_{177} + x_6 \\
&\quad + 8x_{177} \\
R_1 &= -x_1 - x_3x_4 + x_5x_6 \\
R_2 &= -1 + x_0 - x_5x_6 \\
R_3 &= -2x_1x_{205} - x_2x_{17} + x_3 + 2x_{15} \\
R_4 &= -4 - 3x_7 - x_7^2 - 2x_7x_8 \\
&\quad - 4x_8 - x_8^2 + x_{18} \\
R_5 &= x_7 - x_7^2 - 8x_7x_{18} - 4x_7x_{19} \\
&\quad - 16x_{18}^2 - 16x_{18}x_{19} - 4x_{19}^2 + x_{20} \\
R_6 &= 1 + 2x_{20} - 4x_{20}^2 + x_{21} \\
R_7 &= -2 - x_8^2 - 2x_8x_9 - x_9^2 \\
&\quad + x_{22} \\
R_8 &= 1 - x_7^2 - 2x_7x_9 - x_9^2 \\
&\quad + x_{23} \\
R_9 &= -29 - 61x_7 - 35x_7^2 - 96x_7x_9 \\
&\quad - 24x_7x_{18} - 4x_7x_{22} - 24x_7x_{23} + 16x_8 \\
&\quad - 2x_8^2 - 8x_8x_{22} - 96x_9 - 64x_9^2 \\
&\quad - 32x_9x_{18} - 32x_9x_{23} - 28x_{18} - 4x_{18}^2 \\
&\quad - 8x_{18}x_{23} + 8x_{22} - 4x_{22}^2 - 24x_{23} \\
&\quad - 4x_{23}^2 + x_{24} \\
R_{10} &= -4 + 10x_7 - 2x_7^2 - 8x_7x_8 \\
&\quad - 8x_7x_{22} - 4x_{24}^2 + x_{25} - 2x_{42} \\
R_{11} &= -4 - 16x_9^2 - 16x_9x_{22} - 16x_9x_{23} \\
&\quad - 4x_{22}^2 - 8x_{22}x_{23} - 4x_{23}^2 + x_{26} \\
R_{12} &= -7 + 24x_9 - 16x_9^2 - 16x_9x_{18} \\
&\quad - 32x_9x_{23} - 16x_9x_{26} + 10x_{18} - 4x_{18}^2 \\
&\quad - 16x_{18}x_{23} - 8x_{18}x_{26} + 24x_{23} - 16x_{23}^2 \\
&\quad - 16x_{23}x_{26} + 12x_{26} - 4x_{26}^2 + x_{27} \\
R_{13} &= -114 - 127x_7 - 37x_7^2 - 24x_7x_8 \\
&\quad - 112x_7x_9 - 4x_7x_{22} - 24x_7x_{23} - 67x_8 \\
&\quad - 37x_8^2 - 112x_8x_9 - 24x_8x_{22} - 4x_8x_{23} \\
&\quad - 224x_9 - 128x_9^2 - 32x_9x_{22} - 32x_9x_{23} \\
&\quad - 16x_{22} - 4x_{22}^2 - 40x_{23} - 4x_{23}^2 \\
&\quad + x_{28} \\
R_{14} &= -30 - 14x_7 - 2x_7^2 - 24x_7x_8 \\
&\quad - 32x_7x_9 - 8x_7x_{22} - 96x_8 - 72x_8^2 \\
&\quad - 192x_8x_9 - 48x_8x_{22} - 128x_9 - 128x_9^2 \\
&\quad - 64x_9x_{22} - 32x_{22} - 8x_{22}^2 - 4x_{28}^2 \\
&\quad + x_{29} \\
R_{15} &= -74 - 42x_7 + 8x_7x_9 + 16x_7x_{22} \\
&\quad - 84x_8 + 16x_8x_9 + 32x_8x_{22} + 29x_9
\end{aligned}$$

$$
\begin{aligned}
&\quad - 3x_9^2 - 4x_9x_{18} - 8x_9x_{22} + 21x_{18} \\
&\quad - 8x_{18}x_{22} + 40x_{22} - 4x_{22}^2 + x_{30} \\
&\quad - x_{58} \\
R_{16} &= -131 - 184x_7 - 64x_7^2 - 160x_7x_9 \\
&\quad - 64x_7x_{22} - 32x_7x_{23} - 32x_7x_{30} - 226x_9 \\
&\quad - 100x_9^2 - 80x_9x_{22} - 40x_9x_{23} - 40x_9x_{30} \\
&\quad - 88x_{22} - 16x_{22}^2 - 16x_{22}x_{23} - 16x_{22}x_{30} \\
&\quad - 46x_{23} - 4x_{23}^2 - 8x_{23}x_{30} - 44x_{30} \\
&\quad - 4x_{30}^2 + x_{31} \\
R_{17} &= -4x_{18}^2 + x_{19} \\
R_{18} &= x_7 - x_7^2 + 8x_7x_{18} - 4x_7x_{19} \\
&\quad - 16x_{18}^2 + 16x_{18}x_{19} - 4x_{19}^2 + x_{32} \\
R_{19} &= 1 + 2x_{32} - 4x_{32}^2 + x_{33} \\
R_{20} &= -x_8^2 - 4x_8x_{33} - 4x_{33}^2 + x_{34} \\
R_{21} &= -44 - 52x_7 - 16x_7^2 - 64x_7x_8 \\
&\quad + 32x_7x_{18} - 104x_8 - 64x_8^2 + 64x_8x_{18} \\
&\quad + 50x_{18} - 16x_{18}^2 + x_{35} \\
R_{22} &= 8x_8 - 62x_8^2 - 92x_8x_9 - 32x_8x_{22} \\
&\quad - 32x_8x_{35} + 6x_9 - 34x_9^2 - 24x_9x_{22} \\
&\quad - 24x_9x_{35} - 4x_{22}^2 - 8x_{22}x_{35} + 4x_{35} \\
&\quad - 4x_{35}^2 + x_{36} \\
R_{23} &= -90 - 152x_7 - 64x_7^2 - 96x_7x_9 \\
&\quad - 32x_7x_{23} - 32x_7x_{36} - 114x_9 - 36x_9^2 \\
&\quad - 24x_9x_{23} - 24x_9x_{36} - 38x_{23} - 4x_{23}^2 \\
&\quad - 8x_{23}x_{36} - 36x_{36} - 4x_{36}^2 + x_{37} \\
R_{24} &= -x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{38} \\
R_{25} &= -7 + 8x_7 - 3x_7^2 - 8x_7x_8 \\
&\quad - 8x_7x_{22} - 8x_7x_{23} - 4x_7x_{38} - 8x_8x_{23} \\
&\quad - 2x_9^2 - 8x_9x_{38} - 8x_{22}x_{23} + 6x_{23} \\
&\quad - 4x_{23}^2 - 4x_{38}^2 + x_{39} - x_{42} \\
R_{26} &= -16x_7^2 - 36x_7x_9 - 32x_7x_{23} - 40x_7x_{38} \\
&\quad - 16x_7x_{39} - 4x_9 - 22x_9^2 - 32x_9x_{23} \\
&\quad - 48x_9x_{38} - 16x_9x_{39} - 2x_{23} - 16x_{23}^2 \\
&\quad - 32x_{23}x_{38} - 16x_{23}x_{39} - 4x_{38} - 24x_{38}^2 \\
&\quad - 16x_{38}x_{39} - 4x_{39}^2 + x_{40} \\
R_{27} &= x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{26} \\
&\quad - 16x_{23}^2 - 16x_{23}x_{26} - 4x_{26}^2 + x_{41} \\
R_{28} &= -1 + 8x_8 - 4x_8^2 - 8x_8x_{22} \\
&\quad + 8x_{22} - 4x_{22}^2 + x_{42} \\
R_{29} &= -1 - 16x_8^2 - 8x_8x_9 - 32x_8x_{22} \\
&\quad - 16x_8x_{42} - x_9^2 - 8x_9x_{22} - 4x_9x_{42} \\
&\quad - 16x_{22}^2 - 16x_{22}x_{42} - 4x_{42}^2 + x_{43} \\
R_{30} &= 3 + x_8 - 2x_8x_9 - 4x_8x_{43} \\
&\quad - 3x_9^2 - 8x_9x_{43} - x_{22} - 4x_{43}^2 \\
&\quad + x_{44} \\
R_{31} &= -1 - 4x_{44} - 4x_{44}^2 + x_{45} \\
R_{32} &= -1 - x_9^2 - 8x_9x_{44} - 4x_9x_{45} \\
&\quad - 16x_{44}^2 - 16x_{44}x_{45} - 4x_{45}^2 + x_{46} \\
R_{33} &= 2 + x_8 - x_8^2 - 4x_8x_9 \\
&\quad - 4x_8x_{46} - 4x_9^2 - 8x_9x_{46} - 4x_{46}^2 \\
&\quad + x_{47} \\
R_{34} &= 1 - 4x_{41}^2 - 8x_{41}x_{47} - 4x_{47}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ x_{48} \\
R_{35} &= 1 + x_7 - x_7^2 - 8x_7x_{41} \\
&- 4x_7x_{48} - 16x_{41}^2 - 16x_{41}x_{48} - 4x_{48}^2 \\
&+ x_{49} \\
R_{36} &= -1 - 4x_{49} - 4x_{49}^2 + x_{50} \\
R_{37} &= -1 - x_9^2 - 8x_9x_{49} - 4x_9x_{50} \\
&- 16x_{49}^2 - 16x_{49}x_{50} - 4x_{50}^2 + x_{51} \\
R_{38} &= 2 + x_7 - x_7^2 - 4x_7x_9 \\
&- 4x_7x_{51} - 4x_9^2 - 8x_9x_{51} - 4x_{51}^2 \\
&+ x_{52} \\
R_{39} &= -10 - 12x_{52} - 4x_{52}^2 + x_{53} \\
R_{40} &= -19 + 4x_8 - 4x_8^2 - 8x_8x_{22} \\
&- 16x_8x_{52} - 8x_8x_{53} + 4x_{22} - 4x_{22}^2 \\
&- 16x_{22}x_{52} - 8x_{22}x_{53} - 16x_{52} - 24x_{52}^2 \\
&- 16x_{52}x_{53} + 6x_{53} - 4x_{53}^2 + x_{54} \\
R_{41} &= -10 + x_8 - 25x_8^2 - 40x_8x_{22} \\
&- 20x_8x_{54} - 16x_{22}^2 - 16x_{22}x_{54} - 12x_{47} \\
&- 4x_{47}^2 - 4x_{54}^2 + x_{55} \\
R_{42} &= 21 + 24x_{47} - 8x_{47}^2 - 16x_{47}x_{55} \\
&- 2x_{55} - 4x_{55}^2 + x_{56} \\
R_{43} &= -1 + 4x_7 + 8x_8 + x_9 \\
&- x_9^2 - 4x_9x_{56} - 2x_{18} - 4x_{56}^2 \\
&+ x_{57} + 12x_{58} - 4x_{58}^2 \\
R_{44} &= -22 - 38x_7 - 16x_7^2 - 64x_7x_8 \\
&+ 8x_7x_9 + 16x_7x_{18} - 76x_8 - 64x_8^2 \\
&+ 16x_8x_9 + 32x_8x_{18} + 11x_9 - x_9^2 \\
&- 4x_9x_{18} + 19x_{18} - 4x_{18}^2 + x_{58} \\
R_{45} &= -4 + 8x_7 + 16x_8 - 4x_{18} \\
&- 4x_{57}^2 + 24x_{58} - 8x_{58}^2 + x_{59} \\
R_{46} &= -1 - 4x_{59} - 4x_{59}^2 + x_{60} \\
R_{47} &= -x_1x_{12} + x_{61} \\
R_{48} &= x_{11} - x_{12} - 2x_{21}x_{61} - x_{62} \\
R_{49} &= x_{11} - x_{12} - 2x_{25}x_{61} - x_{63} \\
R_{50} &= x_{11} - x_{12} - 2x_{27}x_{61} - 2x_{61} \\
&- x_{64} \\
R_{51} &= x_{11} - x_{12} - 2x_{29}x_{61} - x_{65} \\
R_{52} &= x_{11} - x_{12} - 2x_{31}x_{61} + 2x_{61} \\
&- x_{66} \\
R_{53} &= x_{11} - x_{12} - 4x_{33}x_{61} - 2x_{34}x_{61} \\
&- x_{67} \\
R_{54} &= x_{11} - x_{12} - 2x_{37}x_{61} - x_{68} \\
R_{55} &= 2x_{27}x_{61} - 2x_{40}x_{61} - 2x_{61} + x_{64} \\
&- x_{69} \\
R_{56} &= -x_7x_8 + x_7x_9 + x_8x_9 - x_9^2 \\
&+ x_{70} \\
R_{57} &= 1 + 2x_{10} + x_{10}^2 - x_{70}^2 \\
&+ x_{71} \\
R_{58} &= -x_{11}^2 + 2x_{11}x_{12} + 2x_{11}x_{73} - x_{12}^2 \\
&- 2x_{12}x_{73} - x_{71}^2 + x_{72} - x_{73}^2 \\
R_{59} &= -4x_{59}x_{61} - 2x_{60}x_{61} + x_{73} \\
R_{60} &= -x_8x_{61} - x_{12} + x_{74}x_{75} + x_{76} \\
R_{61} &= -1 + 8x_1 + 4x_1x_7 + 8x_1x_8 \\
&- 2x_1x_{18} + x_{74}
\end{aligned}
$$

$$
\begin{aligned}
R_{62} &= -x_7x_{200} - x_{11} + x_{75} \\
R_{63} &= x_8x_{200} - 2x_{18}x_{61} + 2x_{31}x_{61} + x_{66} \\
&\quad + x_{77} \\
R_{64} &= -x_{11}x_{62} + x_{12}x_{62} + x_{78} \\
R_{65} &= -x_{63}x_{78} + x_{79} \\
R_{66} &= -x_{64}x_{79} + x_{80} \\
R_{67} &= -x_{65}x_{80} + x_{81} \\
R_{68} &= -x_{66}x_{81} + x_{82} \\
R_{69} &= -x_{67}x_{82} + x_{83} \\
R_{70} &= -x_{68}x_{83} + x_{84} \\
R_{71} &= -x_{69}x_{84} + x_{85} \\
R_{72} &= -x_{76}x_{85} + x_{86} \\
R_{73} &= -x_{77}x_{86} + x_{87} \\
R_{74} &= -x_{87}^2 + x_{88} \\
R_{75} &= -x_3 - 2x_3x_{15} + 2x_{15} + x_{89} \\
R_{76} &= -x_{13}x_{89} + x_{90} \\
R_{77} &= -2x_0^2 + 2x_0x_{11} + 2x_0x_{12} + 2x_0x_{90} \\
&\quad + 2x_0x_{91} - x_{11}^2 - 2x_{11}x_{90} - x_{12}^2 \\
&\quad - 2x_{12}x_{91} + x_{16}x_{92} - x_{72}x_{88} - x_{90}^2 \\
&\quad - x_{91}^2 \\
R_{78} &= -x_3x_{14} + x_{14} - x_{14}x_{89} + x_{91} \\
R_{79} &= -x_{93}x_{96} + x_{96} + x_{98} \\
R_{80} &= -x_2x_{119} + x_{94}x_{192} - x_{97} - x_{98}x_{119} \\
R_{81} &= x_{99}x_{192} - x_{100}x_{122} - x_{101} \\
R_{82} &= x_{102}x_{192} - x_{103}x_{122} - x_{104} \\
R_{83} &= x_{105}x_{192} - x_{106}x_{122} - x_{107} \\
R_{84} &= x_{108}x_{192} - x_{109}x_{122} - x_{110} \\
R_{85} &= x_{111}x_{193} - x_{113}x_{122} - x_{114} \\
R_{86} &= x_{115}x_{193} - x_{116}x_{122} - x_{117} \\
R_{87} &= -x_{16}x_{119} + x_{122} \\
R_{88} &= x_{118} - x_{120}x_{122} - x_{121} \\
R_{89} &= -x_{122}x_{124} + x_{123} - x_{125} \\
R_{90} &= -x_{122}x_{127} + x_{126}x_{192} - x_{128} \\
R_{91} &= x_{129} - 2x_{129}x_{131} + x_{132} \\
R_{92} &= -x_{129}x_{132} - x_{132} + x_{133} \\
R_{93} &= -1 + x_{130}^2 + x_{132}^2 - x_{133}^2 \\
R_{94} &= 1 - x_{130}x_{135} - x_{134} + x_{136} \\
R_{95} &= -1 + 2x_{134} - x_{134}^2 + 2x_{134}x_{136} \\
&\quad + x_{135}^2 \\
R_{96} &= 1 + x_2^2 - 2x_2x_{130} + x_{137} \\
R_{97} &= -x_2x_{169} + x_{16} + x_{137}x_{138} - x_{168} \\
R_{98} &= -5 - 8x_1 - 4x_1^2 - 16x_1x_{15} \\
&\quad + 4x_1x_{130} - 16x_{15} - 16x_{15}^2 + 8x_{15}x_{130} \\
&\quad + 4x_{130} - x_{139} \\
R_{99} &= -1 - 2x_1x_{169} - 4x_{15}x_{169} + x_{93} \\
&\quad + x_{139}x_{140} - x_{168} - 2x_{169} \\
R_{100} &= -x_{130}x_{142} - x_{141} + x_{143} \\
R_{101} &= -1 - x_{141}^2 + 2x_{141}x_{143} + x_{142}^2 \\
R_{102} &= -x_{130}x_{145} - x_{144} + x_{146} \\
R_{103} &= -1 - x_{144}^2 + 2x_{144}x_{146} + x_{145}^2 \\
R_{104} &= -x_{130}x_{148} - x_{147} + x_{149} \\
R_{105} &= -1 - x_{147}^2 + 2x_{147}x_{149} + x_{148}^2 \\
R_{106} &= -x_{130}x_{151} - x_{150} + x_{152} \\
R_{107} &= -1 - x_{150}^2 + 2x_{150}x_{152} + x_{151}^2 \\
R_{108} &= -x_{130}x_{154} - x_{153} + x_{155}
\end{aligned}
$$

$$
\begin{aligned}
R_{109} &= -1 - x_{153}^2 + 2x_{153}x_{155} + x_{154}^2 \\
R_{110} &= -x_{130}x_{157} - x_{156} + x_{158} \\
R_{111} &= -1 - x_{156}^2 + 2x_{156}x_{158} + x_{157}^2 \\
R_{112} &= 1 - x_{130}x_{160} - x_{159} + x_{161} \\
R_{113} &= -1 + 2x_{159} - x_{159}^2 + 2x_{159}x_{161} \\
&\quad + x_{160}^2 \\
R_{114} &= -x_{130}x_{163} - x_{162} + x_{164} \\
R_{115} &= -1 - x_{162}^2 + 2x_{162}x_{164} + x_{163}^2 \\
R_{116} &= -x_{130}x_{166} - x_{165} + x_{167} \\
R_{117} &= -1 - x_{165}^2 + 2x_{165}x_{167} + x_{166}^2 \\
R_{118} &= 1 - x_{130}x_{169} - x_{168} + x_{170} \\
R_{119} &= -1 + 2x_{168} - x_{168}^2 + 2x_{168}x_{170} \\
&\quad + x_{169}^2 \\
R_{120} &= -x_{130}x_{172} - x_{171} + x_{173} \\
R_{121} &= -1 - x_{171}^2 + 2x_{171}x_{173} + x_{172}^2 \\
R_{122} &= -x_{130}x_{175} - x_{174} + x_{176} \\
R_{123} &= -1 - x_{174}^2 + 2x_{174}x_{176} + x_{175}^2 \\
R_{124} &= -22 - x_0^2 - x_1^2 - x_2^2 \\
&\quad - x_3^2 - x_7^2 - x_8^2 - x_9^2 \\
&\quad - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 \\
&\quad - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{93}^2 \\
&\quad - x_{94}^2 - x_{95}^2 - x_{99}^2 - x_{102}^2 \\
&\quad - x_{105}^2 - x_{108}^2 - x_{111}^2 - x_{112}^2 \\
&\quad - x_{115}^2 - x_{118}^2 - x_{119}^2 - x_{123}^2 \\
&\quad - x_{126}^2 + x_{129} - x_{177}^2 \\
R_{125} &= -x_{130}x_{179} + x_{188} \\
R_{126} &= -1 + x_{178}^2 + x_{179}^2 - x_{188}^2 \\
R_{127} &= -x_{181}x_{182} + x_{189} \\
R_{128} &= -1 + x_{180}^2 + x_{181}^2 - x_{189}^2 \\
R_{129} &= -x_{175}^2 + x_{190} \\
R_{130} &= x_{179} - 2x_{184}x_{190} - x_{190} \\
R_{131} &= -1 - 4x_{179} - 8x_{179}x_{183} + x_{182} \\
R_{132} &= -x_{130} - x_{178}x_{185} + x_{182} \\
R_{133} &= -x_{176} - x_{178}x_{186} + x_{180} \\
R_{134} &= -2x_{15} - 4x_{175}x_{187} + x_{181} \\
R_{135} &= 25 + x_0^2 - 2x_0x_{169} + x_1^2 \\
&\quad + 2x_1x_{135} - 2x_1x_{142} + x_3^2 - 2x_3x_{160} \\
&\quad + x_7^2 - 2x_7x_{145} + x_8^2 - 2x_8x_{148} \\
&\quad + x_9^2 - 2x_9x_{151} + x_{10}^2 - 2x_{10}x_{154} \\
&\quad + x_{11}^2 - 2x_{11}x_{157} + x_{12}^2 - 2x_{12}x_{163} \\
&\quad + x_{13}^2 - 2x_{13}x_{166} + x_{14}^2 - 2x_{14}x_{172} \\
&\quad - 4x_{15}x_{175} - x_{130}x_{191} - 8x_{135} + x_{135}^2 \\
&\quad - 2x_{135}x_{142} - 16x_{135}x_{177} + x_{136}^2 - 2x_{143} \\
&\quad + x_{143}^2 - 2x_{146} + x_{146}^2 - 2x_{149} \\
&\quad + x_{149}^2 - 2x_{152} + x_{152}^2 - 2x_{155} \\
&\quad + x_{155}^2 - 2x_{158} + x_{158}^2 + x_{161}^2 \\
&\quad - 2x_{164} + x_{164}^2 - 2x_{167} + x_{167}^2 \\
&\quad + x_{170}^2 - 2x_{173} + x_{173}^2 + 2x_{175}^2 \\
&\quad - 2x_{176} + x_{176}^2 + 64x_{177} + 64x_{177}^2 \\
&\quad + x_{191} - x_{194}^2 - x_{195}^2 - x_{196} \\
&\quad - x_{196}^2 \\
R_{136} &= -x_{95}x_{192} + x_{119} \\
R_{137} &= -x_{112}x_{193} + x_{119}
\end{aligned}
$$

$$
\begin{aligned}
R_{138} &= -1 - 4x_{15}^2 - x_{135}^2 - x_{142}^2 \\
&\quad - x_{145}^2 - x_{148}^2 - x_{151}^2 - x_{154}^2 \\
&\quad - x_{157}^2 - x_{160}^2 - x_{163}^2 - x_{166}^2 \\
&\quad - x_{169}^2 - x_{172}^2 + x_{175}^2 - x_{194}^2 \\
&\quad - x_{195}^2 - x_{196} - x_{196}^2 \\
R_{139} &= -3 - 4x_{97}^2 - 4x_{101}^2 - 4x_{104}^2 \\
&\quad - 4x_{107}^2 - 4x_{110}^2 - 4x_{114}^2 - 4x_{117}^2 \\
&\quad + x_{119}^2 - 4x_{121}^2 - 4x_{125}^2 - 4x_{128}^2 \\
&\quad - x_{197}^2 - x_{198}^2 + 3x_{199} - x_{199}^2 \\
R_{140} &= -x_1 x_{11} + x_{200} \\
R_{141} &= x_{93}x_{202} - 2x_{94}x_{142} - 2x_{99}x_{145} - 2x_{102}x_{148} \\
&\quad - 2x_{105}x_{151} - 2x_{108}x_{154} - 2x_{111}x_{157} - 2x_{115}x_{163} \\
&\quad - 2x_{118}x_{166} - 2x_{123}x_{172} - 2x_{126}x_{175} - 2x_{130}x_{202} \\
&\quad + 2x_{142}x_{143} + 2x_{145}x_{146} + 2x_{148}x_{149} + 2x_{151}x_{152} \\
&\quad + 2x_{154}x_{155} + 2x_{157}x_{158} + 2x_{163}x_{164} + 2x_{166}x_{167} \\
&\quad + 2x_{172}x_{173} + 2x_{175}x_{176} + x_{201} \\
R_{142} &= -10 - x_{93}x_{201} + x_{94}^2 - 2x_{94}x_{141} \\
&\quad + x_{99}^2 - 2x_{99}x_{144} + x_{102}^2 - 2x_{102}x_{147} \\
&\quad + x_{105}^2 - 2x_{105}x_{150} + x_{108}^2 - 2x_{108}x_{153} \\
&\quad + x_{111}^2 - 2x_{111}x_{156} + x_{115}^2 - 2x_{115}x_{162} \\
&\quad + x_{118}^2 - 2x_{118}x_{165} + x_{123}^2 - 2x_{123}x_{171} \\
&\quad + x_{126}^2 - 2x_{126}x_{174} + 2x_{141}x_{143} + 2x_{144}x_{146} \\
&\quad + 2x_{147}x_{149} + 2x_{150}x_{152} + 2x_{153}x_{155} + 2x_{156}x_{158} \\
&\quad + 2x_{162}x_{164} + 2x_{165}x_{167} + 2x_{171}x_{173} + 2x_{174}x_{176} \\
&\quad + x_{202} \\
R_{143} &= x_{93}x_{204} - 2x_{95}x_{135} - 2x_{112}x_{160} - 2x_{119}x_{169} \\
&\quad - 2x_{130}x_{204} + 2x_{135} + 2x_{135}x_{136} + 2x_{160} \\
&\quad + 2x_{160}x_{161} + 2x_{169} + 2x_{169}x_{170} + x_{203} \\
&\quad - x_{204} \\
R_{144} &= -2 - x_{93}x_{203} + x_{95}^2 - 2x_{95}x_{134} \\
&\quad + x_{112}^2 - 2x_{112}x_{159} + x_{119}^2 - 2x_{119}x_{168} \\
&\quad + 2x_{134} + 2x_{134}x_{136} + 2x_{159} + 2x_{159}x_{161} \\
&\quad + x_{168}^2 - x_{169}^2 + x_{203} + x_{204} \\
R_{145} &= -1 - x_1 - 2x_1 x_{205} + x_3 \\
R_{146} &= -4x_1 - 8x_1 x_{177} + x_{207} \\
R_{147} &= -1 - 8x_1 x_{207} + x_{206}^2 - 16x_{207}^2 \\
R_{148} &= x_1 - 4096x_{208}x_{209} - 2048x_{209} \\
R_{149} &= -1024 - 2048x_{177} + x_{209}
\end{aligned}
$$

B The exact conservative source for reciprocal elimination

The following is the complete finite definition of the stated polynomial. Its independent coordinates are $x_0, \ldots, x_{208} \in \mathbb{Z}$, and The source is `signed/conservative209/equation.json`. The coordinate namespace is local to this appendix.

$$P(x_0, \ldots, x_{208}) = \sum_{i=0}^{148} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$\begin{aligned}
R_0 &= 16 - 17x_3 - 32x_3x_{177} + x_6 \\
&\quad + 32x_{177} \\
R_1 &= -x_1 - x_3x_4 + x_5x_6 \\
R_2 &= -1 + x_0 - x_5x_6 \\
R_3 &= -2x_1x_{205} - x_2x_{17} + x_3 + 2x_{15} \\
R_4 &= -4 - 3x_7 - x_7^2 - 2x_7x_8 \\
&\quad - 4x_8 - x_8^2 + x_{18} \\
R_5 &= x_7 - x_7^2 - 8x_7x_{18} - 4x_7x_{19} \\
&\quad - 16x_{18}^2 - 16x_{18}x_{19} - 4x_{19}^2 + x_{20} \\
R_6 &= 1 + 2x_{20} - 4x_{20}^2 + x_{21} \\
R_7 &= -2 - x_8^2 - 2x_8x_9 - x_9^2 \\
&\quad + x_{22} \\
R_8 &= 1 - x_7^2 - 2x_7x_9 - x_9^2 \\
&\quad + x_{23} \\
R_9 &= -29 - 61x_7 - 35x_7^2 - 96x_7x_9 \\
&\quad - 24x_7x_{18} - 4x_7x_{22} - 24x_7x_{23} + 16x_8 \\
&\quad - 2x_8^2 - 8x_8x_{22} - 96x_9 - 64x_9^2 \\
&\quad - 32x_9x_{18} - 32x_9x_{23} - 28x_{18} - 4x_{18}^2 \\
&\quad - 8x_{18}x_{23} + 8x_{22} - 4x_{22}^2 - 24x_{23} \\
&\quad - 4x_{23}^2 + x_{24} \\
R_{10} &= -4 + 10x_7 - 2x_7^2 - 8x_7x_8 \\
&\quad - 8x_7x_{22} - 4x_{24}^2 + x_{25} - 2x_{42} \\
R_{11} &= -4 - 16x_9^2 - 16x_9x_{22} - 16x_9x_{23} \\
&\quad - 4x_{22}^2 - 8x_{22}x_{23} - 4x_{23}^2 + x_{26} \\
R_{12} &= -7 + 24x_9 - 16x_9^2 - 16x_9x_{18} \\
&\quad - 32x_9x_{23} - 16x_9x_{26} + 10x_{18} - 4x_{18}^2 \\
&\quad - 16x_{18}x_{23} - 8x_{18}x_{26} + 24x_{23} - 16x_{23}^2 \\
&\quad - 16x_{23}x_{26} + 12x_{26} - 4x_{26}^2 + x_{27} \\
R_{13} &= -114 - 127x_7 - 37x_7^2 - 24x_7x_8 \\
&\quad - 112x_7x_9 - 4x_7x_{22} - 24x_7x_{23} - 67x_8 \\
&\quad - 37x_8^2 - 112x_8x_9 - 24x_8x_{22} - 4x_8x_{23} \\
&\quad - 224x_9 - 128x_9^2 - 32x_9x_{22} - 32x_9x_{23} \\
&\quad - 16x_{22} - 4x_{22}^2 - 40x_{23} - 4x_{23}^2 \\
&\quad + x_{28} \\
R_{14} &= -30 - 14x_7 - 2x_7^2 - 24x_7x_8 \\
&\quad - 32x_7x_9 - 8x_7x_{22} - 96x_8 - 72x_8^2 \\
&\quad - 192x_8x_9 - 48x_8x_{22} - 128x_9 - 128x_9^2 \\
&\quad - 64x_9x_{22} - 32x_{22} - 8x_{22}^2 - 4x_{28}^2 \\
&\quad + x_{29} \\
R_{15} &= -74 - 42x_7 + 8x_7x_9 + 16x_7x_{22} \\
&\quad - 84x_8 + 16x_8x_9 + 32x_8x_{22} + 29x_9
\end{aligned}$$

$$
\begin{aligned}
&\quad - 3x_9^2 - 4x_9x_{18} - 8x_9x_{22} + 21x_{18} \\
&\quad - 8x_{18}x_{22} + 40x_{22} - 4x_{22}^2 + x_{30} \\
&\quad - x_{58} \\
R_{16} &= -131 - 184x_7 - 64x_7^2 - 160x_7x_9 \\
&\quad - 64x_7x_{22} - 32x_7x_{23} - 32x_7x_{30} - 226x_9 \\
&\quad - 100x_9^2 - 80x_9x_{22} - 40x_9x_{23} - 40x_9x_{30} \\
&\quad - 88x_{22} - 16x_{22}^2 - 16x_{22}x_{23} - 16x_{22}x_{30} \\
&\quad - 46x_{23} - 4x_{23}^2 - 8x_{23}x_{30} - 44x_{30} \\
&\quad - 4x_{30}^2 + x_{31} \\
R_{17} &= -4x_{18}^2 + x_{19} \\
R_{18} &= x_7 - x_7^2 + 8x_7x_{18} - 4x_7x_{19} \\
&\quad - 16x_{18}^2 + 16x_{18}x_{19} - 4x_{19}^2 + x_{32} \\
R_{19} &= 1 + 2x_{32} - 4x_{32}^2 + x_{33} \\
R_{20} &= -x_8^2 - 4x_8x_{33} - 4x_{33}^2 + x_{34} \\
R_{21} &= -44 - 52x_7 - 16x_7^2 - 64x_7x_8 \\
&\quad + 32x_7x_{18} - 104x_8 - 64x_8^2 + 64x_8x_{18} \\
&\quad + 50x_{18} - 16x_{18}^2 + x_{35} \\
R_{22} &= 8x_8 - 62x_8^2 - 92x_8x_9 - 32x_8x_{22} \\
&\quad - 32x_8x_{35} + 6x_9 - 34x_9^2 - 24x_9x_{22} \\
&\quad - 24x_9x_{35} - 4x_{22}^2 - 8x_{22}x_{35} + 4x_{35} \\
&\quad - 4x_{35}^2 + x_{36} \\
R_{23} &= -90 - 152x_7 - 64x_7^2 - 96x_7x_9 \\
&\quad - 32x_7x_{23} - 32x_7x_{36} - 114x_9 - 36x_9^2 \\
&\quad - 24x_9x_{23} - 24x_9x_{36} - 38x_{23} - 4x_{23}^2 \\
&\quad - 8x_{23}x_{36} - 36x_{36} - 4x_{36}^2 + x_{37} \\
R_{24} &= -x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{38} \\
R_{25} &= -7 + 8x_7 - 3x_7^2 - 8x_7x_8 \\
&\quad - 8x_7x_{22} - 8x_7x_{23} - 4x_7x_{38} - 8x_8x_{23} \\
&\quad - 2x_9^2 - 8x_9x_{38} - 8x_{22}x_{23} + 6x_{23} \\
&\quad - 4x_{23}^2 - 4x_{38}^2 + x_{39} - x_{42} \\
R_{26} &= -16x_7^2 - 36x_7x_9 - 32x_7x_{23} - 40x_7x_{38} \\
&\quad - 16x_7x_{39} - 4x_9 - 22x_9^2 - 32x_9x_{23} \\
&\quad - 48x_9x_{38} - 16x_9x_{39} - 2x_{23} - 16x_{23}^2 \\
&\quad - 32x_{23}x_{38} - 16x_{23}x_{39} - 4x_{38} - 24x_{38}^2 \\
&\quad - 16x_{38}x_{39} - 4x_{39}^2 + x_{40} \\
R_{27} &= x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{26} \\
&\quad - 16x_{23}^2 - 16x_{23}x_{26} - 4x_{26}^2 + x_{41} \\
R_{28} &= -1 + 8x_8 - 4x_8^2 - 8x_8x_{22} \\
&\quad + 8x_{22} - 4x_{22}^2 + x_{42} \\
R_{29} &= -1 - 16x_8^2 - 8x_8x_9 - 32x_8x_{22} \\
&\quad - 16x_8x_{42} - x_9^2 - 8x_9x_{22} - 4x_9x_{42} \\
&\quad - 16x_{22}^2 - 16x_{22}x_{42} - 4x_{42}^2 + x_{43} \\
R_{30} &= 3 + x_8 - 2x_8x_9 - 4x_8x_{43} \\
&\quad - 3x_9^2 - 8x_9x_{43} - x_{22} - 4x_{43}^2 \\
&\quad + x_{44} \\
R_{31} &= -1 - 4x_{44} - 4x_{44}^2 + x_{45} \\
R_{32} &= -1 - x_9^2 - 8x_9x_{44} - 4x_9x_{45} \\
&\quad - 16x_{44}^2 - 16x_{44}x_{45} - 4x_{45}^2 + x_{46} \\
R_{33} &= 2 + x_8 - x_8^2 - 4x_8x_9 \\
&\quad - 4x_8x_{46} - 4x_9^2 - 8x_9x_{46} - 4x_{46}^2 \\
&\quad + x_{47} \\
R_{34} &= 1 - 4x_{41}^2 - 8x_{41}x_{47} - 4x_{47}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ x_{48} \\
R_{35} &= 1 + x_7 - x_7^2 - 8x_7x_{41} \\
&- 4x_7x_{48} - 16x_{41}^2 - 16x_{41}x_{48} - 4x_{48}^2 \\
&+ x_{49} \\
R_{36} &= -1 - 4x_{49} - 4x_{49}^2 + x_{50} \\
R_{37} &= -1 - x_9^2 - 8x_9x_{49} - 4x_9x_{50} \\
&- 16x_{49}^2 - 16x_{49}x_{50} - 4x_{50}^2 + x_{51} \\
R_{38} &= 2 + x_7 - x_7^2 - 4x_7x_9 \\
&- 4x_7x_{51} - 4x_9^2 - 8x_9x_{51} - 4x_{51}^2 \\
&+ x_{52} \\
R_{39} &= -10 - 12x_{52} - 4x_{52}^2 + x_{53} \\
R_{40} &= -19 + 4x_8 - 4x_8^2 - 8x_8x_{22} \\
&- 16x_8x_{52} - 8x_8x_{53} + 4x_{22} - 4x_{22}^2 \\
&- 16x_{22}x_{52} - 8x_{22}x_{53} - 16x_{52} - 24x_{52}^2 \\
&- 16x_{52}x_{53} + 6x_{53} - 4x_{53}^2 + x_{54} \\
R_{41} &= -10 + x_8 - 25x_8^2 - 40x_8x_{22} \\
&- 20x_8x_{54} - 16x_{22}^2 - 16x_{22}x_{54} - 12x_{47} \\
&- 4x_{47}^2 - 4x_{54}^2 + x_{55} \\
R_{42} &= 21 + 24x_{47} - 8x_{47}^2 - 16x_{47}x_{55} \\
&- 2x_{55} - 4x_{55}^2 + x_{56} \\
R_{43} &= -1 + 4x_7 + 8x_8 + x_9 \\
&- x_9^2 - 4x_9x_{56} - 2x_{18} - 4x_{56}^2 \\
&+ x_{57} + 12x_{58} - 4x_{58}^2 \\
R_{44} &= -22 - 38x_7 - 16x_7^2 - 64x_7x_8 \\
&+ 8x_7x_9 + 16x_7x_{18} - 76x_8 - 64x_8^2 \\
&+ 16x_8x_9 + 32x_8x_{18} + 11x_9 - x_9^2 \\
&- 4x_9x_{18} + 19x_{18} - 4x_{18}^2 + x_{58} \\
R_{45} &= -4 + 8x_7 + 16x_8 - 4x_{18} \\
&- 4x_{57}^2 + 24x_{58} - 8x_{58}^2 + x_{59} \\
R_{46} &= -1 - 4x_{59} - 4x_{59}^2 + x_{60} \\
R_{47} &= -x_1x_{12} + x_{61} \\
R_{48} &= x_{11} - x_{12} - 2x_{21}x_{61} - x_{62} \\
R_{49} &= x_{11} - x_{12} - 2x_{25}x_{61} - x_{63} \\
R_{50} &= x_{11} - x_{12} - 2x_{27}x_{61} - 2x_{61} \\
&- x_{64} \\
R_{51} &= x_{11} - x_{12} - 2x_{29}x_{61} - x_{65} \\
R_{52} &= x_{11} - x_{12} - 2x_{31}x_{61} + 2x_{61} \\
&- x_{66} \\
R_{53} &= x_{11} - x_{12} - 4x_{33}x_{61} - 2x_{34}x_{61} \\
&- x_{67} \\
R_{54} &= x_{11} - x_{12} - 2x_{37}x_{61} - x_{68} \\
R_{55} &= 2x_{27}x_{61} - 2x_{40}x_{61} - 2x_{61} + x_{64} \\
&- x_{69} \\
R_{56} &= -x_7x_8 + x_7x_9 + x_8x_9 - x_9^2 \\
&+ x_{70} \\
R_{57} &= 1 + 2x_{10} + x_{10}^2 - x_{70}^2 \\
&+ x_{71} \\
R_{58} &= -x_{11}^2 + 2x_{11}x_{12} + 2x_{11}x_{73} - x_{12}^2 \\
&- 2x_{12}x_{73} - x_{71}^2 + x_{72} - x_{73}^2 \\
R_{59} &= -4x_{59}x_{61} - 2x_{60}x_{61} + x_{73} \\
R_{60} &= -x_8x_{61} - x_{12} + x_{74}x_{75} + x_{76} \\
R_{61} &= -1 + 8x_1 + 4x_1x_7 + 8x_1x_8 \\
&- 2x_1x_{18} + x_{74}
\end{aligned}
$$

$$
\begin{aligned}
R_{62} &= -x_7x_{200} - x_{11} + x_{75} \\
R_{63} &= x_8x_{200} - 2x_{18}x_{61} + 2x_{31}x_{61} + x_{66} \\
&\quad + x_{77} \\
R_{64} &= -x_{11}x_{62} + x_{12}x_{62} + x_{78} \\
R_{65} &= -x_{63}x_{78} + x_{79} \\
R_{66} &= -x_{64}x_{79} + x_{80} \\
R_{67} &= -x_{65}x_{80} + x_{81} \\
R_{68} &= -x_{66}x_{81} + x_{82} \\
R_{69} &= -x_{67}x_{82} + x_{83} \\
R_{70} &= -x_{68}x_{83} + x_{84} \\
R_{71} &= -x_{69}x_{84} + x_{85} \\
R_{72} &= -x_{76}x_{85} + x_{86} \\
R_{73} &= -x_{77}x_{86} + x_{87} \\
R_{74} &= -x_{87}^2 + x_{88} \\
R_{75} &= -x_3 - 2x_3x_{15} + 2x_{15} + x_{89} \\
R_{76} &= -x_{13}x_{89} + x_{90} \\
R_{77} &= -2x_0^2 + 2x_0x_{11} + 2x_0x_{12} + 2x_0x_{90} \\
&\quad + 2x_0x_{91} - x_{11}^2 - 2x_{11}x_{90} - x_{12}^2 \\
&\quad - 2x_{12}x_{91} + x_{16}x_{92} - x_{72}x_{88} - x_{90}^2 \\
&\quad - x_{91}^2 \\
R_{78} &= -x_3x_{14} + x_{14} - x_{14}x_{89} + x_{91} \\
R_{79} &= -x_{93}x_{96} + x_{96} + x_{98} \\
R_{80} &= -x_2x_{119} + x_{94}x_{192} - x_{97} - x_{98}x_{119} \\
R_{81} &= x_{99}x_{192} - x_{100}x_{122} - x_{101} \\
R_{82} &= x_{102}x_{192} - x_{103}x_{122} - x_{104} \\
R_{83} &= x_{105}x_{192} - x_{106}x_{122} - x_{107} \\
R_{84} &= x_{108}x_{192} - x_{109}x_{122} - x_{110} \\
R_{85} &= x_{111}x_{193} - x_{113}x_{122} - x_{114} \\
R_{86} &= x_{115}x_{193} - x_{116}x_{122} - x_{117} \\
R_{87} &= -x_{16}x_{119} + x_{122} \\
R_{88} &= x_{118} - x_{120}x_{122} - x_{121} \\
R_{89} &= -x_{122}x_{124} + x_{123} - x_{125} \\
R_{90} &= -x_{122}x_{127} + x_{126}x_{192} - x_{128} \\
R_{91} &= x_{129} - 2x_{129}x_{131} + x_{132} \\
R_{92} &= -x_{129}x_{132} - x_{132} + x_{133} \\
R_{93} &= -1 + x_{130}^2 + x_{132}^2 - x_{133}^2 \\
R_{94} &= 1 - x_{130}x_{135} - x_{134} + x_{136} \\
R_{95} &= -1 + 2x_{134} - x_{134}^2 + 2x_{134}x_{136} \\
&\quad + x_{135}^2 \\
R_{96} &= 1 + x_2^2 - 2x_2x_{130} + x_{137} \\
R_{97} &= -x_2x_{169} + x_{16} + x_{137}x_{138} - x_{168} \\
R_{98} &= -5 - 8x_1 - 4x_1^2 - 16x_1x_{15} \\
&\quad + 4x_1x_{130} - 16x_{15} - 16x_{15}^2 + 8x_{15}x_{130} \\
&\quad + 4x_{130} - x_{139} \\
R_{99} &= -1 - 2x_1x_{169} - 4x_{15}x_{169} + x_{93} \\
&\quad + x_{139}x_{140} - x_{168} - 2x_{169} \\
R_{100} &= -x_{130}x_{142} - x_{141} + x_{143} \\
R_{101} &= -1 - x_{141}^2 + 2x_{141}x_{143} + x_{142}^2 \\
R_{102} &= -x_{130}x_{145} - x_{144} + x_{146} \\
R_{103} &= -1 - x_{144}^2 + 2x_{144}x_{146} + x_{145}^2 \\
R_{104} &= -x_{130}x_{148} - x_{147} + x_{149} \\
R_{105} &= -1 - x_{147}^2 + 2x_{147}x_{149} + x_{148}^2 \\
R_{106} &= -x_{130}x_{151} - x_{150} + x_{152} \\
R_{107} &= -1 - x_{150}^2 + 2x_{150}x_{152} + x_{151}^2 \\
R_{108} &= -x_{130}x_{154} - x_{153} + x_{155}
\end{aligned}
$$

$$
\begin{aligned}
R_{109} &= -1 - x_{153}^2 + 2x_{153}x_{155} + x_{154}^2 \\
R_{110} &= -x_{130}x_{157} - x_{156} + x_{158} \\
R_{111} &= -1 - x_{156}^2 + 2x_{156}x_{158} + x_{157}^2 \\
R_{112} &= 1 - x_{130}x_{160} - x_{159} + x_{161} \\
R_{113} &= -1 + 2x_{159} - x_{159}^2 + 2x_{159}x_{161} \\
&\quad + x_{160}^2 \\
R_{114} &= -x_{130}x_{163} - x_{162} + x_{164} \\
R_{115} &= -1 - x_{162}^2 + 2x_{162}x_{164} + x_{163}^2 \\
R_{116} &= -x_{130}x_{166} - x_{165} + x_{167} \\
R_{117} &= -1 - x_{165}^2 + 2x_{165}x_{167} + x_{166}^2 \\
R_{118} &= 1 - x_{130}x_{169} - x_{168} + x_{170} \\
R_{119} &= -1 + 2x_{168} - x_{168}^2 + 2x_{168}x_{170} \\
&\quad + x_{169}^2 \\
R_{120} &= -x_{130}x_{172} - x_{171} + x_{173} \\
R_{121} &= -1 - x_{171}^2 + 2x_{171}x_{173} + x_{172}^2 \\
R_{122} &= -x_{130}x_{175} - x_{174} + x_{176} \\
R_{123} &= -1 - x_{174}^2 + 2x_{174}x_{176} + x_{175}^2 \\
R_{124} &= -1064 - x_0^2 - 40x_1 - 6x_1^2 \\
&\quad - 16x_1x_{15} - 64x_1x_{177} - x_2^2 - x_3^2 \\
&\quad - x_7^2 - x_8^2 - x_9^2 - x_{10}^2 \\
&\quad - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\
&\quad - 16x_{15} - 20x_{15}^2 - x_{16}^2 + 2x_{93} \\
&\quad - 2x_{93}^2 - x_{94}^2 - x_{95}^2 - x_{99}^2 \\
&\quad - x_{102}^2 - x_{105}^2 - x_{108}^2 - x_{111}^2 \\
&\quad - x_{112}^2 - x_{115}^2 - x_{118}^2 - x_{119}^2 \\
&\quad - x_{123}^2 - x_{126}^2 + x_{129} - 4160x_{177} \\
&\quad - 4096x_{177}^2 \\
R_{125} &= -x_{130}x_{179} + x_{188} \\
R_{126} &= -1 + x_{178}^2 + x_{179}^2 - x_{188}^2 \\
R_{127} &= -x_{181}x_{182} + x_{189} \\
R_{128} &= -1 + x_{180}^2 + x_{181}^2 - x_{189}^2 \\
R_{129} &= -x_{175}^2 + x_{190} \\
R_{130} &= x_{179} - 2x_{184}x_{190} - x_{190} \\
R_{131} &= -1 - 4x_{179} - 8x_{179}x_{183} + x_{182} \\
R_{132} &= -x_{130} - x_{178}x_{185} + x_{182} \\
R_{133} &= -x_{176} - x_{178}x_{186} + x_{180} \\
R_{134} &= -2x_{15} - 4x_{175}x_{187} + x_{181} \\
R_{135} &= 521 + x_0^2 - 2x_0x_{169} + 32x_1 \\
&\quad + x_1^2 - 2x_1x_{142} + 64x_1x_{177} + x_3^2 \\
&\quad - 2x_3x_{160} + x_7^2 - 2x_7x_{145} + x_8^2 \\
&\quad - 2x_8x_{148} + x_9^2 - 2x_9x_{151} + x_{10}^2 \\
&\quad - 2x_{10}x_{154} + x_{11}^2 - 2x_{11}x_{157} + x_{12}^2 \\
&\quad - 2x_{12}x_{163} + x_{13}^2 - 2x_{13}x_{166} + x_{14}^2 \\
&\quad - 2x_{14}x_{172} - 4x_{15}x_{175} - x_{130}x_{191} - 32x_{135} \\
&\quad - 64x_{135}x_{177} + x_{136}^2 - 32x_{142} - 64x_{142}x_{177} \\
&\quad - 2x_{143} + x_{143}^2 - 2x_{146} + x_{146}^2 \\
&\quad - 2x_{149} + x_{149}^2 - 2x_{152} + x_{152}^2 \\
&\quad - 2x_{155} + x_{155}^2 - 2x_{158} + x_{158}^2 \\
&\quad + x_{161}^2 - 2x_{164} + x_{164}^2 - 2x_{167} \\
&\quad + x_{167}^2 + x_{170}^2 - 2x_{173} + x_{173}^2 \\
&\quad + 2x_{175}^2 - 2x_{176} + x_{176}^2 + 2048x_{177} \\
&\quad + 2048x_{177}^2 + x_{191} - x_{194}^2 - x_{195}^2 \\
&\quad - x_{196} - x_{196}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{136} &= -x_{95}x_{192} + x_{119} \\
R_{137} &= -x_{112}x_{193} + x_{119} \\
R_{138} &= -1 - 4x_{15}^2 - x_{135}^2 - x_{142}^2 \\
&\quad - x_{145}^2 - x_{148}^2 - x_{151}^2 - x_{154}^2 \\
&\quad - x_{157}^2 - x_{160}^2 - x_{163}^2 - x_{166}^2 \\
&\quad - x_{169}^2 - x_{172}^2 + x_{175}^2 - x_{194}^2 \\
&\quad - x_{195}^2 - x_{196} - x_{196}^2 \\
R_{139} &= -3 - 4x_{97}^2 - 4x_{101}^2 - 4x_{104}^2 \\
&\quad - 4x_{107}^2 - 4x_{110}^2 - 4x_{114}^2 - 4x_{117}^2 \\
&\quad + x_{119}^2 - 4x_{121}^2 - 4x_{125}^2 - 4x_{128}^2 \\
&\quad - x_{197}^2 - x_{198}^2 + 3x_{199} - x_{199}^2 \\
R_{140} &= -x_1x_{11} + x_{200} \\
R_{141} &= x_{93}x_{202} - 2x_{94}x_{142} - 2x_{99}x_{145} - 2x_{102}x_{148} \\
&\quad - 2x_{105}x_{151} - 2x_{108}x_{154} - 2x_{111}x_{157} - 2x_{115}x_{163} \\
&\quad - 2x_{118}x_{166} - 2x_{123}x_{172} - 2x_{126}x_{175} - 2x_{130}x_{202} \\
&\quad + 2x_{142}x_{143} + 2x_{145}x_{146} + 2x_{148}x_{149} + 2x_{151}x_{152} \\
&\quad + 2x_{154}x_{155} + 2x_{157}x_{158} + 2x_{163}x_{164} + 2x_{166}x_{167} \\
&\quad + 2x_{172}x_{173} + 2x_{175}x_{176} + x_{201} \\
R_{142} &= -10 - x_{93}x_{201} + x_{94}^2 - 2x_{94}x_{141} \\
&\quad + x_{99}^2 - 2x_{99}x_{144} + x_{102}^2 - 2x_{102}x_{147} \\
&\quad + x_{105}^2 - 2x_{105}x_{150} + x_{108}^2 - 2x_{108}x_{153} \\
&\quad + x_{111}^2 - 2x_{111}x_{156} + x_{115}^2 - 2x_{115}x_{162} \\
&\quad + x_{118}^2 - 2x_{118}x_{165} + x_{123}^2 - 2x_{123}x_{171} \\
&\quad + x_{126}^2 - 2x_{126}x_{174} + 2x_{141}x_{143} + 2x_{144}x_{146} \\
&\quad + 2x_{147}x_{149} + 2x_{150}x_{152} + 2x_{153}x_{155} + 2x_{156}x_{158} \\
&\quad + 2x_{162}x_{164} + 2x_{165}x_{167} + 2x_{171}x_{173} + 2x_{174}x_{176} \\
&\quad + x_{202} \\
R_{143} &= x_{93}x_{204} - 2x_{95}x_{135} - 2x_{112}x_{160} - 2x_{119}x_{169} \\
&\quad - 2x_{130}x_{204} + 2x_{135} + 2x_{135}x_{136} + 2x_{160} \\
&\quad + 2x_{160}x_{161} + 2x_{169} + 2x_{169}x_{170} + x_{203} \\
&\quad - x_{204} \\
R_{144} &= -2 - x_{93}x_{203} + x_{95}^2 - 2x_{95}x_{134} \\
&\quad + x_{112}^2 - 2x_{112}x_{159} + x_{119}^2 - 2x_{119}x_{168} \\
&\quad + 2x_{134} + 2x_{134}x_{136} + 2x_{159} + 2x_{159}x_{161} \\
&\quad + x_{168}^2 - x_{169}^2 + x_{203} + x_{204} \\
R_{145} &= -1 - x_1 - 2x_1x_{205} + x_3 \\
R_{146} &= -16x_1 - 32x_1x_{177} + x_{207} \\
R_{147} &= -1 - 8x_1x_{207} + x_{206}^2 - 16x_{207}^2 \\
R_{148} &= -8388608 + x_1 - 16777216x_{177} - 33554432x_{177}x_{208} \\
&\quad - 16777216x_{208}
\end{aligned}
$$

C Exact reciprocal ports and substitution ledgers

This appendix, together with the complete conservative residual array in Appendix B, defines every selected reciprocal polynomial without a search or an implicit coefficient. All x_i, R_i in the port tables refer to that appendix, not to the principal quartic.

C.1 The common principal pairs and power requests

Put $A = x_{130}$ and $d = A - 1$. The thirteen pairs have the following literal ports. The norm and multiplication rows remain part of the source.

Pair	k_i	X_i	Y_i	Z_i	Gate	Norm
0	$16 + 32x_{177}$	$1 + x_{136}$	x_{135}	$1 - x_{134} + x_{136}$	94	95
2	$16 + x_1 + 32x_{177}$	x_{143}	x_{142}	$-x_{141} + x_{143}$	100	101
3	x_7	x_{146}	x_{145}	$-x_{144} + x_{146}$	102	103
5	x_8	x_{149}	x_{148}	$-x_{147} + x_{149}$	104	105
6	x_9	x_{152}	x_{151}	$-x_{150} + x_{152}$	106	107
7	x_{10}	x_{155}	x_{154}	$-x_{153} + x_{155}$	108	109
8	x_{11}	x_{158}	x_{157}	$-x_{156} + x_{158}$	110	111
9	x_3	$1 + x_{161}$	x_{160}	$1 - x_{159} + x_{161}$	112	113
10	x_{12}	x_{164}	x_{163}	$-x_{162} + x_{164}$	114	115
11	x_{13}	x_{167}	x_{166}	$-x_{165} + x_{167}$	116	117
12	x_0	$1 + x_{170}$	x_{169}	$1 - x_{168} + x_{170}$	118	119
13	x_{14}	x_{173}	x_{172}	$-x_{171} + x_{173}$	120	121
14	$2x_{15}$	x_{176}	x_{175}	$-x_{174} + x_{176}$	122	123

The four base blocks and all fifteen (pair, m) requests are:

Block	Base n_j	Requests (i, m)
0	x_2	$(12, x_{16})$
1	$2 + 2x_1 + 4x_{15}$	$(12, -1 + x_{93})$
2	$-1 + x_{93}$	$(0, x_{95})$, $(9, x_{112})$, $(12, x_{119})$
3	x_{93}	$(2, x_{94})$, $(3, x_{99})$, $(5, x_{102})$
		$(6, x_{105})$, $(7, x_{108})$, $(8, x_{111})$
		$(10, x_{115})$, $(11, x_{118})$, $(13, x_{123})$
		$(14, x_{126})$

Define $T_j, H_j, S_j, E_j, \Omega$ from precisely these ports as in Section 16. Thus Z_i is the displayed polynomial port, not a new independent variable. To define the literal reduced source chart, put $\widehat{E}_0 = E_0$, $\widehat{E}_1 = E_1$ and

$$\widehat{E}_2 = E_2 + R_{95} + R_{113}, \qquad \widehat{E}_3 = E_3 + \sum_{r \in \mathcal{J}} R_r,$$

where $\mathcal{J} = \{101, 103, 105, 107, 109, 111, 115, 117, 121, 123\}$. These exact retained-norm corrections are necessary for the literal coefficient arrays, although they vanish at source zeros. Use $\widehat{E}_j$, not E_j, in every seed pool below. The source orientation identity is

$$\Omega - (R_{135} + dx_{191}) = -R_{138}.$$

This identity is exact in the conservative source coordinates, including their existing row chart. The joint numerator is exactly $\Omega' = \Omega + R_{124}$.

C.2 The seeds before elimination

For a fixed ordered block set I and a new variable z_I, write

$$\mathcal{P}_I = \sum_{j\in I} \widehat{E}_j \prod_{h\in I\setminus\{j\}} T_h - z_I \prod_{j\in I} T_j.$$

The sparse 114 seed retains all source rows except $96, 97, 98, 99, 141, 142, 143, 144$, deletes coordinates $137, 138, 139, 140, 201, 202, 203, 204$, and appends $\mathcal{P}_{\{0,1\}}$ followed by $\mathcal{P}_{\{2,3\}}$. The other retained rows preserve ascending original order. First retain coordinates in ascending original order, then append the two new pool variables in that same pool order. This gives 203 coordinates and 143 rows. The joint 113 seed additionally deletes row 135 and coordinate 191. Its appended rows, in order, are

$$\Omega' T_0 + d\mathcal{P}_{\{0\}}, \qquad \mathcal{P}_{\{1,2,3\}}.$$

Use the same ascending retained-coordinate order and then the two new pool variables. This gives 202 coordinates and 142 rows. The matched unfused control instead keeps row 135 and coordinate 191 and appends $\mathcal{P}_{\{0\}}, \mathcal{P}_{\{1,2,3\}}$. The 138-variable endpoint starts directly from the complete 209-coordinate, 149-row conservative source, with no new pool seed. Its 71 monic substitutions are therefore generic exact eliminations, not a gain credited to the new joint theorem.

C.3 How to read the complete ordered tables

In each following table, (j, r, v, ϵ) means: at ordinal j, the current row with original seed label r has the form $\epsilon u_v + f$, with no other u_v term. Replace u_v by $-\epsilon f$ in every other current row and delete row r. Row and coordinate labels remain the initial seed labels throughout; do not compact after a step. At the end retain the undeleted row labels in ascending order, and compact the undeleted coordinate labels in ascending order. The sum of squares of those remaining rows is the literal final polynomial. Thus the tables and fixed seeds alone specify every coefficient, with no optimizer or choice left to execute. The adjacent `NATIVE_DEFINITION.json` files also give every resulting pivot polynomial explicitly; independent replay verifies each unit coefficient and the complete final residual arrays. The tables below are parsed independently against those files.

C.4 The 138 variable degree eight polynomial

The fixed seed has 209 coordinates and 149 rows. Perform all 71 steps in ordinal order, reading each four-column block downward, then the next block.

j	r	v	ϵ	j	r	v	ϵ	j	r	v	ϵ
0	68	82	1	24	0	6	1	48	104	147	-1
1	54	68	-1	25	72	86	1	49	100	141	-1
2	55	64	1	26	39	53	1	50	112	159	-1
3	74	88	1	27	125	188	1	51	102	144	-1
4	48	62	-1	28	133	180	1	52	116	165	-1
5	129	190	1	29	36	50	1	53	122	174	-1
6	63	77	1	30	86	117	-1	54	106	150	-1
7	20	34	1	31	88	121	-1	55	31	45	1
8	51	65	-1	32	89	125	-1	56	14	29	1
9	49	63	-1	33	82	104	-1	57	78	91	1
10	96	137	1	34	81	101	-1	58	118	168	-1
11	132	182	1	35	90	128	-1	59	17	19	1
12	79	98	1	36	84	110	-1	60	34	48	1
13	50	69	-1	37	85	114	-1	61	12	27	1
14	76	90	1	38	10	25	1	62	23	37	1
15	62	75	1	39	143	203	1	63	137	119	1
16	46	60	1	40	134	181	1	64	42	56	1
17	52	66	-1	41	141	201	1	65	26	40	1
18	98	139	-1	42	108	153	-1	66	56	70	1
19	146	207	1	43	83	107	-1	67	28	42	1
20	6	21	1	44	110	156	-1	68	21	35	1
21	61	74	1	45	114	162	-1	69	15	30	1
22	58	72	1	46	120	171	-1	70	44	58	1
23	92	133	1	47	94	134	-1				

The final count is 138 coordinates and 78 rows; all are live. The ordered complement of the table's deleted indices agrees exactly with the supplied retained-index arrays.

C.5 The sparse 114 variable degree sixteen polynomial

The fixed seed has 203 coordinates and 143 rows. Perform all 89 steps in ordinal order, reading each four-column block downward, then the next block.

j	r	v	ϵ	j	r	v	ϵ	j	r	v	ϵ
0	74	88	1	30	92	133	1	60	91	132	1
1	125	186	1	31	130	177	1	61	23	37	1
2	68	82	1	32	86	117	-1	62	78	91	1
3	55	69	-1	33	90	128	-1	63	14	29	1
4	6	21	1	34	121	184	1	64	76	90	1
5	51	65	-1	35	36	50	1	65	80	97	-1
6	49	63	-1	36	48	62	-1	66	24	38	1
7	72	86	1	37	57	71	1	67	21	35	1
8	79	98	1	38	71	85	1	68	126	175	1
9	50	64	-1	39	10	25	1	69	87	122	1
10	62	75	1	40	102	146	-1	70	16	31	1
11	128	178	1	41	34	48	1	71	63	77	1
12	75	89	1	42	19	33	1	72	123	185	1
13	31	45	1	43	118	170	-1	73	114	166	1
14	46	60	1	44	17	19	1	74	32	46	1
15	54	68	-1	45	42	56	1	75	37	51	1
16	52	66	-1	46	100	143	-1	76	58	72	1
17	20	34	1	47	108	155	-1	77	2	0	1
18	129	176	1	48	106	152	-1	78	44	58	1
19	138	199	1	49	61	74	1	79	40	54	1
20	88	121	-1	50	116	167	-1	80	132	119	1
21	82	104	-1	51	28	42	1	81	11	26	1
22	89	125	-1	52	112	161	-1	82	137	3	1
23	84	110	-1	53	110	158	-1	83	64	78	1
24	81	101	-1	54	96	137	-1	84	29	43	1
25	39	53	1	55	98	140	-1	85	13	28	1
26	85	114	-1	56	104	149	-1	86	15	30	1
27	136	196	1	57	94	134	-1	87	25	39	1
28	0	6	1	58	12	27	1	88	43	57	1
29	83	107	-1	59	59	73	1				

The final count is 114 coordinates and 54 rows; all are live. The ordered complement of the table's deleted indices agrees exactly with the supplied retained-index arrays.

C.6 The joint 113 variable degree sixteen polynomial

The fixed seed has 202 coordinates and 142 rows. Perform all 89 steps in ordinal order, reading each four-column block downward, then the next block.

j	r	v	ϵ	j	r	v	ϵ	j	r	v	ϵ
0	71	85	1	30	89	125	-1	60	72	86	1
1	46	60	1	31	90	128	-1	61	58	72	1
2	49	63	-1	32	81	101	-1	62	32	46	1
3	6	21	1	33	82	104	-1	63	37	51	1
4	74	88	1	34	84	110	-1	64	44	58	1
5	54	68	-1	35	86	117	-1	65	75	89	1
6	66	80	1	36	83	107	-1	66	40	54	1
7	128	178	1	37	39	53	1	67	9	24	1
8	79	98	1	38	85	114	-1	68	104	149	-1
9	52	66	-1	39	57	71	1	69	98	140	-1
10	125	186	1	40	78	91	1	70	100	143	-1
11	62	75	1	41	19	33	1	71	106	152	-1
12	20	34	1	42	17	19	1	72	96	137	-1
13	0	6	1	43	61	74	1	73	118	170	-1
14	63	77	1	44	34	48	1	74	112	161	-1
15	50	64	-1	45	28	42	1	75	116	167	-1
16	76	90	1	46	42	56	1	76	102	146	-1
17	121	184	1	47	59	73	1	77	110	158	-1
18	129	176	1	48	91	132	1	78	94	134	-1
19	69	83	1	49	16	31	1	79	114	166	1
20	14	29	1	50	80	97	-1	80	108	155	-1
21	135	195	1	51	24	38	1	81	11	26	1
22	92	133	1	52	136	3	1	82	13	28	1
23	31	45	1	53	21	35	1	83	2	0	1
24	55	69	-1	54	12	27	1	84	29	43	1
25	64	78	1	55	23	37	1	85	131	119	1
26	137	198	1	56	48	62	-1	86	15	30	1
27	36	50	1	57	126	175	1	87	25	39	1
28	130	177	1	58	87	122	1	88	43	57	1
29	88	121	-1	59	123	185	1				

The final count is 113 coordinates and 53 rows; all are live. The ordered complement of the table's deleted indices agrees exactly with the supplied retained-index arrays.

D The complete retained Start only native proof

This appendix retains the complete incoming ordinary native proof, dated 25 October 2026. Its fifteen-positive P66 equation is an inherited bridge, not an additional new endpoint. Historical relative paths refer to the retained native source subtree. The retained Round22 ten-positive transfer is proved in Section 20.

This note is an ordinary mathematical construction, composed with the retained K215 / native / positive-P66 / syntactic-refutation calibration. It changes all four fixed integer coefficients coherently. It does not assert consistency, global minimality, an end-to-end proof-assistant certificate, or worldwide novelty. The idea of trying a Start-only marker is explicitly suggested by the active repository SIZE_REDUCTION.md; this note completes a particular shorter layout, its complete proof and its finite controller certificate.

D.1 Exact source and result

The immutable comparison commit is `fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb` of `VladimirReshetnikov/ProveIt`. The immediate whole ordinary source is `Computability/DiophantineConsistency/research/zero-weight-flag-independent/PROOF.md`. The full 943-tile language, 430 actions, 436-character numeral and decoder are exactly those printed in `Computability/DiophantineConsistency/equation.tex` at that commit. Local copies are retained in `inherited/`.

The same fifteen-variable P66 free-port polynomial now has a coherent tuple with cell length 4127 instead of 6007 and binary cell size

```
d = 47 * 4127 = 193969,    B = 2^193969,    beta = B-1.
```

Its four minimal unsigned big-endian coefficient payloads occupy 97046 bytes, versus 141226, a reduction of 44180 bytes (about 31.28%). There are 66 live integer-leaf arithmetic operations and exact degree 157, as before. The new DC kernel has 1957 nonzero rho-adic coefficients, versus 2899: it deletes 942 unit-kernel terms. These constant-recipe counts are distinct from the 66-gate evaluation cost. No unknown or equation is hidden in the coefficient recipe.

Every assertion below uses the new tuple. Old6007 K/controller residues are never reused.

D.2 The whole language, support and coefficients

Put rho=2^47, M=2^44 and mu=M-2. Retain the old tile list `t_j=(w_j,n_j,p_j,s_j)`, j=0,...,942. Colors interpreted by the native clauses are the positive complements of these old labels: eight-bit east/west and nine-bit north/south. In particular the forced Start tile is `(2,435,3,1)`, so its complemented south label is 510. Its bits 1 through8 are all one. All genuine complemented colors are positive; empty cells have zero colors.

The physical positions in one rho-cell are exactly

```
E = {0,...,942} union {943,...,951} union {973,...,981}
    union {992,...,999} union {1002,...,1009} union {990,991,1031}.
```

There are 980 positions. The low dummy 990 can carry digits 0,...,3; all other physical digits are Boolean. Position 1031 is a separate zero-weight physical flag. The flag has no clause weight. Define every coefficient c_e, for 0<=e<=3296, by

```
c_j = 1 + 2^10(1+w_j+2^8 n_j+2^17 p_j+2^25 s_j), 0<=j<=942;
c_(1002+j)=2^(10+j), c_(992+j)=2^(27+j), 0<=j<8;
c_(973+j)=2^(18+j), c_(943+j)=2^(35+j), 0<=j<9;
all remaining coefficients are zero.
```

Thus c_990=c_991=c_1031=0. These3297 coefficients are pointwise identical to the old zero-weight6007 coefficient vector: changing the physical flag does not change that vector. There are977 nonzero coefficients, each<M, and the34 label coefficients sum to M-1024. The checker independently decodes every action and tile from the436-character numeral before comparing all 3297 weights with the frozen parent vector.

Let G_m(x)=sum_(j=0)^(m-1)x^j and define

```
Psi(x,t)=x^992(x^10+t^17)G_8(tx)
         +t^8 x^943(x^30+t^17)G_9(tx)
         +t^34(2-t)x^1031,
Phi(x)=sum_(j=0)^942 c_j x^j + 2^10 Psi(x,2),
Lambda=rho^2042+rho^3074.
```

At t=1, Psi specifies35 physical label/flag positions. At t=2, its flag term vanishes and it gives the same weighted Phi as before. Physical support must not be inferred from nonzero Phi coefficients.

D.3 Four fixed integers, completely specified

Take L=4127 and B=rho^L. Define

```
DC = rho^20 + rho^30 + rho^1031 + Lambda*Phi(1/rho),
K = DC + B*rho^10,
MC = beta + rho - (rho+3)rho^990 - G_943(rho) - Psi(rho,1),
MF = B+3+(M-2)Lambda + rho^973 G_9(rho)
     +rho^1012 G_8(rho)+rho^1031
     +2^29 rho^4124-rho^4106.
```

Every reflected exponent is nonnegative: the first reflected coefficient is at2042-1009=1033. Hence the apparently rational expression in DC is an integer specified by a finite sum. In the P66 graph use the four ports `beta,K0=K,MC,MF`. This MF is the source mask, already shifted once by beta. Write MF_native=MF-beta only in the proof.

The core mask consists of two43-bit clause masks,17 label-copy tests, one flag-copy test and the origin test, for105 bits. The zero padding is 18 complete rho-fields4106,...,4123 and29 bits at4124, for875 more bits. The positive padding identity is

```
2^29 rho^4124-rho^4106
  = (rho-1)sum_(j=4106)^4123 rho^j + (2^29-1)rho^4124.
```

The full finite integer computation verifies

```
0<DC<B,    0<MC<beta,    0<MF_native<beta,
pc(MF_native)=980,    pc(MC)=d-980,
MC=2 mod4,    MF_native=4 mod8.
```

No carries or borrows in these explicit mask calculations are left to an informal picture. They are checked on the complete integers. These are the same numerical interface conditions used by the inherited outer arithmetic.

D.4 The initial clause lemma and bounded cyclic carries

The zero-weight clause lemma of the immediate source is unchanged. For a selector population n, the complemented-label word W and selected tile words u_j, write

```
Q = 2^34 n + W - sum_selected u_j.
```

The clause raw value is n+2^10 Q. With incoming carry in0,...,29 and the mask mu, the source's integer inequalities force n=0 or1. For n=0 all labels are zero; for n=1 exactly one selected tile determines all34 complemented labels. A flag is not recovered or used by this lemma.

The new DC has three bare unit terms, at20,30,1031, and two reflected coefficient blocks. The block separation is1032, larger than max(E)=1031. Therefore a given output has at most one predecessor for each reflected coefficient, even though two full blocks occur in DC. The shifted low-dummy extra0/2 is treated separately. It meets at most one reflected block, since their separation is again absent from E. Its unit hit2021 precedes every reflected extra-dummy hit, whose first possible field is2023. Its bare and spatial extra hits are1010,1020,1000, all distinct.

One may use the inherited hand bound on the full coefficient sum and obtain the same29 carry bound. The finite certificate also supplies an independent stronger direct check: form the full nonnegative convolution DC*1_E, add twice the translated DC at dummy 990, and take the exact maximum Hmax. This is a finite array containing every permitted non-temporal raw value. The complete arithmetic check is

```
Hmax + 3 + 3rho/4 + 29 < 30rho.
```

Here3 bounds the spatial source digit; 3rho/4 bounds the temporal digit in any fractional binary phase. At phase46 the only extra odd spill is from low dummy 990 into991, already in E. This bound is independent of alignment, the flag value, or a solution to the clause tests.

The monotone cyclic carry map preserves the finite box0,...,29. Iterating from zero stabilizes and telescoping identifies its digit word with the transport residue. The strict representative supplied by the inherited outer inequalities rules out the zero/all-ones ambiguity, exactly as in the source. Since the clause centers differ by1032, they cannot both have a temporal physical predecessor in any cell or fractional phase. At least one clean clause applies. Thus every cell has zero or one selector and its correct complemented labels before any alignment assertion.

D.5 A nine-target alignment lemma

The new singleton rho^1031 copies only selector0 (Start) to field1031. It does not copy all 943 selectors. Below1033 there is no reflected term. There are at most three Boolean bare contributions and the bounded dummy contribution; the safe bound9 suffices. Hence

```
9 + 3rho/4 + 29 < rho
```

makes every outgoing carry in fields0,...,1032 zero. This includes every incoming carry at targets974,...,981 and1031.

In the forced Start cell, the eight targets974,...,981 each receive one from the corresponding south label. Target1031 also receives one from the Start selector. Each of their low-bit tests requires an odd temporal contribution. Fractional phases1,...,45 have only even temporal digits. Phase46 has at most one odd low-dummy spill, and cannot meet nine targets. Thus the fractional phase is zero.

Let r be the remaining rho-digit offset moduloL, and put u=974-r moduloL. The eight temporal preimages u,...,u+7 must be physical. They cannot wrap across the long unsupported top interval. Within each predecessor cell at most one of them may be a selector, by the already established clause lemma. The complete possible starts are therefore exactly

```
u in {942,943,944,973,974,990,991,992,1002}.
```

This is also immediate by inspecting the five physical intervals: at most one selector permits only942,...,944 at the end of0,...,951. The ninth temporal preimage is u+57. Among the nine displayed starts, only u=942 and u=974 have u+57 in E, namely999 and1031. The first case would require selector942 to be one and also the south low bit at943 to be one in the same predecessor cell. The clean clause already forces that cell's labels to be those of tile 942. The complete decoder

gives tile 942=(4,437,2,437), whose complemented south label is511-437=74. Its low bit is zero, a contradiction. Therefore u=974 and r=0 moduloL. No value of the physical flag was assumed, and no uniqueness of Start outside the forced cell was assumed. The finite checker additionally tests every r=0,...,L-1, partitioning possible wrapped preimages by their actual predecessor cells. Support and selector capacity leave offsets0,32; the decoded labels exclude32, leaving only0.

D.6 Exact copies, all resets and soundness

The full convolution identities are

```
target973+j: current943+j, 0<=j<9;
target1012+j: current992+j plus spatial1002+j, 0<=j<8;
target1031: currentselector0;
targets2042 and3074: sum_e c_e current_e.
```

The singleton cannot contaminate a clause: its would-be predecessors are 1011 and2043, both absent. Its would-be predecessor at2041 is1010, also absent. Reflection starts above all copy tests. The bare/spatial sources at all clause targets and predecessors are absent. These identities are checked as complete coefficient lists, not just at intended witnesses.

After alignment, the17 ordinary copies force every temporal/spatial color equality. The flag copy gives precisely

```
flag_(temporal predecessor of i) = StartSelector_i.
```

Since temporal rotation is a permutation, this uniquely determines all flags from Start selectors. Occupation is instead forced by color equality: empty cells have zero colors, genuine tiles have positive complemented colors, and spatial one-cell rotation propagates occupancy from the forced Start to every cell. Thus every cell contains exactly one genuine tile. The flag is never required to equal occupancy.

Let Aadj=sum_(e,e+1 in E)c_(e+1)x_e. The two complete clause predecessor values are

```
field2041: Aadj;
field3073: Aadj+c_0*x_1031.
```

The extra term comes from the unique pair1031-0=1031; difference1033 is absent. With at most one selected tile, Aadj<=2M-1024, including selector-to-label adjacency942/943 and dummy-to- label adjacency991/992. The flag has no adjacent nonzero coefficient. Since c_0<M, both predecessors plus incoming29 are<rho and reset carry. The actual occupied clause value is M+1<rho.

The maximum raw product field is3074+1031=4105. Its entire value is c_0*x_1031<M; it resets incoming29. All fields4106,...,4126 are raw zero, and aligned temporal/spatial support is absent there. This proves every875 padding bit and a terminal zero field. Each B-cell origin is reset. Origin behavior is identical to the source: DC and the spatial shift have positive exponents, so only the temporal Start selector can meet an origin.

The forced Start's complemented north76 differs from its south510, so the temporal stride is nonzero. Inverse complementation now gives exactly the retained genuine marked blank computation and syntactic ZFC refutation. Soundness uses no finite controller or prime-existence assumption.

D.7 Fresh canonical encoding and strict inequalities

From a finite syntactic refutation, the retained helical-padding theorem gives genuine one-Start rectangles with independently cofinal width and height. Complement the colors and encode all cells on the new E. Define each flag by the unique permutation rule in Section6. Supply both

dummies initially zero. Every Boolean subset of safe low990 dummy bits leaves all selectors, colors and the flag relation unchanged.

The same bounded cyclic carry construction is valid before choosing a strict representative. The exact aligned resets proved above apply to every such subset. Occupied clauses make F positive, and the terminal zero field gives

```
0<F<q/rho<q/7,
0<C<q/rho^(L-1032),
2C+F<q*(1/7+2/rho^(L-1032))<q/3.
```

Thus alpha=q-F-C is positive, and0<F<q-1. The flags have been encoded afresh; the old 6007 word with all flags one is not reused.

D.8 Actual rank and the new controller

Set t=dN, q=2^t, J=(q-1)/beta and

```
S'=(C-1)+qF,
Tmask=(MC*J+1)+q*(MF_native*J-1).
```

The inherited outer mask theorem gives disjoint S' and Tmask belowq^2. The explicit populations and origin changes give pc(Tmask)=t+2. With exactly one source-mask shift, the actual P66 rank is

```
R=[q(q-F)-C](q^2-1)+(MC+qMF)J
  =(q^2-1-S')q^2+(S'+Tmask).
```

It follows that pc(R)=3t+2. The unchanged origin residues C=1 mod4, F=0 mod4 and J=1 mod4 give R=3 mod4.

A safe bit at cellj changes C by rho^990 B^j. The complete DC/dummy maximum exponent is3074+990=4064. Its other later-cell exponents are1000 (spatial) and990 (temporal). All remain belowL with at least three free rho-digits. Different later cells or distinct later exponents prevent coefficient collisions. Exactly as in the source, for j+1<N and j+width<N one obtains a positive ordinary increment deltaRaw<q/7 and, uniformly over intermediate subsets, F_old+deltaRaw<2q/7<q-1. Thus no modular wrap is hidden in

```
deltaR=-Gamma B^j,
Gamma=rho^990(q^2-1)[1+q(K+B^width)].
```

The new finite certificate gives the following exact controller data:

```
L=4127 prime, d=193969,
ord_47(2)=ord_47(B)=23,
ord_4127(2)=ord_4127(B)=2063,
spacing=lcm(23,2063)=47449,
lambda=474491=10*47449+1 prime > d.
```

All orders are checked by their full exponent and every distinct prime factor of that exponent. Both orders are coprime to d. Also B^2!=1 modulo each of47,4127,lambda, and the certificate gives

```
H0=1+B*K+B^2 = 9 mod47,
H0 = 3487 mod4127,
H0 = 153928 mod474491,
B^(lambda-1) = 1+135285*lambda mod lambda^2,
0 < 135285 < lambda.
```

The precise last residue is in `start_only_certificate.json`. Primality here is established by full trial division up to the square root, so no probabilistic primality or unsupported prime assertion is used. The dense Phi evaluation, sparse DC evaluation and complete integer K agree at all three primes.

For completeness choose independent large width=lambda^r_w and height=lambda^r_h, with N=width*height=lambda^n, N>d*lambda^2 and height>lambda. Put T_orbit=N/lambda and C0=d*lambda. Every power of lambda is1 modulo lambda-1, so q=B^N=B^width=B at every prime divisor ofd*lambda. The new units above make Gamma a unit modulodN. The valuation-one lift gives exact orderT_orbit ofB^(lambda-1) modulo lambda^n, while that power is1 modulod. CRT gives its orbit as1+C0*z, 0<=z<T_orbit.

For clarity, the source's subset construction is complete with these new parameters: normalize a target byB^-2 and choose epsilon in{0,1} so that subtracting epsilon*B leaves a nonzero residue modulo lambda. Its least representative a0 moduloC0 lies in1,...,C0-1 and is a unit moduloT_orbit. Write the target a0+C0*u and solve

```
a0*z0+a0*(a0-1)/2 = u modT_orbit.
```

The a0 consecutive distinct orbit coordinates sum to the target. Use the safe bit indices2+(lambda-1)j and the additional index3 exactly when epsilon=1. Their largest index is N-T_orbit-lambda+3; adding1 or width stays belowN. Applying this subset to (R0-d*width)*Gamma^-1 modulo dN gives

```
R=d*width mod dN.
```

All masks and inequalities are preserved. The source's rank growth yields R>q(q^2-1)>dN+d*width. Hence the required positive outer power and transport witnesses are constructed exactly as before with the actual new K.

D.9 P66 and the final consistency equivalence

Use precisely the fifteen independent positive integer variables

```
J,F,alpha,z,f,h,i,T,s,w,tau,eta,zeta,y,gamma.
```

All other names in the following display are polynomial abbreviations:

```
q_-=beta*J, q=q_-+1, C=q-F-alpha,
X=wq, Y=sq^3, E=XY, k=eta+zeta,
ell=EkY, c=kY+eta, a=E+Y, H=4a+3,
Delta=a^2+H, D=X+ac+gamma*H,
R=[q_-(q-F)+alpha](q^2-1)+(MC+qMF)J,
V=f^2(cT-R)-c, S=i*Delta*c^2,
N0=tau^2-ell(ell+k), N1=D^2-Delta*c^2,
Na=S^2(V^2-y^2)+y^2, Nk=k-hE-R,
Nt=(K+w)C+(q-F)-z*q_-, Ns=Delta*f^2-S^2,
P=N0*N1*Na*Nk*Nt*Ns-Delta.
```

The single Diophantine equation is P=0. The complete source's Section12 establishes both P66 witness maps for any coherent native tuple with the scalar boxes, masks, genuine native soundness and cofinal controller completeness proved in Sections3–8 above. None of its Pell constructions depends on the old flag location or on setting all flags one. Its proof uses the exact rank identities pc(R)=3t+2, R=3 mod4, its canonical bounds and the actual native transport; all have just been reconstructed for this tuple. Its signed unit analysis, main/strong/auxiliary index recovery, half-binomial projection and positive completeness maps therefore apply unchanged, with the new four coefficients together.

Consequently ordinary ZFC proves

```
(there exist fifteen positive integers with P=0)
   iff (there exists a finite syntactic ZFC refutation).
```

Negating yields unsolvability iff Con(ZFC). The ordinary proof does not decide either side. It composes the retained explicit foundational and machine calibration, rather than replacing them by the numerical box conditions alone.

The literal free-port P66 graph is unchanged. Its66 live integer gates, fifteen live positive inputs and exact factor degrees (22,18,62,7,2,46) are retained. The full leading form has the nonzero monomial coefficient8*beta^92, so the output degree remains exactly157 at the new positive beta. Different fixed tuples give different specialized polynomials; no equality with the old 6007 specialization is claimed.

D.10 Verification and useful next questions

The previously retained 4159 construction uses physical flag 1042 and a support-only nine-target alignment argument. This 4127 refinement additionally uses the actual decoded color of tile 942; it saves 752 further raw bytes. The two coefficient tuples and their controller certificates are distinct.

`python3 start_only.py -write` deterministically creates the complete four binary coefficients, all 943 tiles, all 3297 weights and the finite certificate. `python3 start_only.py` reconstructs them without writing. Python optimization does not remove guards: every check uses an explicit exception. The checker reads the frozen coefficient vector as data, but imports no inherited implementation and separately decodes the whole seed. It checks full convolution lists at every designated field, every possible digit-offset alignment, the actual carry envelope, all mask bits, every controller order/unit/lift and every coefficient byte.

The unbounded proof is Sections 4–9, not the finite tests. An independent review should especially challenge the order: clean clauses first, then at-most-one-selector, then nine-target alignment, then flag interpretation and occupied-cell propagation. Treating the flag as occupancy before alignment would reinstate the unnecessary 943-term kernel.

Further genuine opportunities are a shorter separated label layout, a smaller guaranteed-zero padding cost, or another physical alignment pattern with fewer marker tests. They require complete new geometric and controller proofs. This note supplies no global lower bound for cell length, coefficient storage, polynomial degree, variable count or arithmetic operations.